

Practical - 5

AIM: Implementation of Windows security using firewalls and other tools.

Tools: Windows Defender Firewall

Description:

Windows Firewall is a security feature that helps to protect your device by filtering network traffic that enters and exits your device. This traffic can be filtered based on several criteria, including source and destination IP address, IP protocol, or source and destination port number. Windows Firewall can be configured to block or allow network traffic based on the services and applications that are installed on your device. This allows you to restrict network traffic to only those applications and services that are explicitly allowed to communicate on the network.

The default behavior of Windows Firewall is to:

- block all incoming traffic, unless solicited or matching a rule
- allow all outgoing traffic, unless matching a rule

Firewall rules

Firewall rules identify allowed or blocked network traffic, and the conditions for this to happen. The rules offer an extensive selection of conditions to identify traffic, including:

- Application, service or program name
- Source and destination IP addresses
- Can make use dynamic values, like default gateway, DHCP servers, DNS servers and local subnets
- Protocol name or type. For transport layer protocols, TCP and UDP, you can specify ports or port ranges. For custom protocols, you can use a number between 0 and 255 representing the IP protocol
- Interface type
- ICMP/ICMPv6 traffic type and code

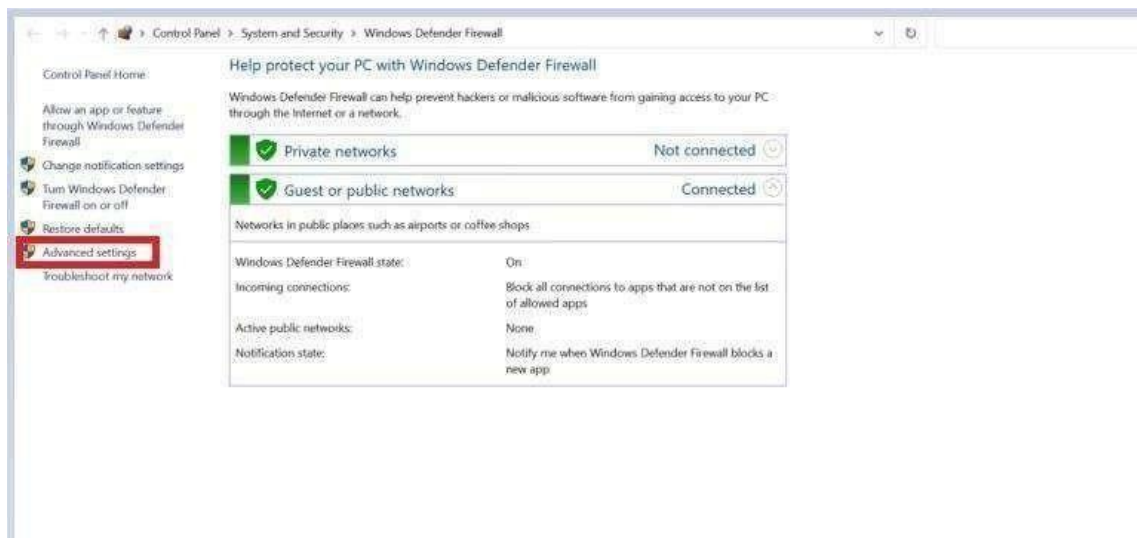
Implementation:

How Can You block a Program?

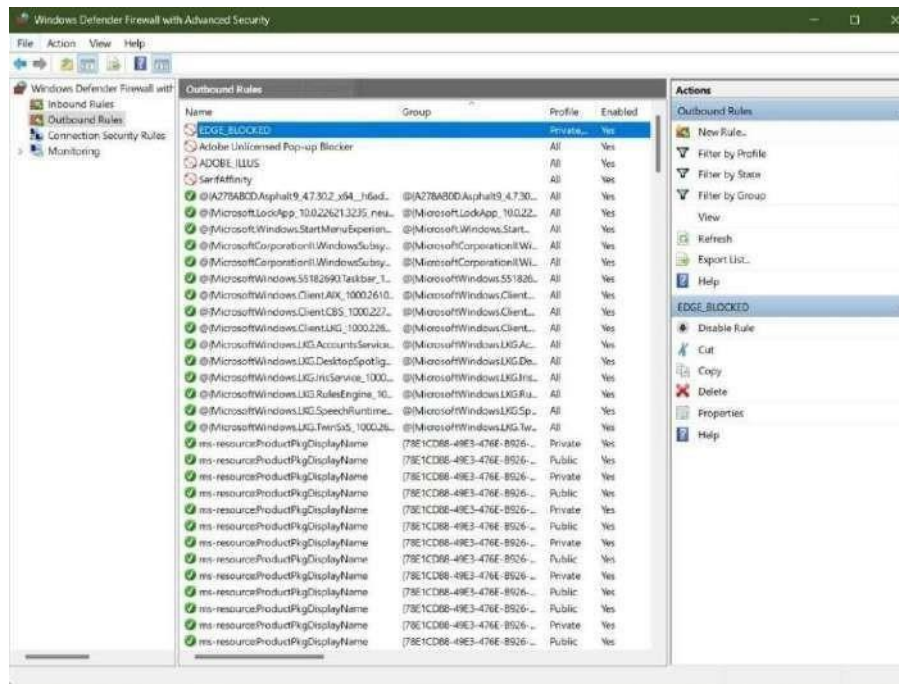
Step: 1 Open Windows defender Firewall



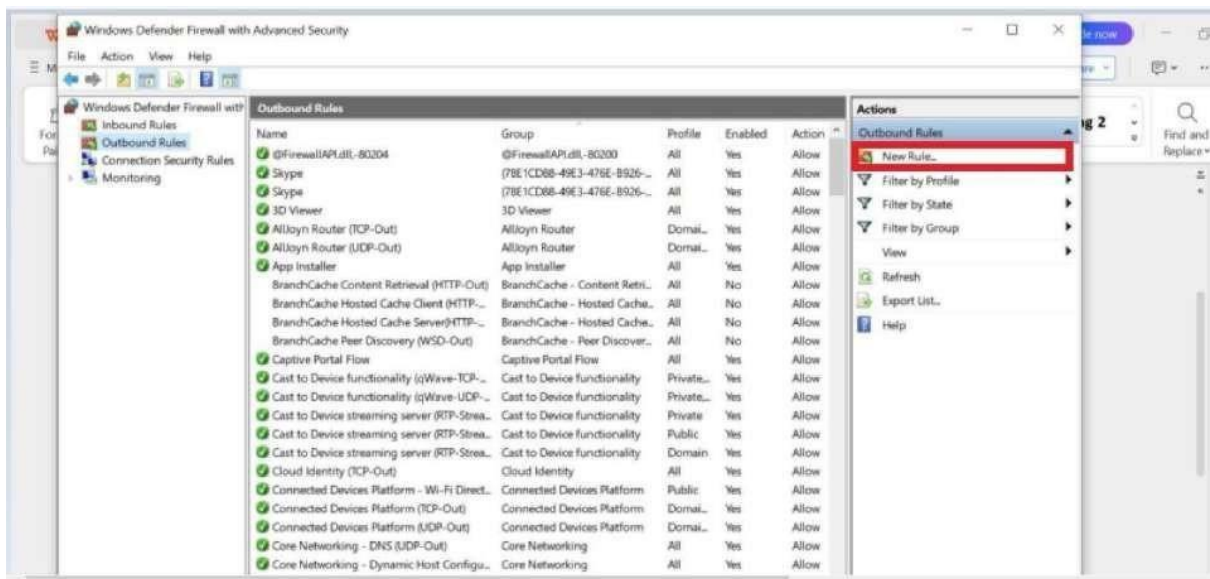
Step: 2 Select Advance Settings



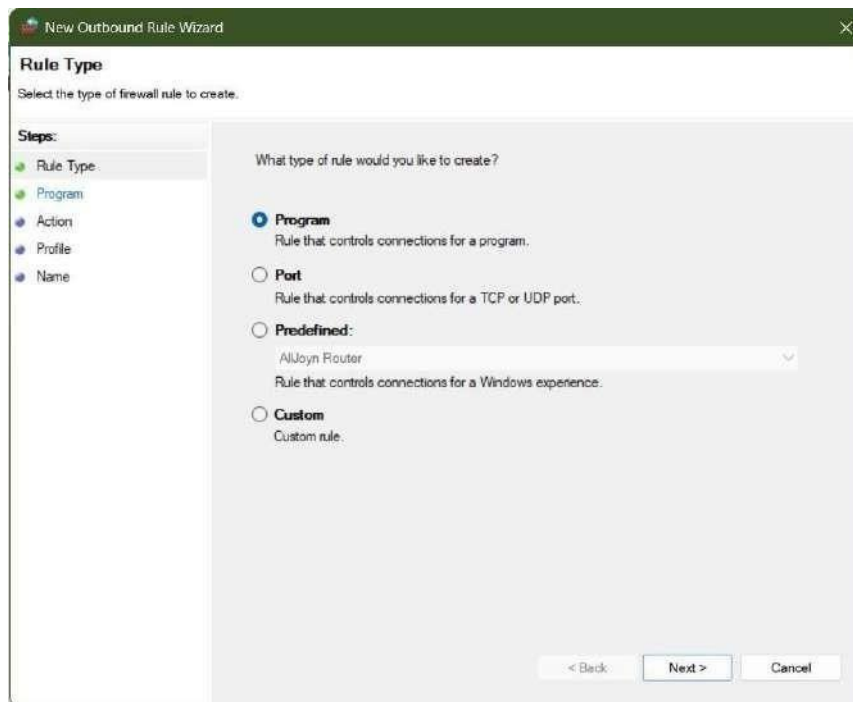
Step : 3 Select outbound Rules



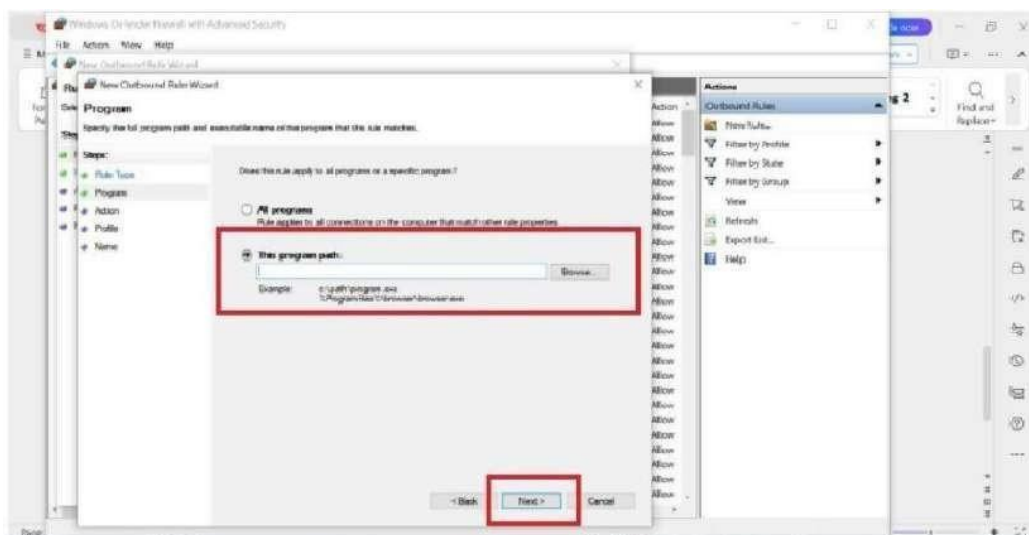
Step : 4 Select New Rule



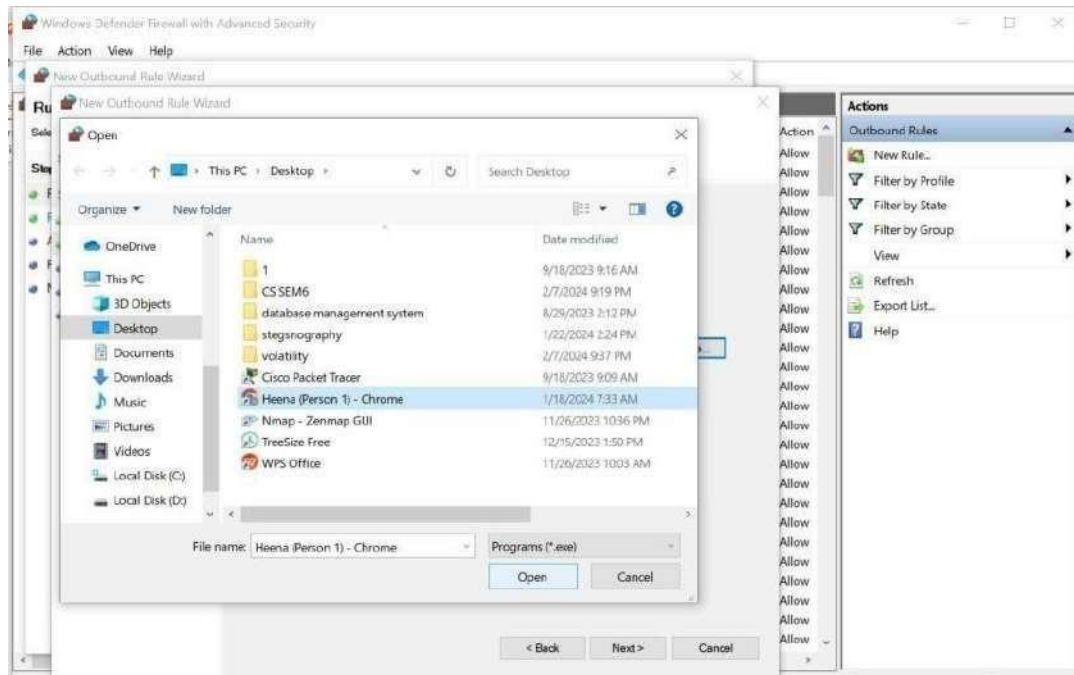
Step : 5 Select Program and Click on a next Button



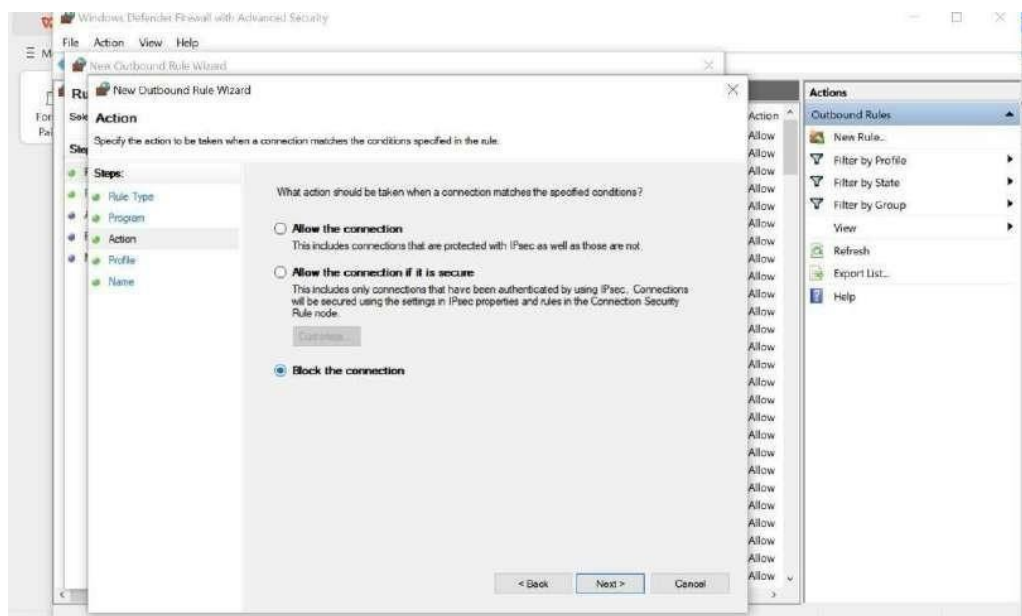
Step : 6 Give the path of Program which you want to Block



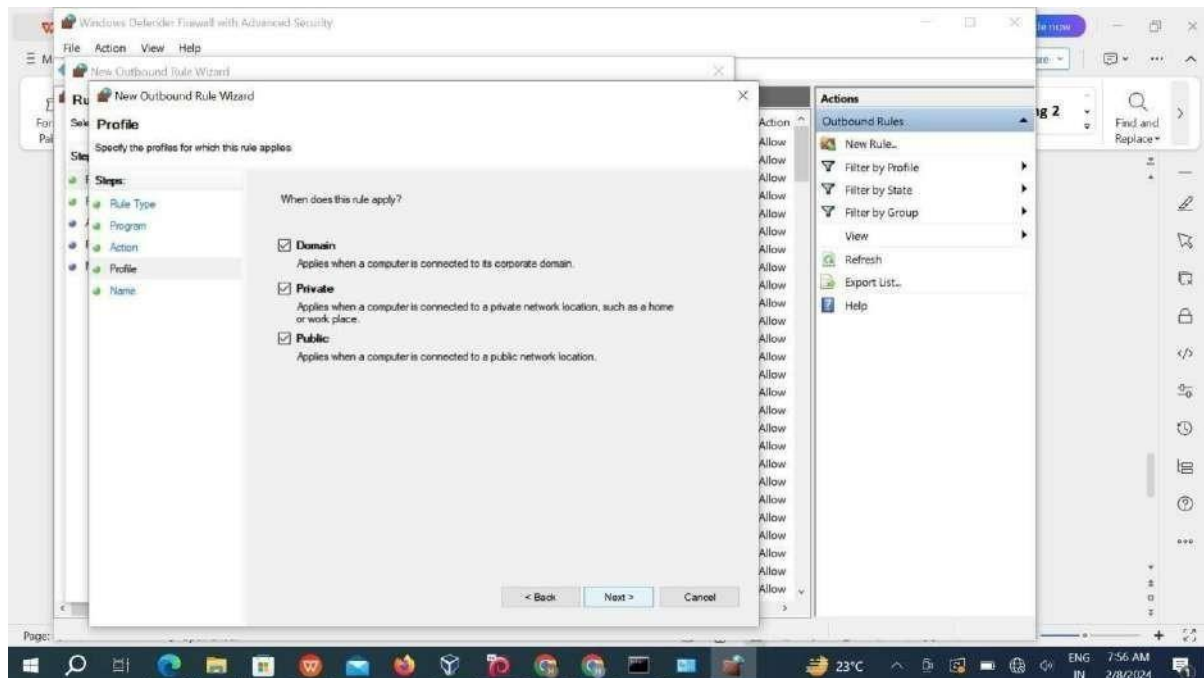
For Example edge Browser



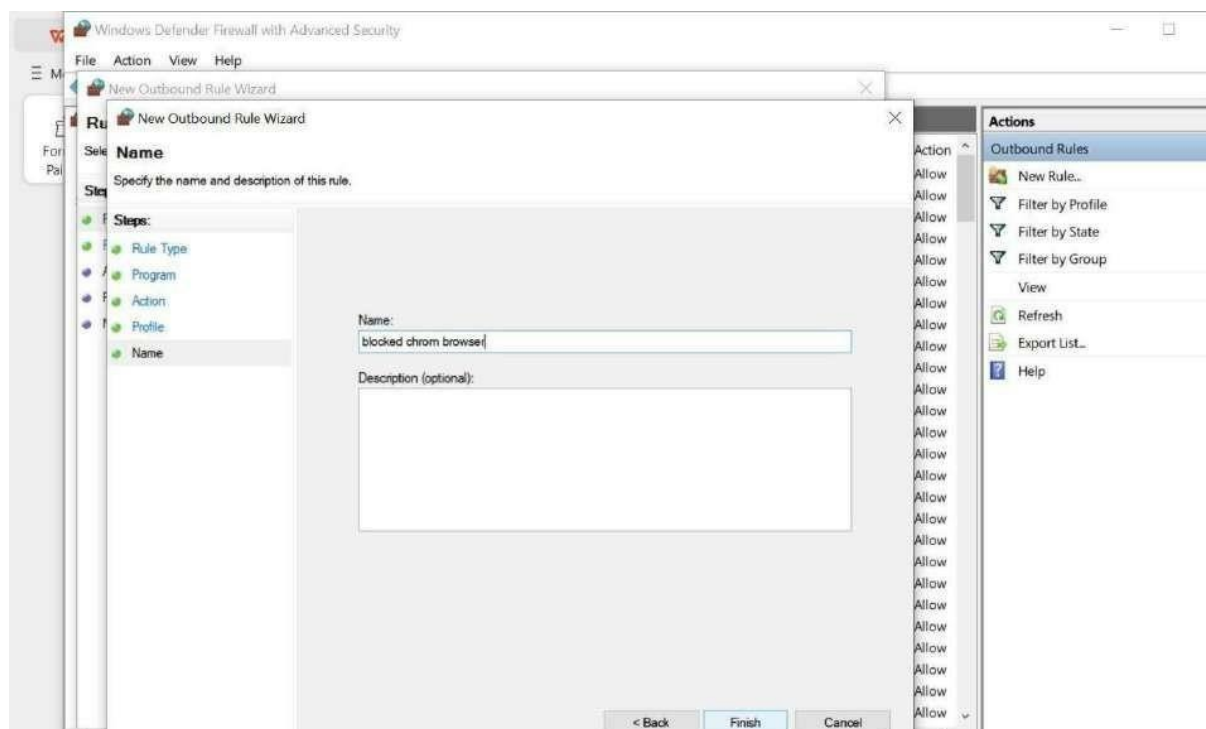
Step 7 : Select Block Connection Option

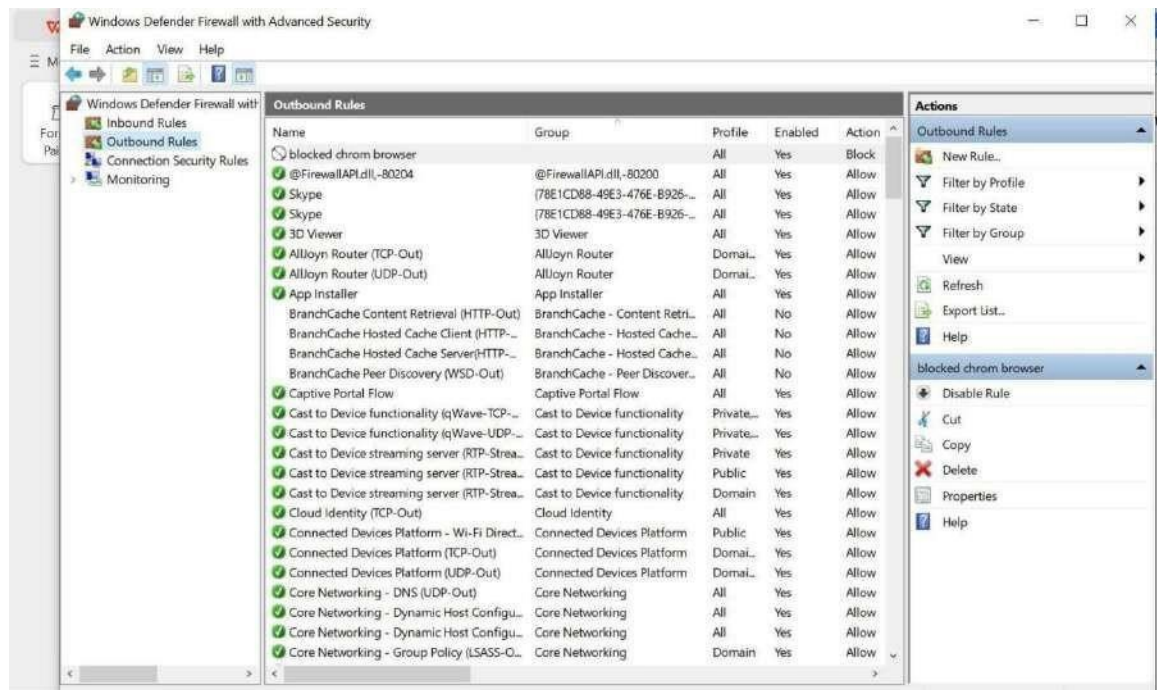


Select all 3 boxes and Click on a next button

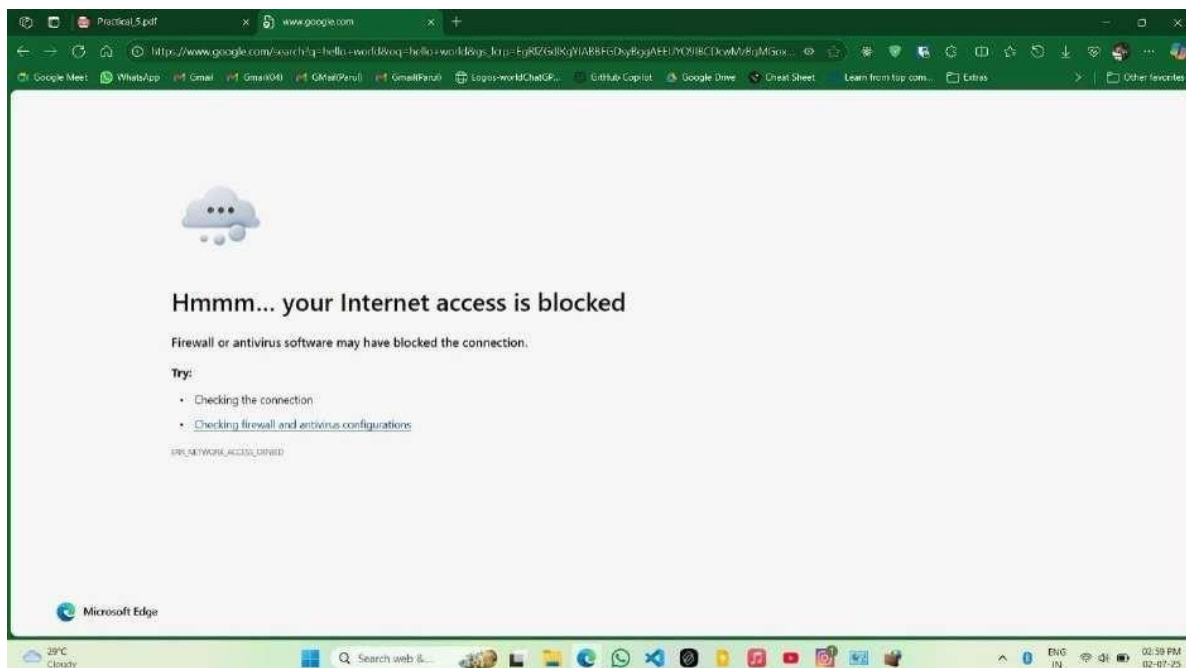


Step : 8 Give a Name to your Created rule and Click on a finish Button
You can see that rule is created





Now go to the Chrome Browser and Search you are not able to access edge browser.



Practical - 6

AIM: Implementation to identify web vulnerabilities, using OWASP project.

Tools: OWASP ZAPProxy (windows and kali)

Description

Introducing ZAP:-

- Zed Attack Proxy (ZAP) is a free, open-source penetration testing tool being maintained under the umbrella of the Open Web Application Security Project (OWASP). ZAP is designed specifically for testing webapplications and is both flexible and extensible. At its core, ZAP is what is known as a “man-in-the-middle proxy.” It stands between the tester’s browser and the web application so that it can intercept and inspect messages sent between browser and web application, modify the contents if needed, and then forward those packets on to the destination. It can be used as a stand-alone application, and as a daemon process. If there is another network proxy already in use, as in many corporate environments, ZAP can be configured to connect to that proxy.
- ZAP provides functionality for a range of skill levels – from developers, to testers new to security testing, to security testing specialists. ZAP has versions for each major OS and Docker, so you are not tied to a single OS.



- Additional functionality is freely available from a variety of add-ons in the ZAP Marketplace, accessible from within the ZAP client. Because ZAP is open-source, the source code can be examined to see exactly how the functionality is implemented. Anyone can volunteer to work on ZAP, fix bugs, add features, create pull requests to pull fixes into the project, and author add-ons to support specialized situations.

Install and Configure ZAP

ZAP has installers for Windows, Linux, and Mac OS/X. There are also Docker images available on the

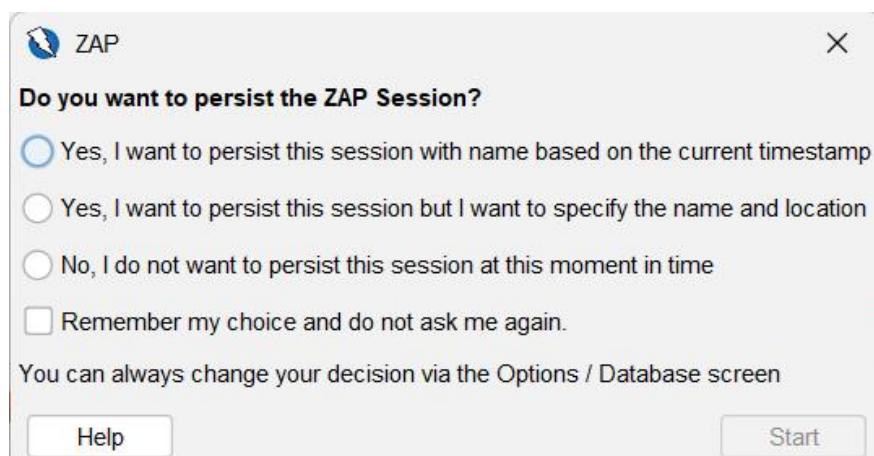
download site listed below.

Install ZAP The first thing to do is install ZAP on the system you intend to perform pentesting on.

Download the appropriate installer from ZAP's download location at <https://www.zaproxy.org/download/> and execute the installer.

Note that ZAP requires Java 8+ in order to run. The Mac OS/X installer includes an appropriate version of Java but you must install Java 8+ separately for Windows, Linux, and Cross-Platform versions.

- The Docker versions do not require you to install Java.
- Once the installation is complete, launch ZAP and read the license terms. Click Agree if you accept the terms, and ZAP will finish installing, then ZAP will automatically start.
- Persisting a Session When you first start ZAP, you will be asked if you want to persist the ZAP session.
- By default, ZAP sessions are always recorded to disk in a HSQLDB database with a default name and location.
- If you do not persist the session, those files are deleted when you exit ZAP.



- If you choose to persist a session, the session information will be saved in the local database so you can access it later, and you will be able to provide custom names and locations for saving the files

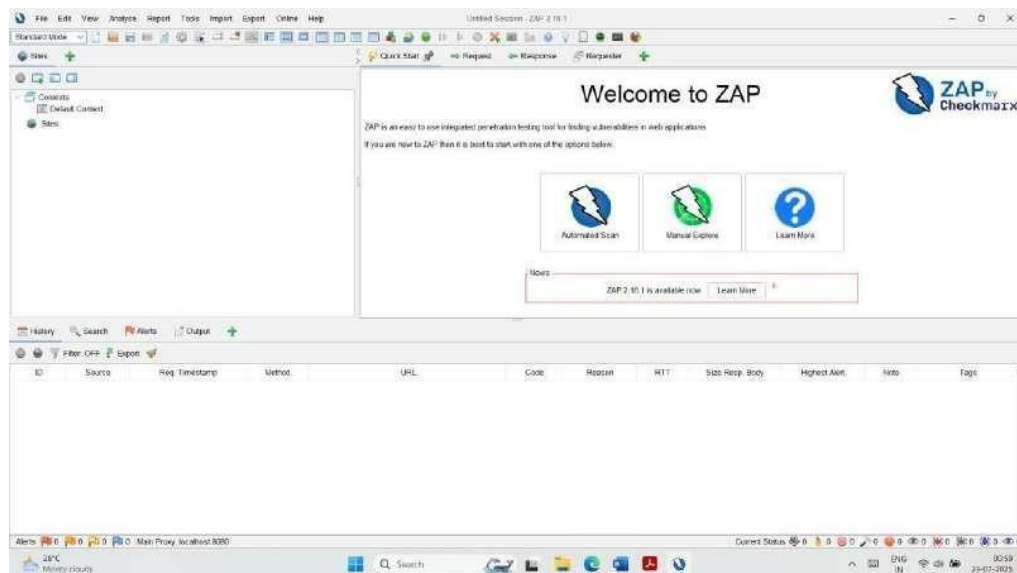
For now, select No, I do not want to persist this session at this moment in time, then click Start. The ZAP

sessions will not be persisted for now.

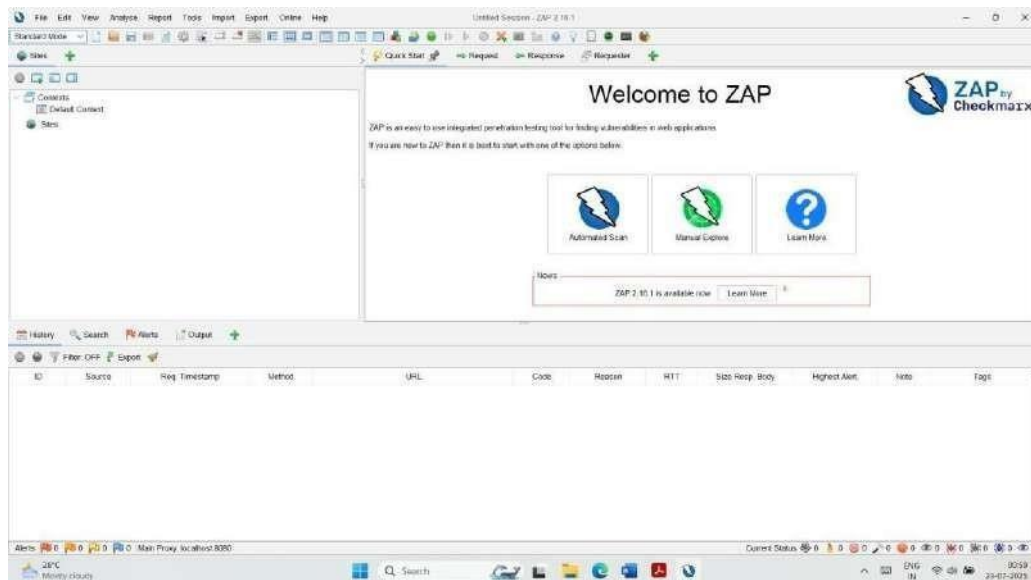
ZAP Desktop UI:

The ZAP Desktop UI is composed of the following elements:

1. Menu Bar – Provides access to many of the automated and manual tools.



2. Toolbar – Includes buttons which provide easy access to most commonly used features.
3. Tree Window – Displays the Sites tree and the Scripts tree.
4. Workspace Window – Displays requests, responses, and scripts and allows you to edit them.
5. Information Window – Displays details of the automated and manual tools.
6. Footer – Displays a summary of the alerts found and the status of the main automated tools.

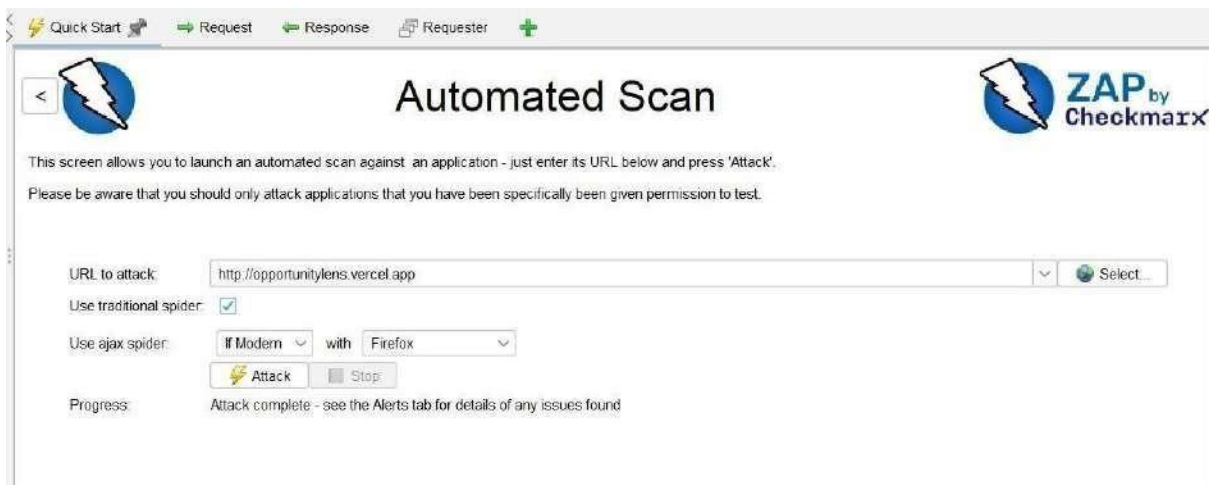


Running an Automated Scan

The easiest way to start using ZAP is via the Quick Start tab. Quick Start is a ZAP add-on that is included automatically when you installed ZAP.

To run a Quick Start Automated Scan :

1. Start ZAP and click the Quick Start tab of the Workspace Window.
2. Click the large Automated Scan button.
3. In the URL to attack text box, enter the full URL of the web application you want to attack.
4. Click the Attack button.



- ZAP will proceed to crawl the web application with its spider and passively scan each page it finds. Then ZAP will use the active scanner to attack all of the discovered pages, functionality, and parameters.
- ZAP provides 2 spiders for crawling web applications, you can use either or both of them from this screen.
- The traditional ZAP spider which discovers links by examining the HTML in responses from the web application. This spider is fast, but it is not always effective when exploring an AJAX web application that generates links using JavaScript.
- For AJAX applications, ZAP's AJAX spider is likely to be more effective. This spider explores the web application by invoking browsers which then follow the links that have been generated. The AJAX spider is slower than the traditional spider and requires additional configuration for use in a "headless" environment.
- ZAP will passively scan all of the requests and responses proxied through it. So far ZAP has only carried out passive scans of your web application. Passive scanning does not change responses in any way and is considered safe. Scanning is also performed in a background thread to not slow down exploration.
- Passive scanning is good at finding some vulnerabilities and as a way to get a feel for the basic security state of a web application and locate where more investigation may be warranted. Active scanning, however, attempts to find other vulnerabilities by using known attacks against the selected targets.
- Active scanning is a real attack on those targets and can put the targets at risk, so do not use active scanning against targets you do not have permission to test.

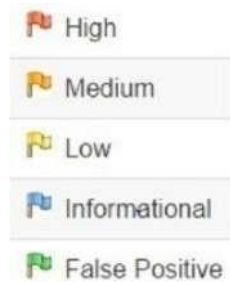
See Explored Pages

To examine a tree view of the explored pages, click the Sites tab in the Tree Window. You

can expand the nodes to see the individual URLs accessed.

View Alerts and Alert Details

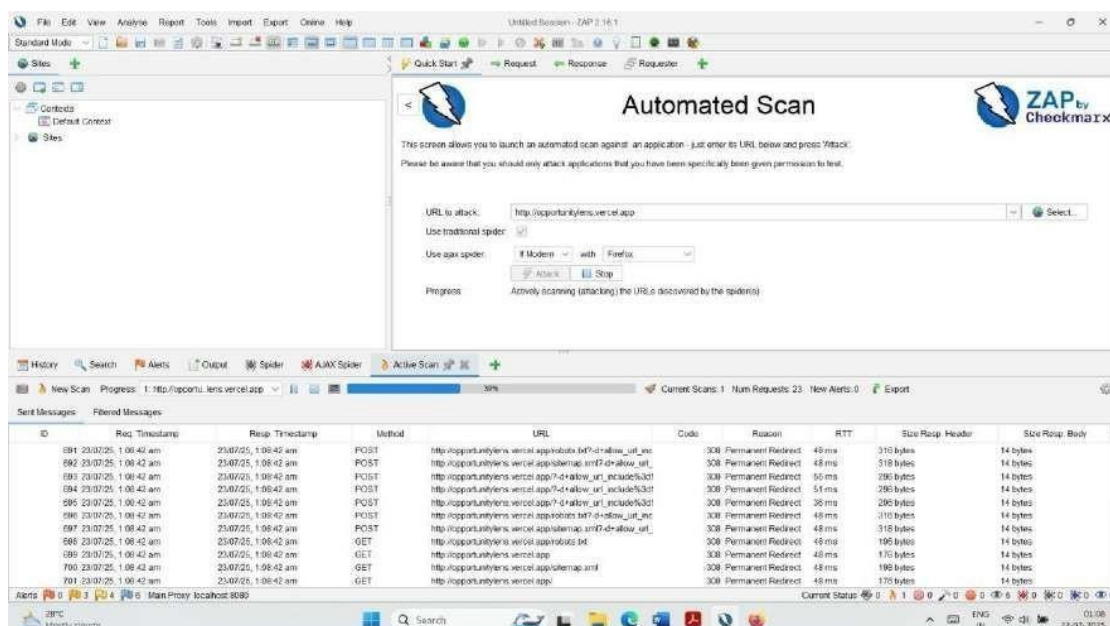
The left-hand side of the Footer contains a count of the Alerts found during your test, Broken out into risk categories. These risk categories are:



To view the alerts created during your test:

1. Click the Alerts tab in the Information Window.
2. Click each alert displayed in that window to display the URL and the vulnerability detected in the right side of the Information Window.
3. In the Workspace Windows, click the Response tab to see the contents of the header and body of the response. The part of the response that generated the alert will be highlighted.

Here are the Outputs of a Sample bank website follow the images below step by step



The screenshot displays the Burp Suite application window. The top menu bar includes File, Edit, View, Analyze, Report, Tools, Import, Export, and Online Help. The main workspace is divided into several panes:

- Left Pane:** Contains 'Sites' (Default Context, Sites), 'History', 'Search', 'Alerts' (with a list of detected issues), 'Output', 'Spider', 'AJAX Spider', and 'Active Scan'.
- Top Right Pane:** Shows the 'Quick Start' tab with 'Request' and 'Response' sub-tabs. The 'Response' tab is active, displaying the HTML content of the response from the login page.
- Bottom Left Pane:** The 'Alerts' pane, which lists various security issues detected by Burp Suite. The issue 'Session Management Response Identified' is highlighted at the bottom of the list.
- Bottom Right Pane:** The 'Other Info' section for the selected alert. It provides details about the identified session management response, including the URL, risk level, confidence, and source.

The 'Response' tab shows the following HTML content:

```
HTTP/1.1 302 Found
Cache-Control: private
Content-Type: text/html; charset=utf-8
Location: /login.aspx
Server: Microsoft-IIS/10.0
Set-Cookie: ASP.NET_SessionId=qhjg2as5ewcnp4tasnau; path=/; secure; httpOnly; SameSite=Lax
X-AspNet-Version: 6.0.30319
X-Powered-By=IIS/10.0
X-XSS-Protection: 1; mode=block
X-Content-Type-Options: nosniff
Strict-Transport-Security: max-age=31536000; includeSubDomains
<html><head><title>Object moved.</title></head><body>
<h2>Object moved to <a href="/login.aspx">here</a>.</h2>
</body></html>
```

The 'Alerts' pane lists the following issues:

- Missing Anti-clickjacking Header (5)
- Vulnerable JS Library (5)
- Cross-Domain JavaScript Source File Inclusion (7)
- Server Lease Version Information via "Server" HTTP Response
- Strict-Transport-Security Header Not Set (7)
- Timestamp Disclosure - Unix (15)
- X-AspNet-Version Response Header (12)
- X-Content-Type-Options Header Missing (3)
- Authentication Request Identified (5)
- Information Disclosure - Suspicious Comments (8)
- Modern Web Application (3)
- Re-examine Cache-control Directives (3)
- Retrieved from Cache (6)
- Session Management Response Identified (2)

The 'Other Info' section for the 'Session Management Response Identified' alert provides the following details:

- URL:** https://kims.pau.ac.uk/
- Risk:** Informational
- Confidence:** Medium
- Parameter:** ASP.NET_SessionId
- Attack:**
- Evidence:** ASP.NET_SessionId
- CWE ID:**
- WASC ID:**
- Source:** Passive (10112 - Session Management Response identified)
- Input Vector:** Description

The description states: "The given response has been identified as containing a session management token. The 'Other Info' field contains a set of header tokens that can be used in the Header Based Session Management Method. If the response is in a context where there is a Session Management Method set to 'Auto-Detect' then this rule will change the session management to use the tokens identified."

The screenshot displays the Burp Suite interface during a CSRF attack simulation. The top pane shows the 'Request' tab of an HTTP request to `http://10.10.10.10:8080/login.aspx`. The request body contains a `<form>` tag with a `<input type='text' value='[redacted]'>` field and a `<input type='submit' value='Login'>` button. The 'Alerts' tab on the left shows a detected CSRF attack alert. The alert details include the request URL, the CSRF token used, and the target URL. The 'Input Vector' section shows the request body with the CSRF token highlighted. The 'Description' section explains that a cross-site request forgery is an attack that involves forcing a victim to send an HTTP request to a target destination without their knowledge or intent. The 'Other Info' section lists the CSRF token and the request body. The 'Solution' section suggests using a vetted library or framework that does not allow this weakness to occur or provides constructs that make this weakness easier to avoid. The 'Reference' section provides a link to the OWASP CSRF Cheat Sheet.

Practical - 8

AIM : Ram dumps and using it for social engineering attacks.

RAM dumps

A memory dump is the process of taking all information content in **RAM** and writing it to a storage drive as a **memory** dump file (*.RAW format).

Volatile memory, or RAM, is used to store data currently used by a running process: whether it is a user application or a system service. This type of memory is much quicker than a regular hard drive but unlike files permanently stored on a drive (unless deleted), data from RAM may disappear instantly. At the same time, it may store data crucial for your case, including passwords in raw format without encryption or encoding, decrypted data otherwise kept encrypted on a drive, decryption keys for various services, apps and WDE, remote sessions data, chats in social networks, malware code, cryptocurrency transactions, various system info such as loaded registry branches, and so on. This is why it is not argued that capturing RAM contents must be one of the first steps in seizing a running computer or laptop.

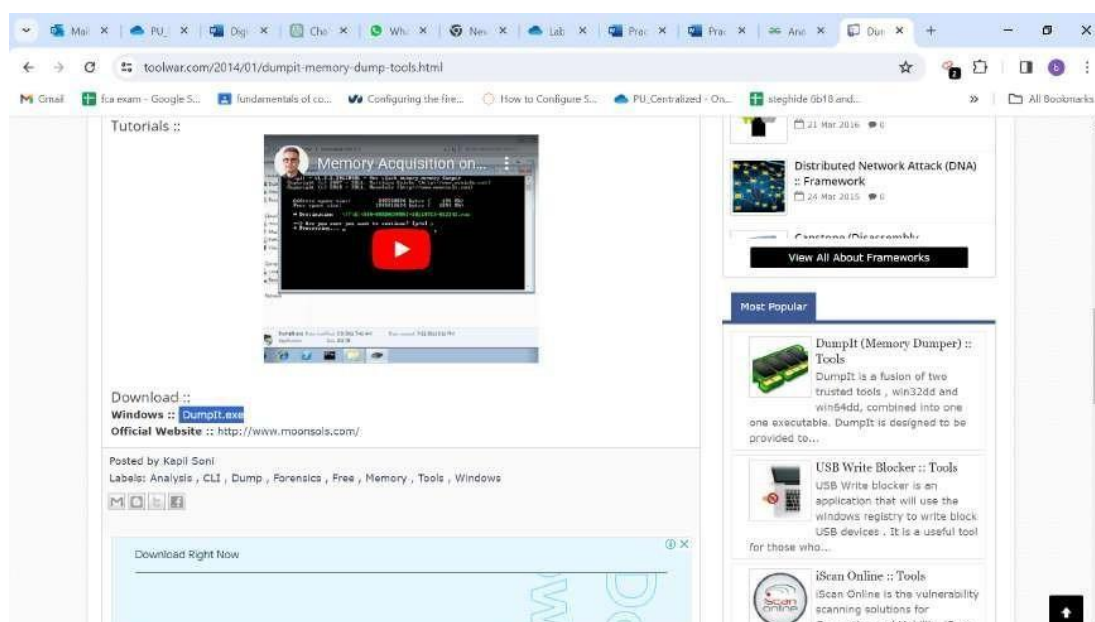
There are various tools that can be used for memory dump. Some of them are:

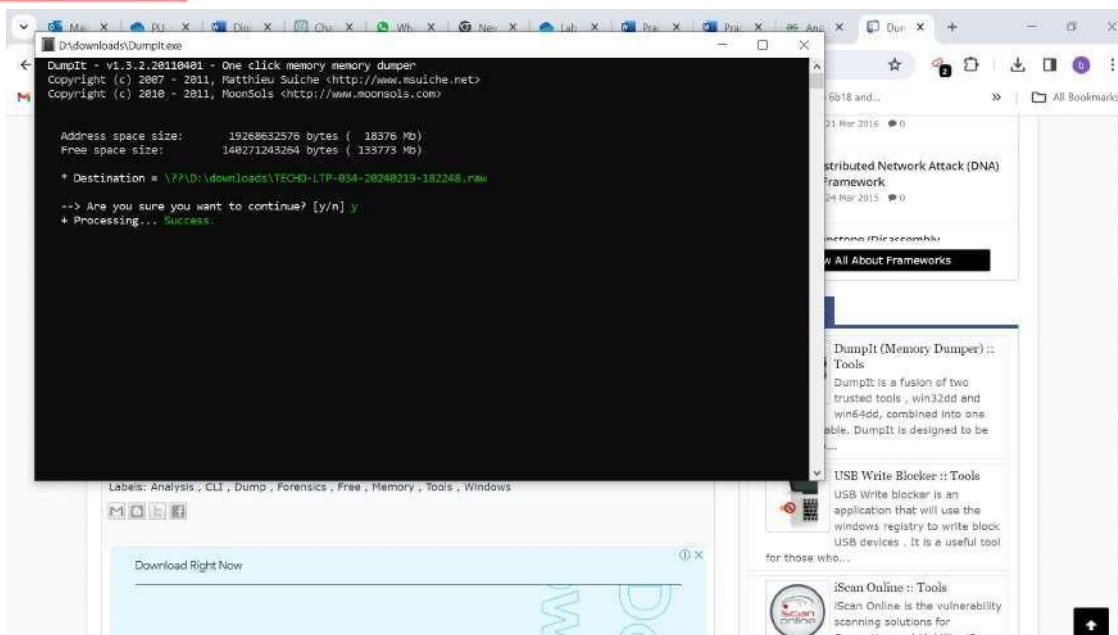
1. Autopsy
2. Dump-IT

Procedure

Creating RAM dump using Dump-IT

1. Download **DumpIT** tool from toolwar website
2. Open Dump-IT.exe





Autopsy performs operations onto disk images which can be created using tools like FTK Imager. Here an already created image is used. You may download Autopsy from [here](#) and the disk image used in this article from [here](#).

1. Getting Started

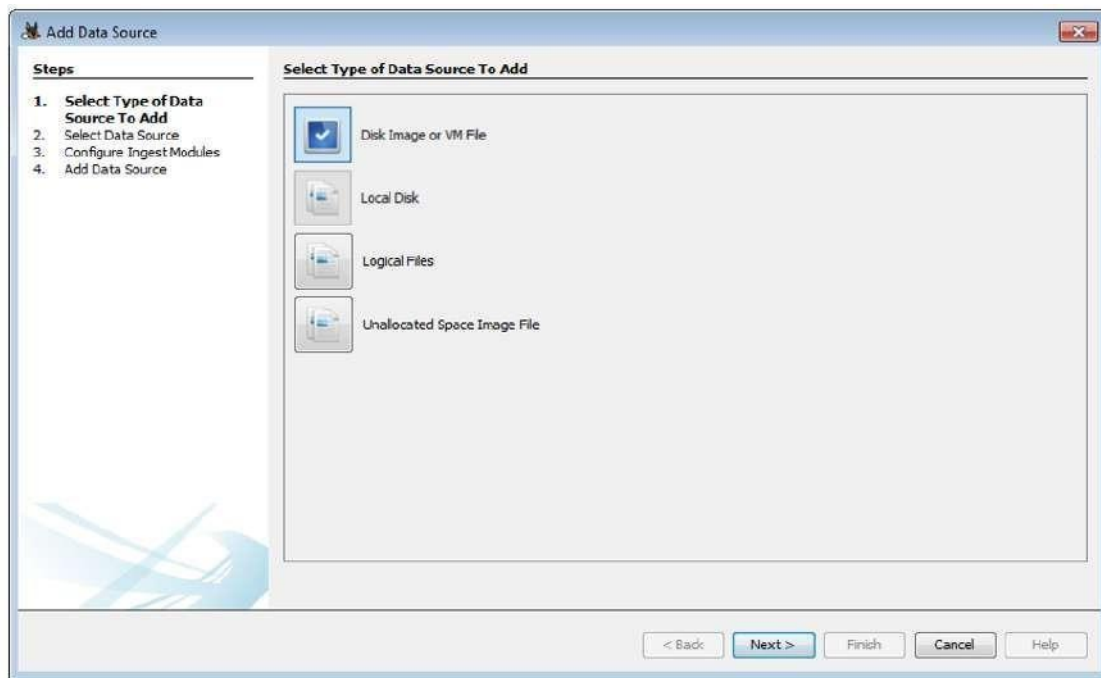
Open Autopsy and create a new case.



Click on **Finish** after completing both the steps.

2. Add a data source.

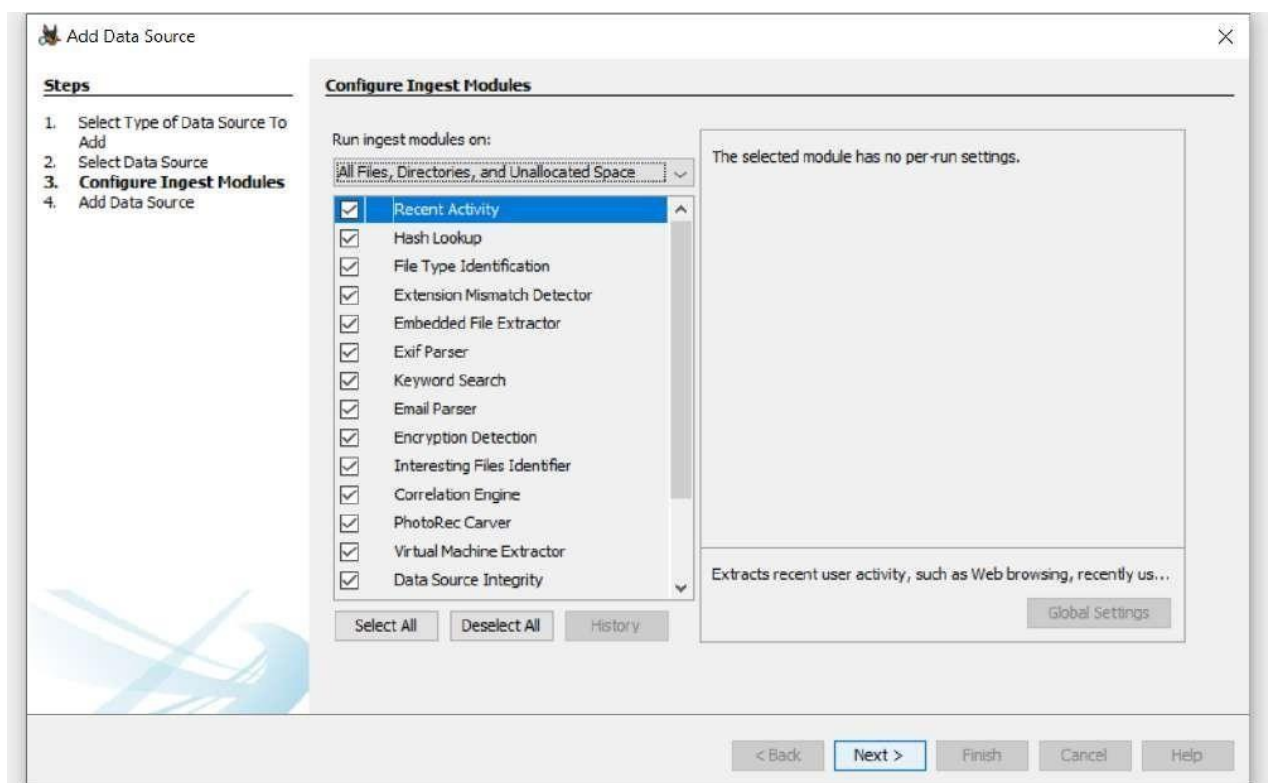
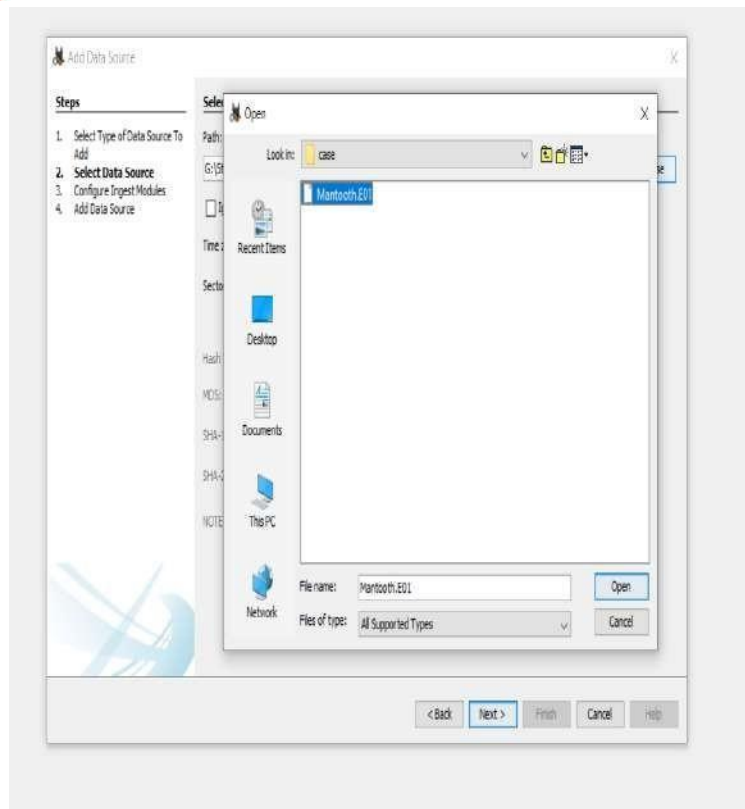
Select the appropriate data source type.



- **Disk Image or VM file:** Includes images that are an exact copy of a hard drive or media card, or a virtual machine image.
- **Local Disk:** Includes Hard disk, Pen drive, memory card, etc.
- **Logical Files.** : Includes local folders or files.
- **Unallocated Space Image File:** Includes files that do not contain a file system but need to run through ingest.

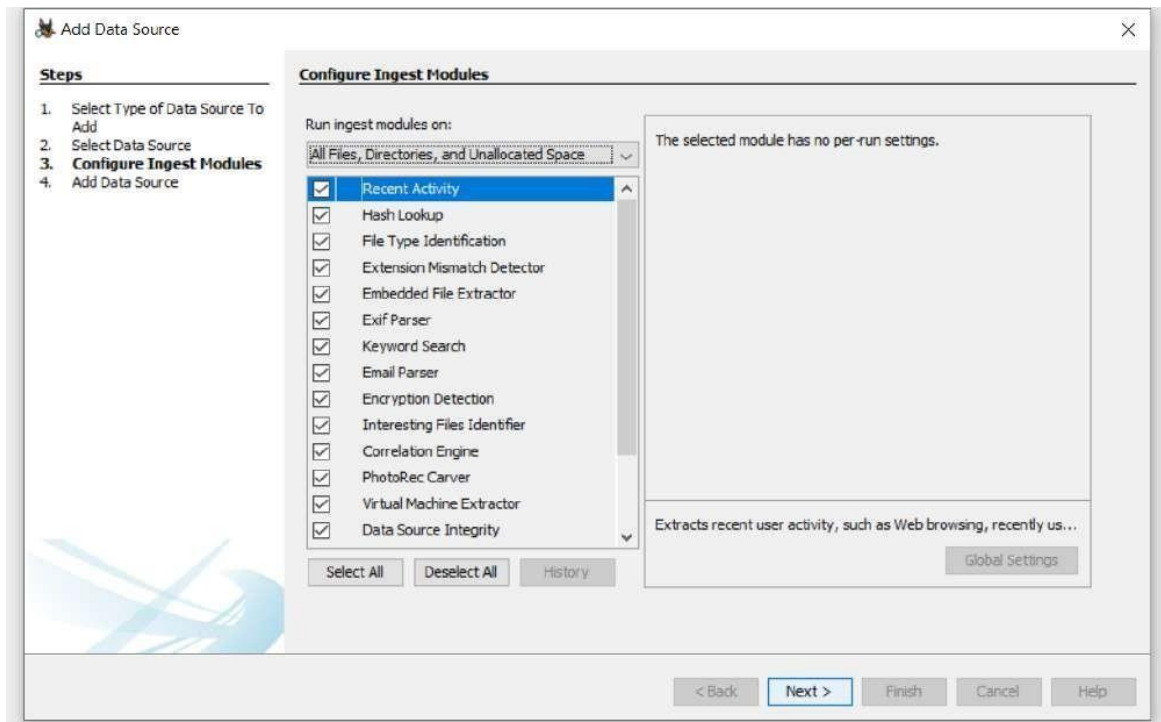
The data source used here is a disk image. Add the data source destination.

Configure ingest modules.



The ingest modules determine factors for which the data in the data source is to be analyzed. Here is a brief overview of each of them.

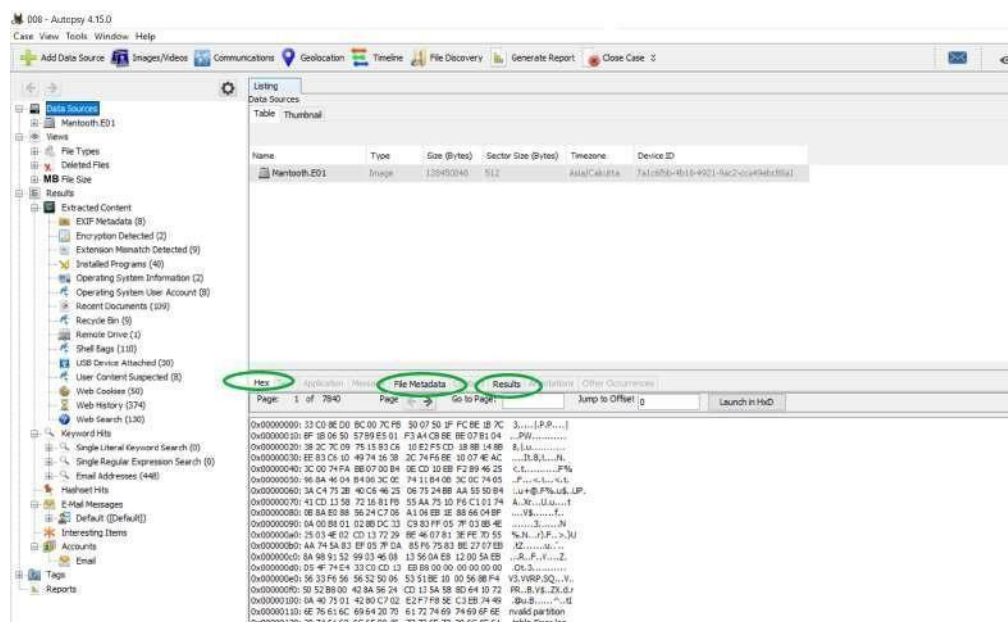
- **Recent Activity:** Discover the recent operations performed on the disk, for example, the files that were last viewed.
- **Hash Lookup:** Identify files using hash values.
- **File Type Identification:** Identify files based on their internal signatures rather than just file extensions.
- **Extension Mismatch Detector:** Identify files whose extensions are tampered with/changed possibly to hide evidence.
- **Embedded File Extractor:** It extracts embedded files such as .zip, .rar, etc. and uses the derived file for analysis. Another example could be a PNG image saved inside a doc to make it appear as a document and thus hide crucial information.
- **EXIF (Exchangeable Image File Format) Parser:** It is used to retrieve metadata about the files, for example, date of creation, geolocation, etc.
- **Keyword Search:** Search for a particular keyword/pattern in the data source.
- **Email Parser:** If the disk holds any form of email database, for example, pst/ost files of outlook then information from these files can be extracted using an email parser.
- **Encryption Detection:** Detects and identifies encrypted / password-protected files.
- **Interesting File Identifier:** Let's set custom rules regarding the filtering of data. Examiner is notified when results pertaining to these rules are found.
- **Correlation Engine:** Allows saving properties in and then retrieved from the central repository. It helps in displaying correlated properties.
- **PhotoRec Carver:** Recover files, photos, etc. from the unallocated space.
- **Virtual Machine Extractor:** Extract and analyze any Virtual machine found on the data source.
- **Data Source Integrity:** Calculates the hash values and stores them in the database in case they aren't already present. Otherwise, it will verify the hash values associated with the database.
- **Plaso:** Extract timestamp for various types of files.
- **Android Analyzer:** Analyze SQLite and other files retrieved from an Android device.



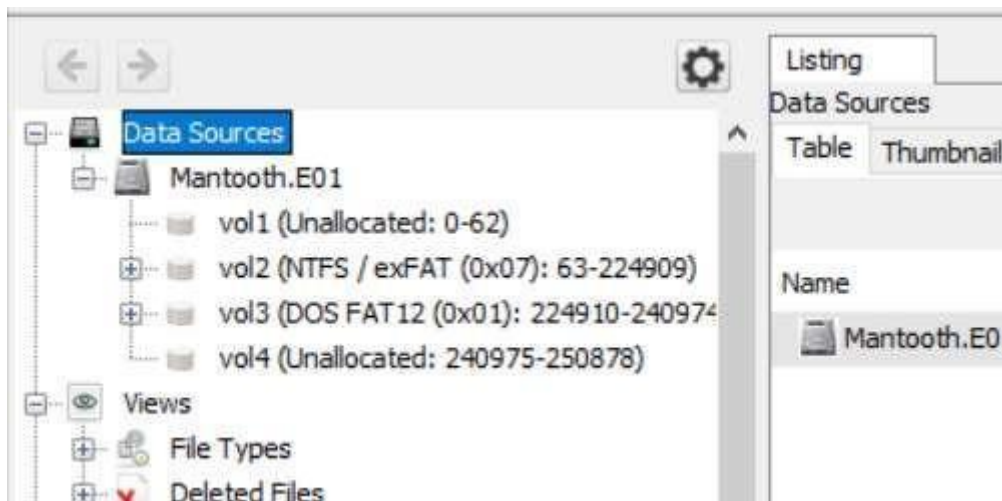
Select all that will serve the purpose of your investigation and click Next. Once the data source is added, click Finish. It will take some buffer time to extract and analyze the data depending upon the size of the Data Source.

3. Exploring the data source:

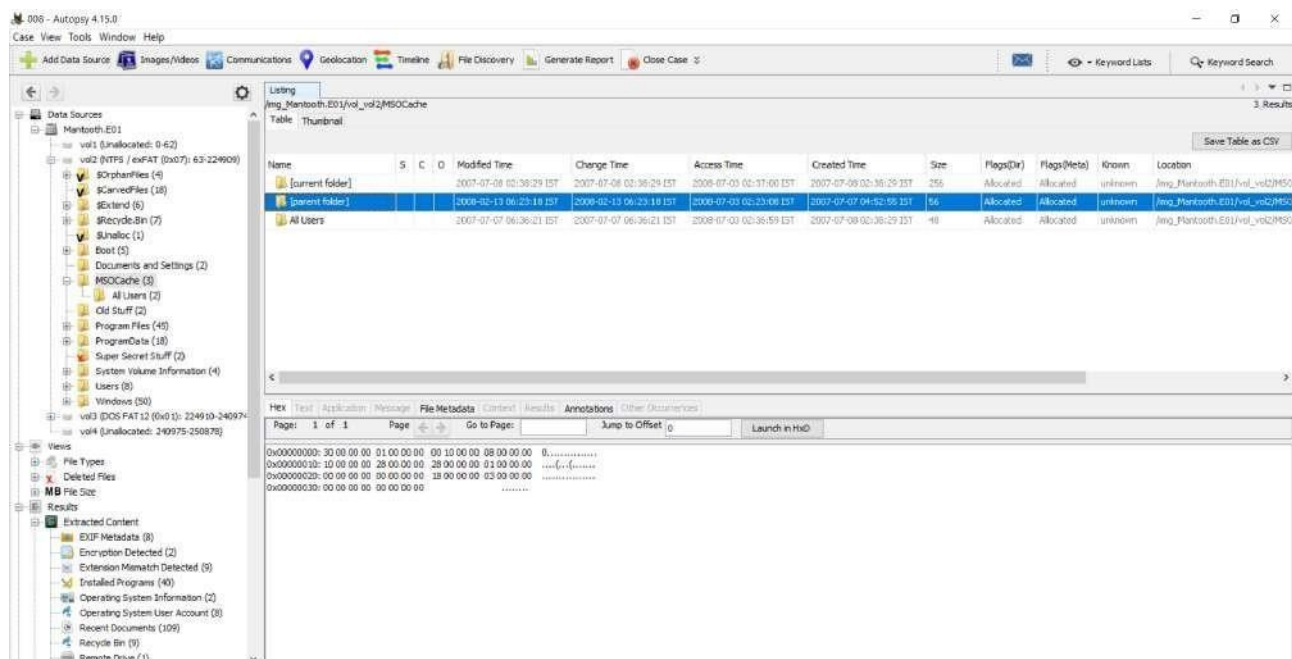
The Data Source information: Here the basic metadata is shown. A detailed analysis is displayed in the bottom section. These details can be extracted in the form of Hex values, Results, File Metadata, etc.



The disk image is then broken down based upon its volume partitions.

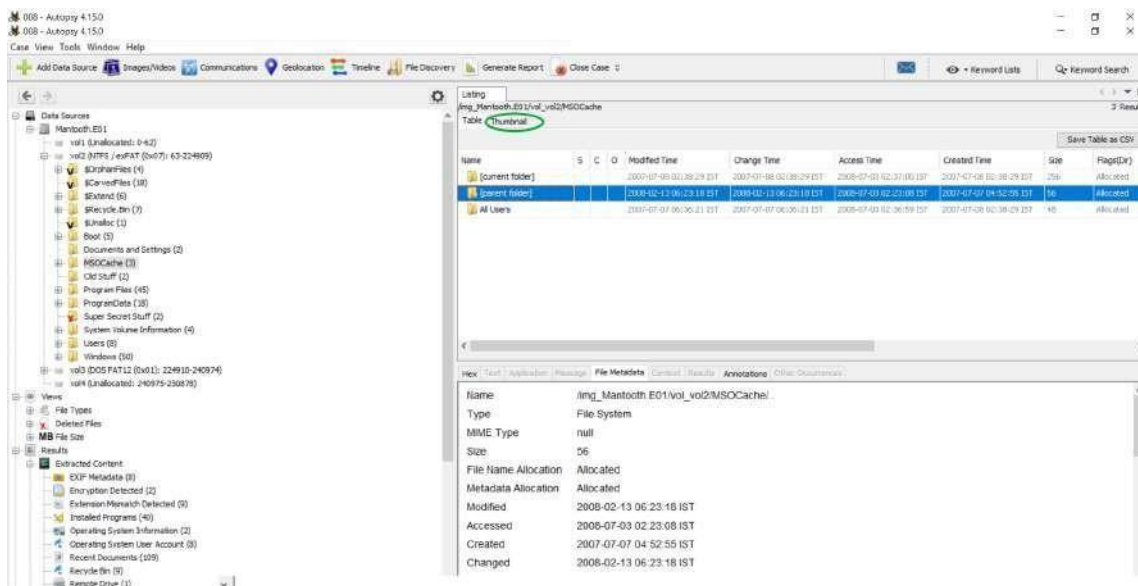


Each volume can be browsed for its contents, results for which are displayed in the section at the bottom. For example, the content shown below belongs to Data Sources -> Mantooth.E01 -> MSOCache-> [Parent Folder].

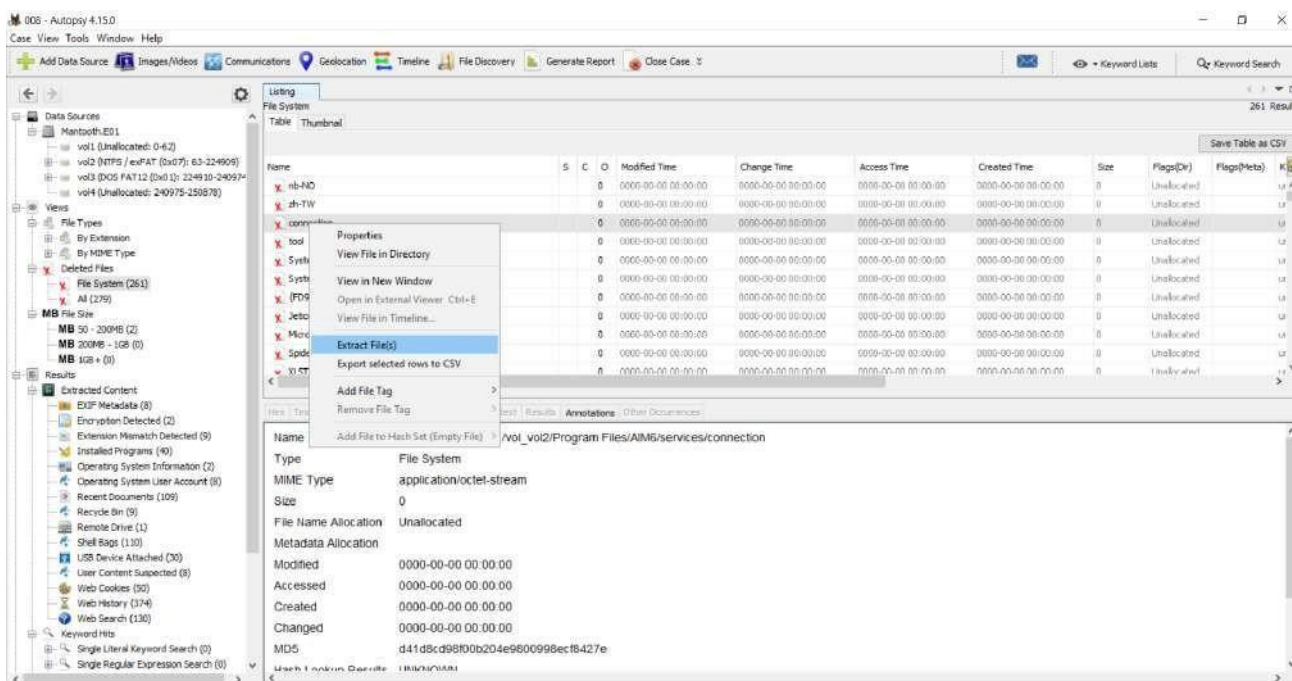


Views (Determines the factor of file classification)

- File Type:** Here the files are categorized based upon their type. The classification can be done either on the basis of file.extension or MIME type. While both of these provide a hint about how to deal with a file, file extensions are commonly used by the OS to decide what program shall be used to open a file and MIME types are used by the browser to decide about how to present the data (or by the server on how to interpret the data received). Files displayed here also include the deleted files.



- **Deleted Files:** Here information about the files that were specifically deleted can be found. These deleted files can be recovered as well: Right-click on the file to be recovered -> click on Extract File(s). -> Save the file in an appropriate destination.



- **MB Size Files:** Here files are classified based upon their size. The range starts from 50MB. This enables the examiner to determine exclusively large files.

Note: It is usually advised to not scan or extract any suspected files/ disks such as payload files, etc. in the main system, rather scan them in safe environments such as a virtual machine, and then extract the data, as they hold the possibility of being corrupt and may infect the examiner's system with viruses.

Results:

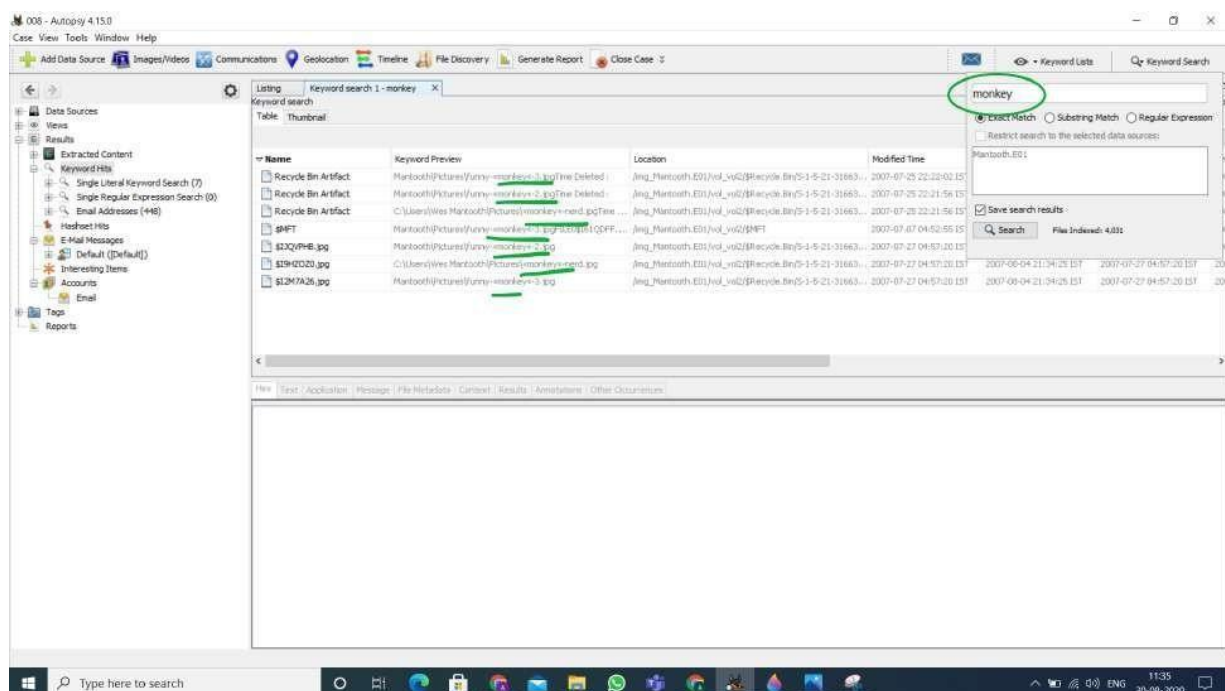
All the extracted data is viewed in *Views/ Data Source*. In *Results*, we get the information about this data.

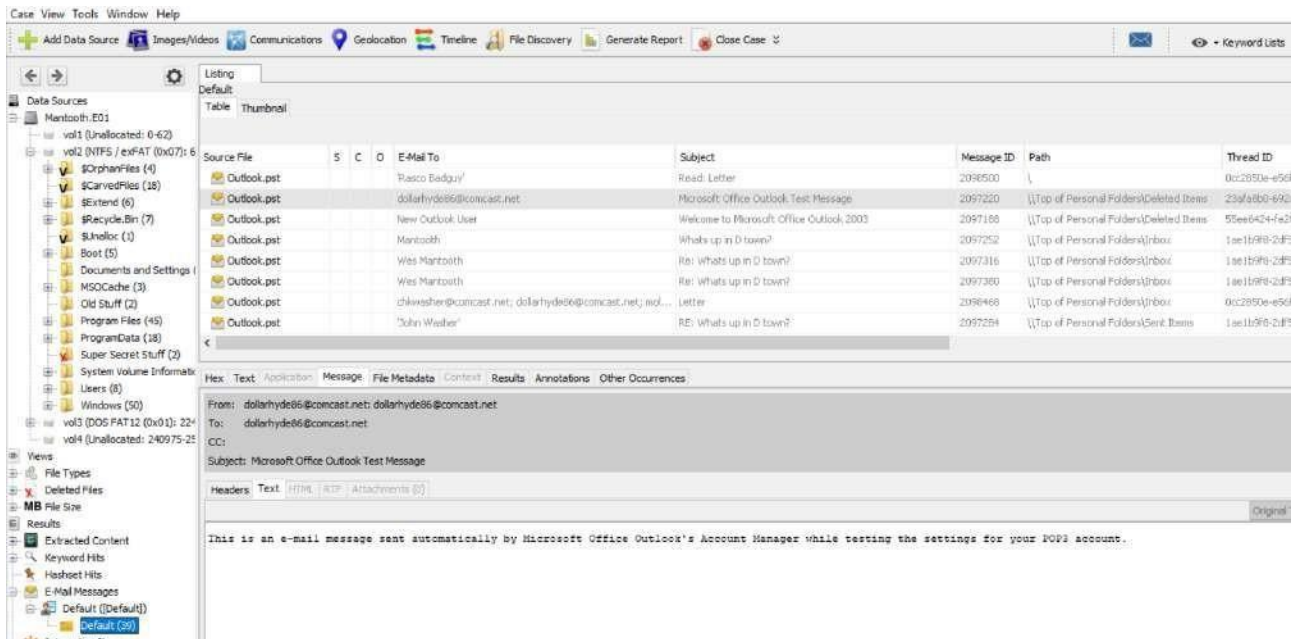
- **Extracted Content:** Each Extracted Content displayed below can be further explored. The following briefly explains each of them.



- **EXIF Metadata:** It contains all the .jpg images that have EXIF Metadata associated with them, this Metadata can be analyzed further.
- **Encryption Detection:** It detects files that are password protected/ encrypted.
- **Extension Mismatch Detection:** As explained above, it Identifies the files whose extensions do not match their MIME types and thus they may be suspicious.
- **Installed Programs:** It gives details about the software used by the user. This information is extracted with the help of the Software Registry hive.
- **Operating System Information:** It gives information about the OS with the help of the Windows Registry hive and the Software Registry hive.
- **Operating System User Account:** It lists information about all the user accounts, for example, accounts belonging to the device are extracted from the Software Hive and the accounts associated with the Internet Explorer using index.data files.

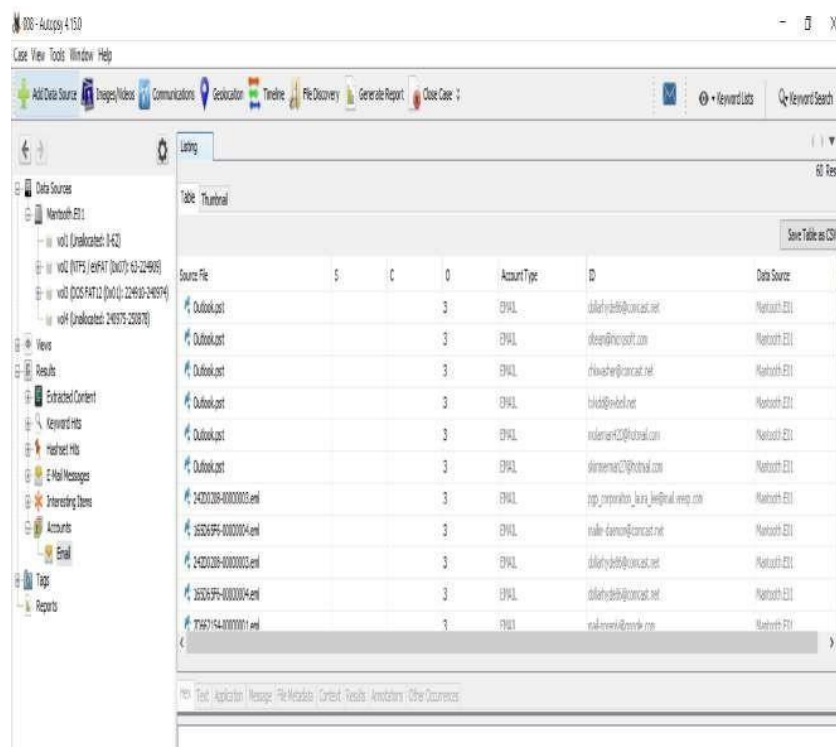
- **Recent documents:** Lists all the documents that were accessed nearby the time the disk image was captured.
- **Recycle Bin:** Files that are temporarily stored on the system before being permanently deleted are visible here.
- **Remote Drive:** Shows information about all the remote drives accessed using the system.
- **Shell bags:** A shell bag is a set of registry keys that stores details about a folder being viewed, such as its position, icon, and size. All the Shell bags from the system can be viewed here.
- **USB Device attached:** All the information about the external devices attached to the system is displayed here. This data is extracted from Windows Registry which is actually a maintained database about all the activities taking place on the system.
- **Web Cookies:** Cookies saves the user information from the sites and thus provide a lot of information about the user's online activities.
- **Web History:** All the details about the browser history is shown here.
- **Web Searches:** Details about the web searches made are displayed here.
- **Keyword Hits:** Here specific keywords can be looked for in the image of the disk. Multiple data sources can be selected for the lookup. The search can be restricted to Exact match, Substring match and Regular expression, for example, emails/ IP Addresses, etc.

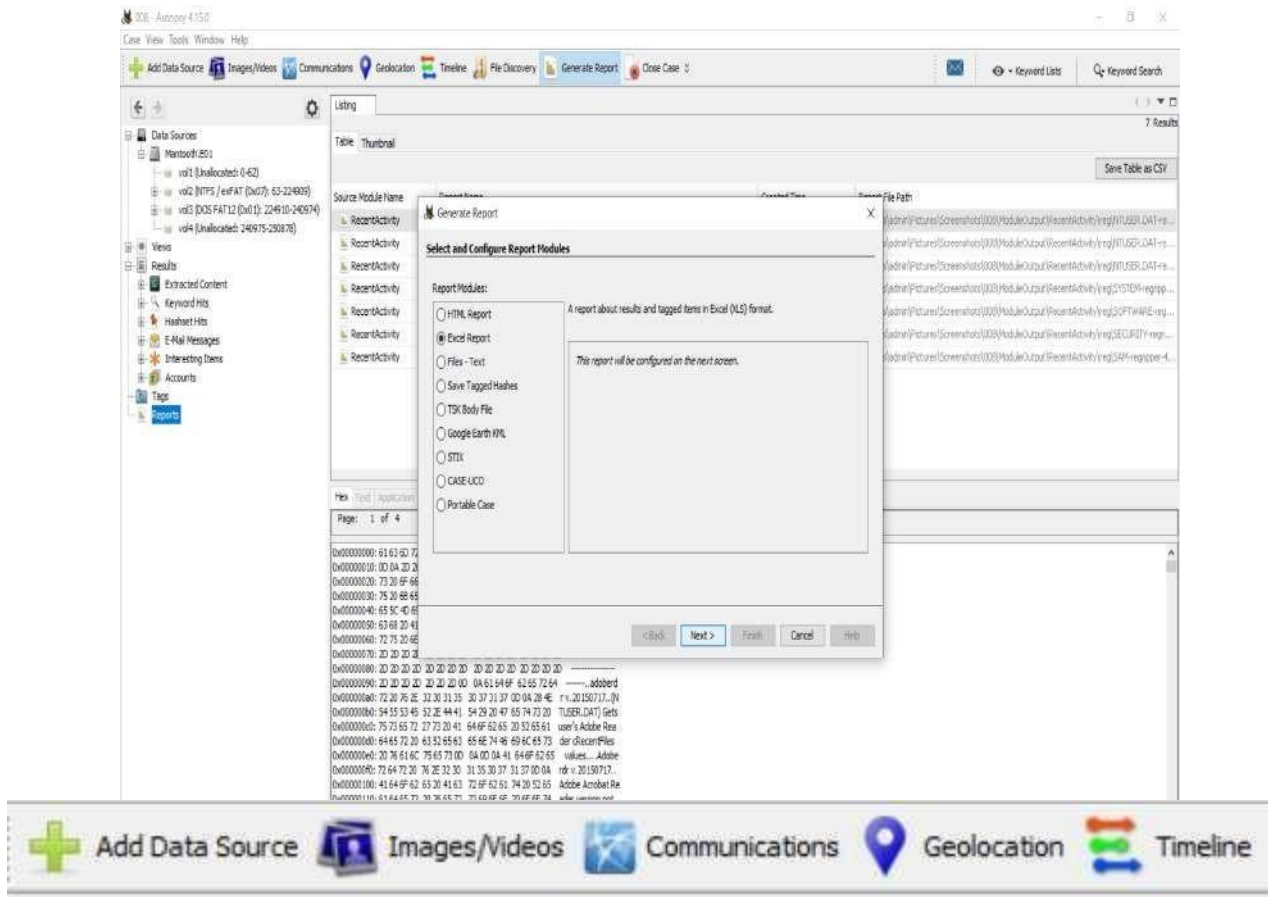




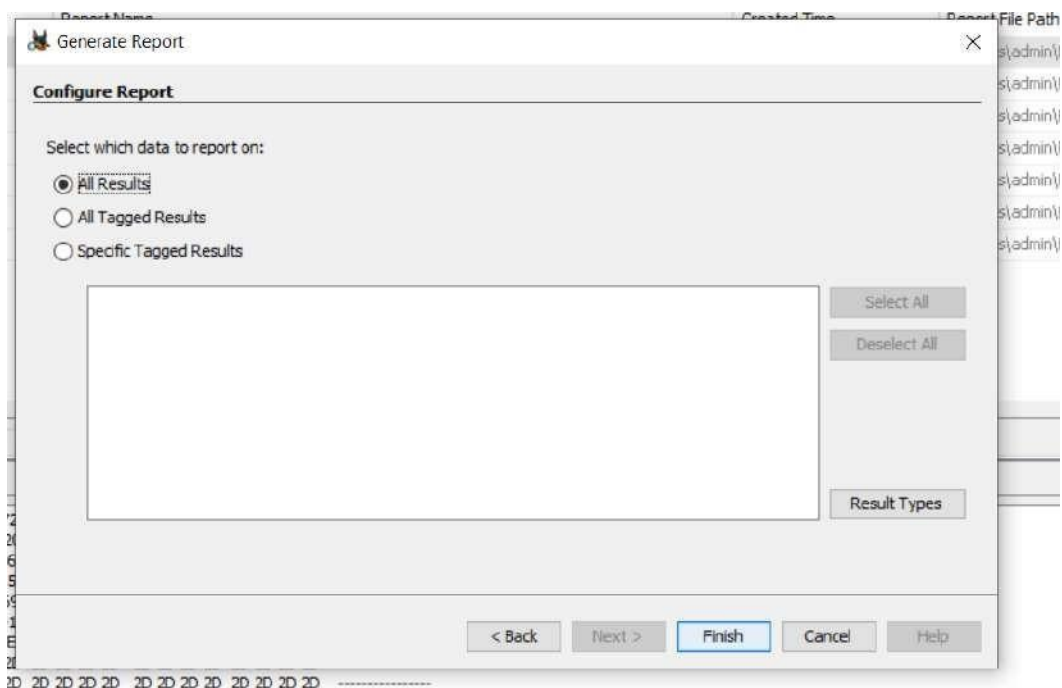
- **HashSet Hits:** Here the search can be made using hash values.
- **E-mail Messages:** Here all the *outlook.pst* files can be explored.
- **Interesting Items:** As discussed before, these are the file results based upon the custom rules set by the examiner.
- **Accounts:** Here all the details regarding the accounts present on the disk are shown. This disk has the following EMAIL accounts.
- **Reports:** Reports about the entire analysis of the data source can be generated and exported in many formats.

Additional Features:





- **Add a Data Source:** Each case can hold multiple Data Sources.
- **Images/Videos:** Images/ Videos in the data source can be viewed in Gallery View. The information here is displayed in the form of attribute-value pairs.



- **Communications:** All the communications made using the source device are displayed here. This device had communications only in the form of emails.

- The screenshot shows a Microsoft Excel spreadsheet with two columns, A and B. Column A contains a list of email addresses, and Column B contains a list of names and a welcome message. The Excel ribbon is visible at the top, showing the 'Home' tab with options for font, alignment, and styles.

	A	B
40	New Outlook User	Outlook 2003 Team: olteam@microsoft.com
41	Wes Mantoith	John Washer: chkwasher@comcast.net
42	Wes Mantoith	John Washer: chkwasher@comcast.net
43	chkwasher@comcast.net;	smee.rox@gmail.com;
44	chkwasher@comcast.net;	tkidd@swbell.net;
45	chkwasher@comcast.net;	tkidd@swbell.net;
46	chkwasher@comcast.net; dollarhyde86@comcast.net;	tkidd@swbell.net;
47	chkwasher@comcast.net; dollarhyde86@comcast.net;	tkidd@swbell.net;
48	chkwasher@comcast.net; dollarhyde86@comcast.net; isolarman420@hotmail.com; skimmerman27@hotmail.com;	Rasco Radeby: tkidd@swbell.net
49	chkwasher@comcast.net; tkidd@swbell.net;	dollarhyde86@comcast.net;
50	dollarhyde86@comcast.net;	dollarhyde86@comcast.net; dollarhyde86@comcast.net
51	dollarhyde86@comcast.net;	PGP_Corporation_Laura_Lee@mail.vresp.com;
52	dollarhyde86@comcast.net;	PGP_Corporation_Laura_Lee@mail.vresp.com;
53	dollarhyde86@comcast.net;	mailer-daemon@comcast.net;
54	dollarhyde86@comcast.net;	skimmerman27@hotmail.com;
55	dollarhyde86@comcast.net;	tkidd@swbell.net;
56	dollarhyde86@comcast.net; smee.rox@gmail.com;	chkwasher@comcast.net;
57	dollarhyde86@comcast.net; smee.rox@gmail.com;	chkwasher@comcast.net;
58	msoe@microsoft.com;	msoe@microsoft.com;
59	smee.rox@gmail.com;	chkwasher@comcast.net;
60	smee.rox@gmail.com;	dollarhyde86@comcast.net;
61	smee.rox@gmail.com;	dollarhyde86@comcast.net;
62	smee.rox@gmail.com;	mail-noreply@google.com;
63	smee.rox@gmail.com;	mail-noreply@google.com;
64	toothfairy@mentaldent.com;	dollarhyde86@comcast.net;
65	trialwareorderconfirmation@sgp.com; dollarhyde86@comcast.net;	TrialSoftwareOrder@sgp.com;
66	tkidd@swbell.net;	chkwasher@comcast.net;
67		
68		
69		

- [illegible]



Practical - 9

Aim : Using mobsf to check apk files for Vulnerability.

Mobsf

MobSF stands for Mobile Security Framework.

It's an automated, open-source mobile application security testing framework that helps in analyzing Android, iOS, and Windows apps for security vulnerabilities.

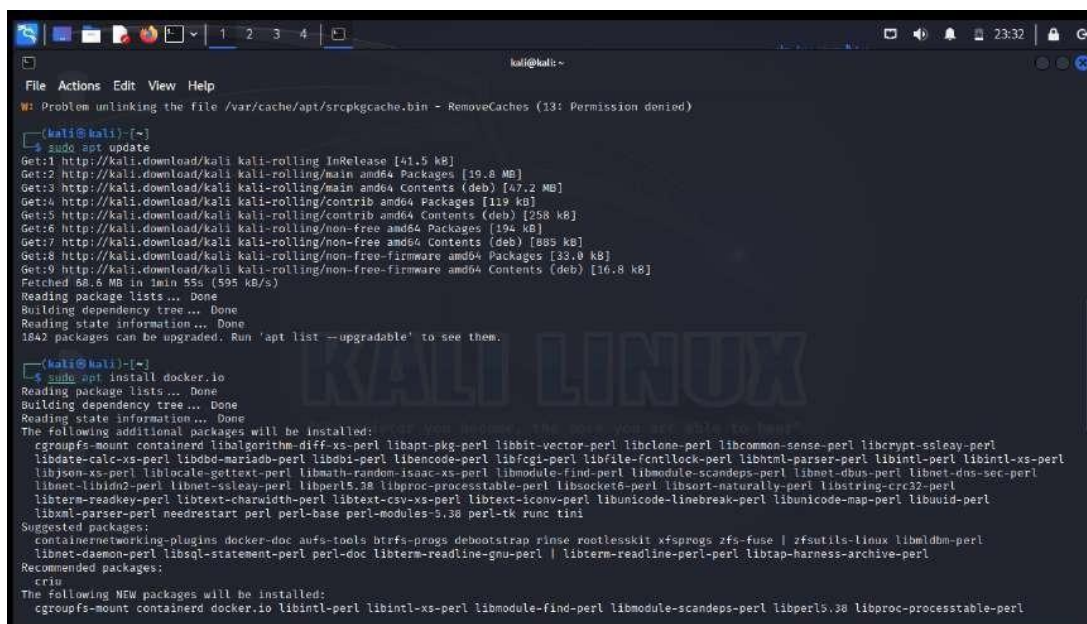
Steps

1. Update APT:

sudo apt update

2. Install docker:

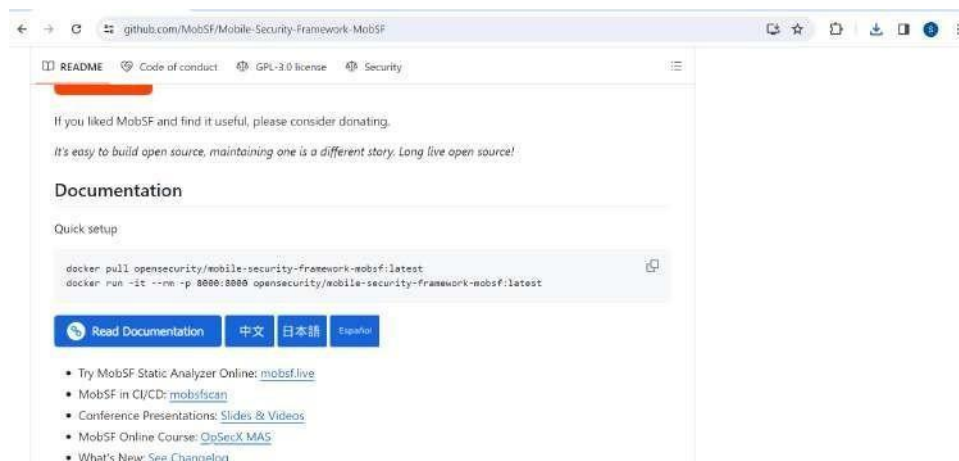
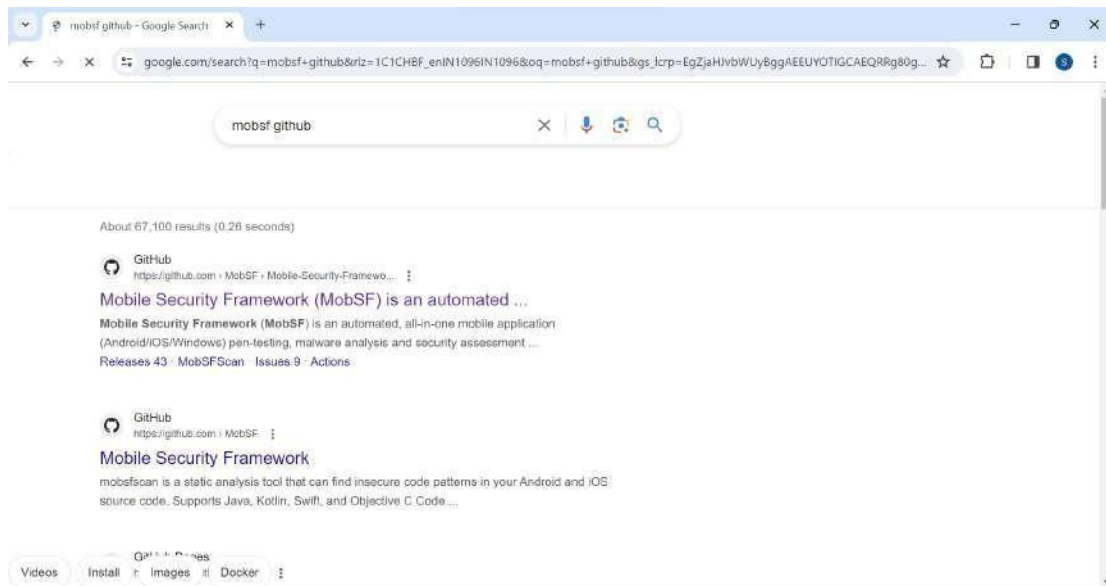
sudo apt install docker.io



```
kali@kali:~$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [19.8 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [47.2 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [119 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [258 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [194 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [885 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [33.0 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [16.8 kB]
Fetched 88.5 MB in 1min 55s (595 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
1842 packages can be upgraded. Run 'apt list --upgradable' to see them.

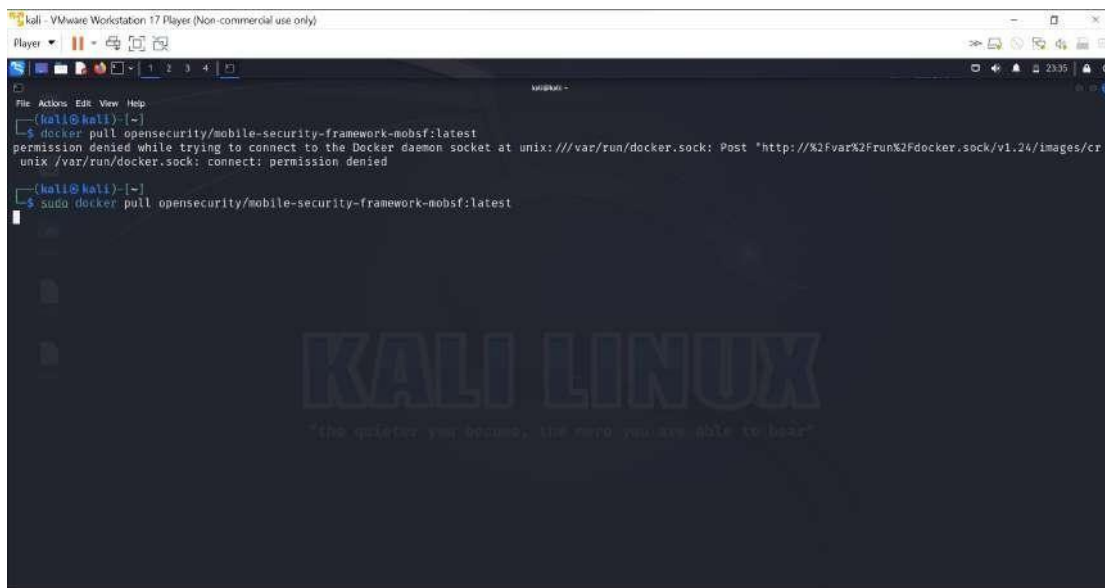
kali@kali:~$ sudo apt install docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  cgroupfs-mount contained libalgorithm-diff-xs-perl libant-pkg-perl libbit-vector-perl libclone-perl libcommon-sense-perl libcrypt-ssleay-perl
  libdate-calc-xs-perl libdbd-mariadb-perl libdbi-perl libencode-perl libfcgi-perl libfile-fcntllock-perl libhtml-parser-perl libhtml-xs-perl
  libjson-xs-perl liblocale-gettext-perl libmath-random-isaac-xs-perl libmodule-find-perl libmodule-scandeps-perl libnet-dbus-perl libnet-dns-sec-perl
  libnet-libidn2-perl libnet-ssleay-perl libperl5.38 libproc-processtable-perl libsocket6-perl libsort-naturally-perl libstring-crc32-perl
  libterm-readkey-perl libtext-charwidth-perl libtext-csv-xs-perl libtext-iconv-perl libunicode-linebreak-perl libunicode-map-perl libuuid-perl
  libxml-parser-perl needrestart perl perl-base perl-modules-5.38 perl-tk runc tini
Suggested packages:
  containerd-plugins docker-doc aufs-tools btrfs-progs debootstrap rinse rootlesskit xfsprogs zfs-fuse | zfsutils-linux libldb-perl
  libnet-daemon-perl libsql-statement-perl perl-doc libterm-readline-gnu-perl | libterm-readline-perl-perl libtap-harness-archive-perl
Recommended packages:
  criu
The following NEW packages will be installed:
  cgroupfs-mount contained docker.io libintl-perl libintl-xs-perl libmodule-find-perl libmodule-scandeps-perl libperl5.38 libproc-processtable-perl
```

3. Search mobsf github on your web browser.



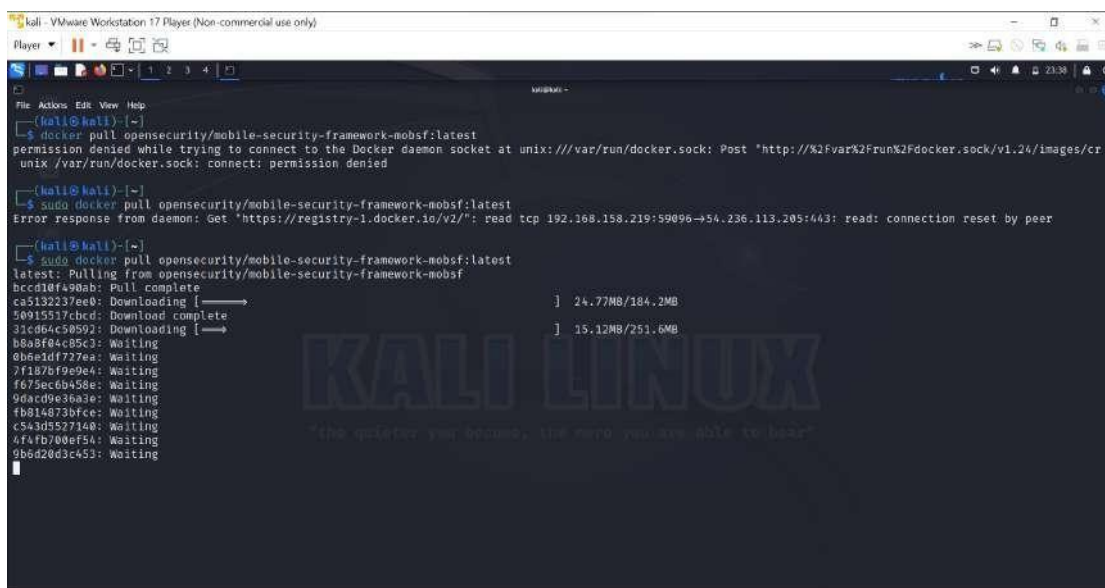
Next, download MobSF Docker image from <https://hub.docker.com/r/opensecurity/mobile-security-framework-mobsf/> with the following command:

```
docker pull opensecurity/mobile-security-framework-mobsf:latest
```

```
kali - VMware Workstation 17 Player (Non-commercial use only)
Player
File Actions Edit View Help
kali@kali:~$ docker pull opensecurity/mobile-security-framework-mobsf:latest
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Post "http://%2Fvar%2Frun%2Fdocker.sock/v1.24/images/cr
unix /var/run/docker.sock: connect: permission denied
kali@kali:~$ sudo docker pull opensecurity/mobile-security-framework-mobsf:latest
```

Once you issue the command, you would notice the following output on your console:



```
kali - VMware Workstation 17 Player (Non-commercial use only)
Player
File Actions Edit View Help
kali@kali:~$ docker pull opensecurity/mobile-security-framework-mobsf:latest
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Post "http://%2Fvar%2Frun%2Fdocker.sock/v1.24/images/cr
unix /var/run/docker.sock: connect: permission denied
kali@kali:~$ sudo docker pull opensecurity/mobile-security-framework-mobsf:latest
Error response from daemon: Get "https://registry-1.docker.io/v2/": read tcp 192.168.158.219:59096->54.236.113.205:443: read: connection reset by peer
kali@kali:~$ sudo docker pull opensecurity/mobile-security-framework-mobsf:latest
latest: Pulling from opensecurity/mobile-security-framework-mobsf
bccc10f490ab: Pull complete
ca5132237ee0: Downloading [====>] 24.77MB/184.2MB
50915517cbcd: Download complete
31cd64c50592: Downloading [====>] 15.12MB/251.6MB
b8a8f04c85c3: Waiting
0b6e1df727ea: Waiting
7f187b19e9e4: Waiting
f679ec6b450e: Waiting
90ac9e36a3e: Waiting
fb814873bfce: Waiting
c543d5527140: Waiting
4f4fb700ef54: Waiting
9b6d20d3c453: Waiting
```

This signifies that the docker image for MobSF is being downloaded. Once completed, the following message will appear:

Now that the docker image is downloaded, the image can be run with the following command:

`docker run -it -p 8000:8000 opensecurity/mobile-security-framework-mobsf`

```
kali - VMware Workstation 17 Player (Non-commercial use only)
Player
File Actions Edit View Help
kali@kali:~$ sudo docker pull opensecurity/mobile-security-framework-mobsf:latest
[sudo] password for kali:
latest: Pulling from opensecurity/mobile-security-framework-mobsf
bced10f490ab: Pull complete
ca5132237ee0: Pull complete
50915517cbcd: Pull complete
11cd64c58992: Pull complete
b8a8f8cc85c3: Pull complete
0b6e1df727ea: Pull complete
7f187bf9e9ea: Pull complete
f675ecb458e: Pull complete
9dad9930a3e: Pull complete
fb814073bfce: Pull complete
c543d5527140: Pull complete
4f4fb780ef5a: Pull complete
9bed20d3c453: Pull complete
Digest: sha256:0c8e302b0962a01a8f6f867991ec64f90a7dea78fabd044d2b47b65fad9e411
Status: Downloaded newer image for opensecurity/mobile-security-framework-mobsf:latest
docker.io/opensecurity/mobile-security-framework-mobsf:latest

kali@kali:~$ sudo docker run -it --rm -p 8000:8000 opensecurity/mobile-security-framework-mobsf:latest
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Post "http://K2PvarK2FrunK2Fdocker.sock/v1.24/c
ontainers/create?": dial unix /var/run/docker.sock: connect: permission denied.
See 'docker run --help'.

kali@kali:~$ sudo !!

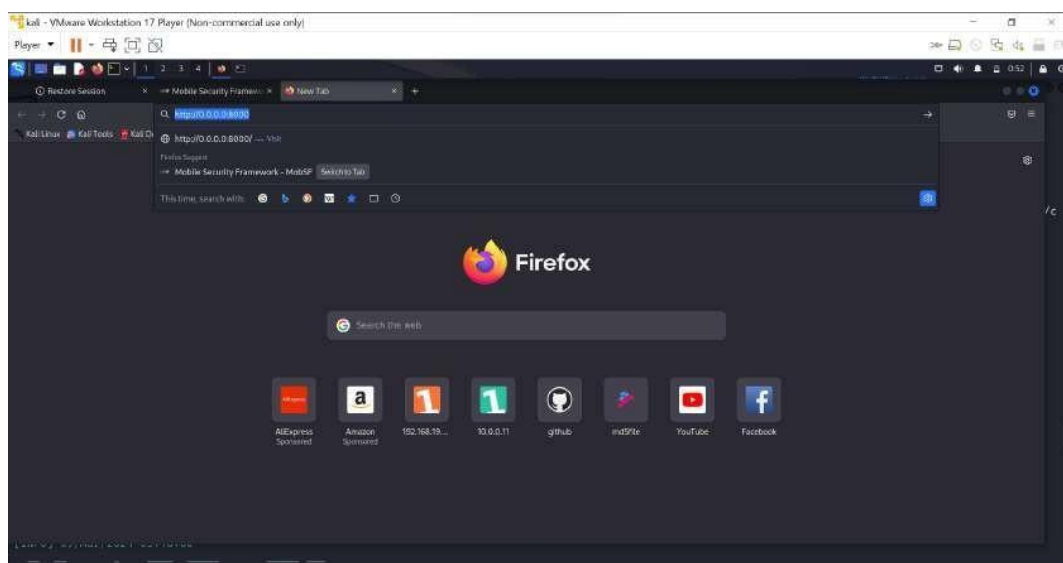
kali@kali:~$ sudo docker run -it --rm -p 8000:8000 opensecurity/mobile-security-framework-mobsf:latest
[sudo] password for kali:
[INFO] 09/Mar/2024 05:47:53 -
[INFO] 09/Mar/2024 05:47:53 - Author: Ajin Abraham | opensecurity.in
[INFO] 09/Mar/2024 05:47:53 - Mobile Security Framework v3.9.5 Beta
[INFO] 09/Mar/2024 05:47:53 - 02 Environment: Linux (Ubuntu 22.04 Jammy Jellyfish) Linux-6.1.0-kali9-amd64-x86_64-with-glibc2.35
[INFO] 09/Mar/2024 05:47:53 - MobSF Basic Environment Check
No changes detected
[INFO] 09/Mar/2024 05:47:54 - Checking for Update.
[INFO] 09/Mar/2024 05:47:55 - No updates available.
[INFO] 09/Mar/2024 05:48:00 -
```

```
kali - VMware Workstation 17 Player (Non-commercial use only)
Player
File Actions Edit View Help
kali@kali:~$ sudo docker run -it --rm -p 8000:8000 opensecurity/mobile-security-framework-mobsf:latest
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Post "http://K2PvarK2FrunK2Fdocker.sock/v1.24/c
ontainers/create?": dial unix /var/run/docker.sock: connect: permission denied.
See 'docker run --help'.

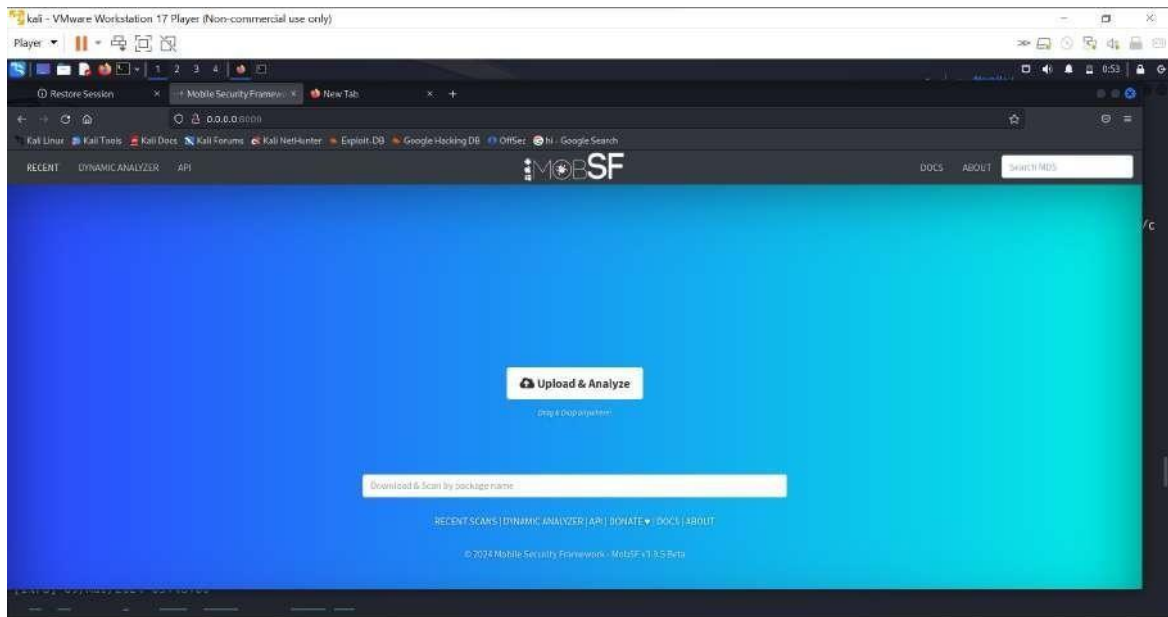
kali@kali:~$ sudo !!

kali@kali:~$ sudo docker run -it --rm -p 8000:8000 opensecurity/mobile-security-framework-mobsf:latest
[sudo] password for kali:
[INFO] 09/Mar/2024 05:47:53 -
[INFO] 09/Mar/2024 05:47:53 - Author: Ajin Abraham | opensecurity.in
[INFO] 09/Mar/2024 05:47:53 - Mobile Security Framework v3.9.5 Beta
[INFO] 09/Mar/2024 05:47:53 - 02 Environment: Linux (Ubuntu 22.04 Jammy Jellyfish) Linux-6.1.0-kali9-amd64-x86_64-with-glibc2.35
[INFO] 09/Mar/2024 05:47:53 - MobSF Basic Environment Check
No changes detected
[INFO] 09/Mar/2024 05:47:54 - Checking for Update.
[INFO] 09/Mar/2024 05:47:55 - No updates available.
[INFO] 09/Mar/2024 05:48:00 -
```

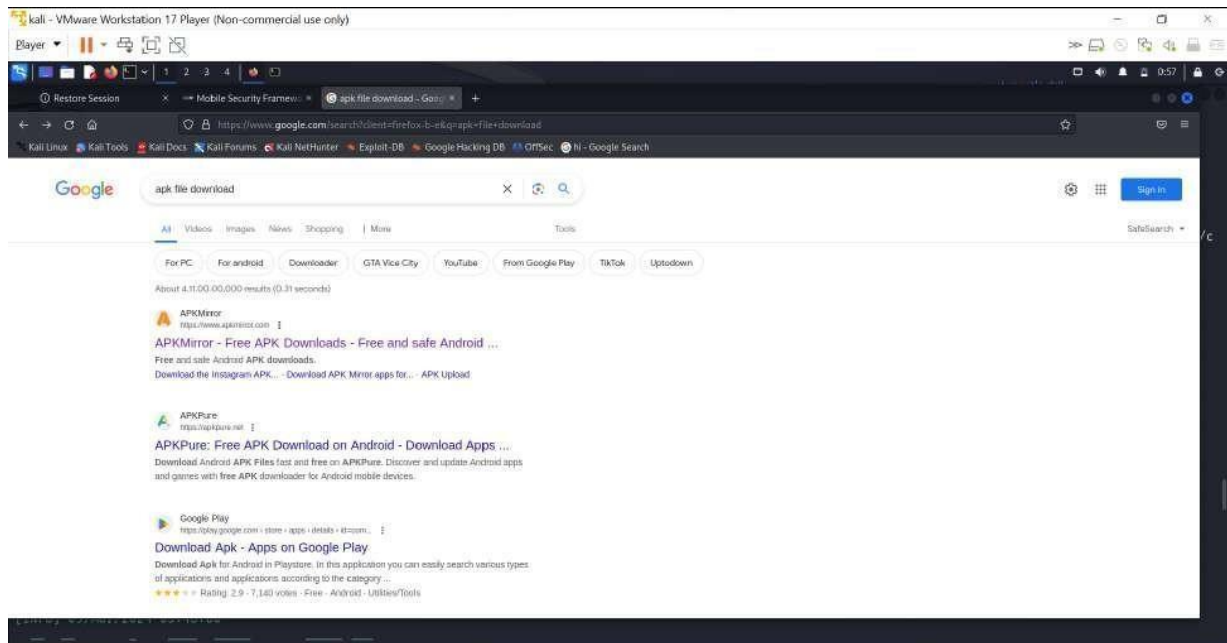
Once done, you would see message “Listening at: http://0.0.0.0:8000”. This signifies the URL from which one can access MobSF

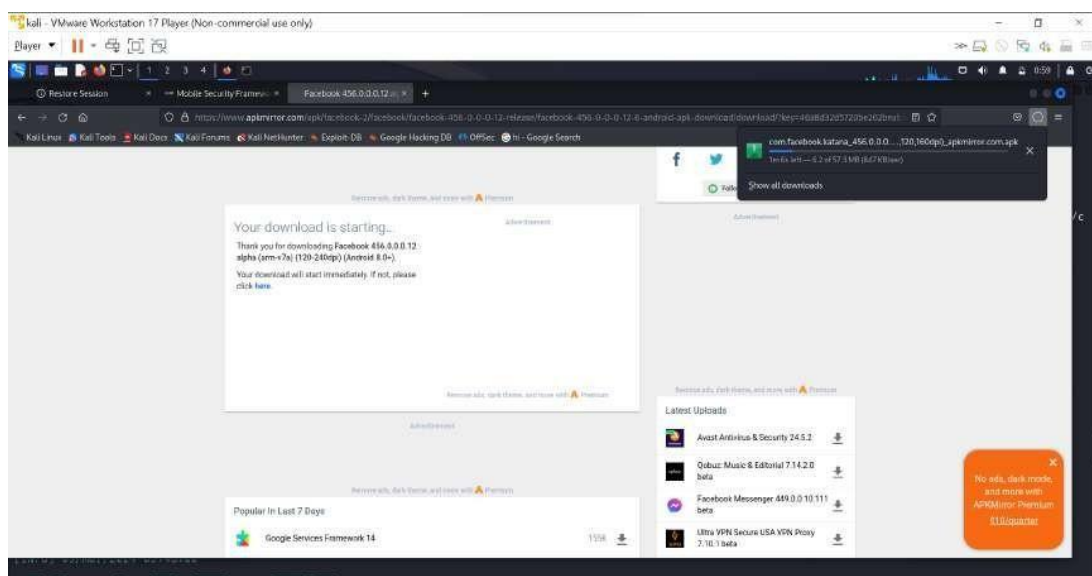
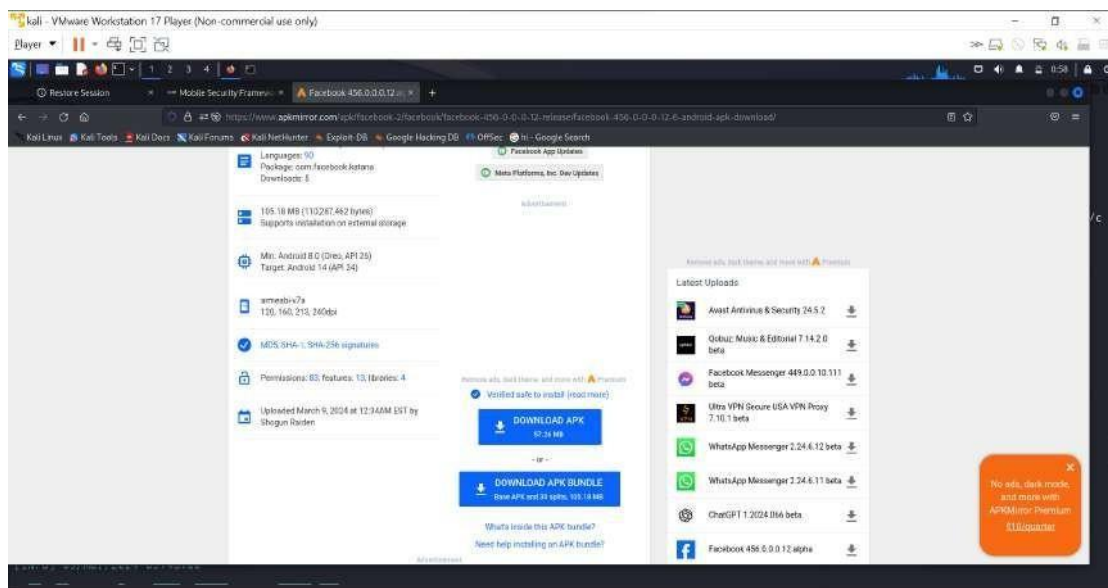
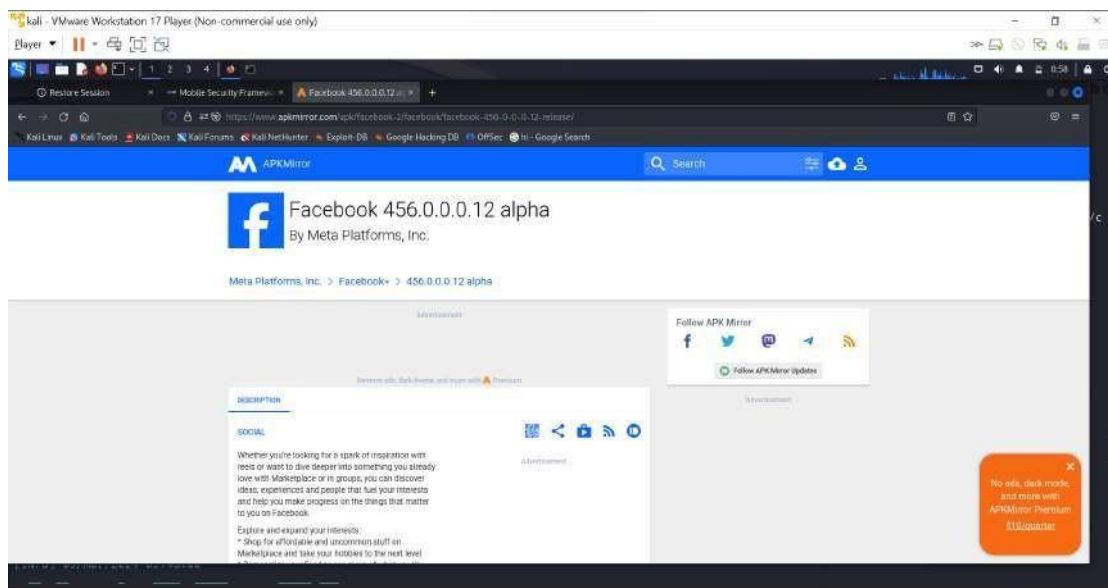


Our MobSF framework is ready for us to conduct static analysis for APK files.

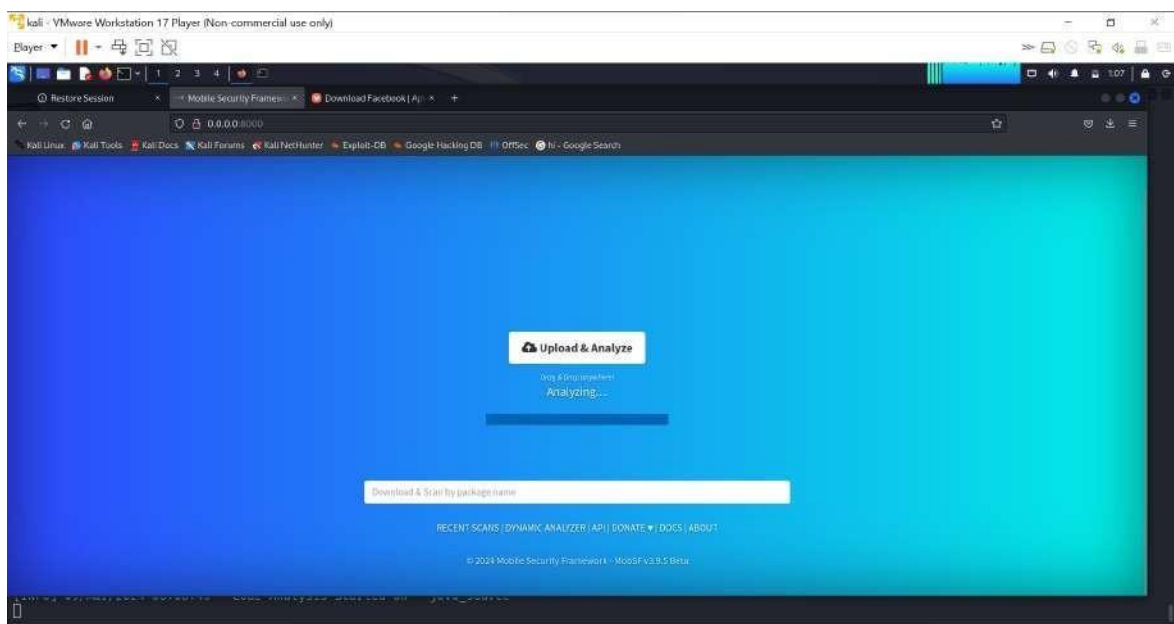
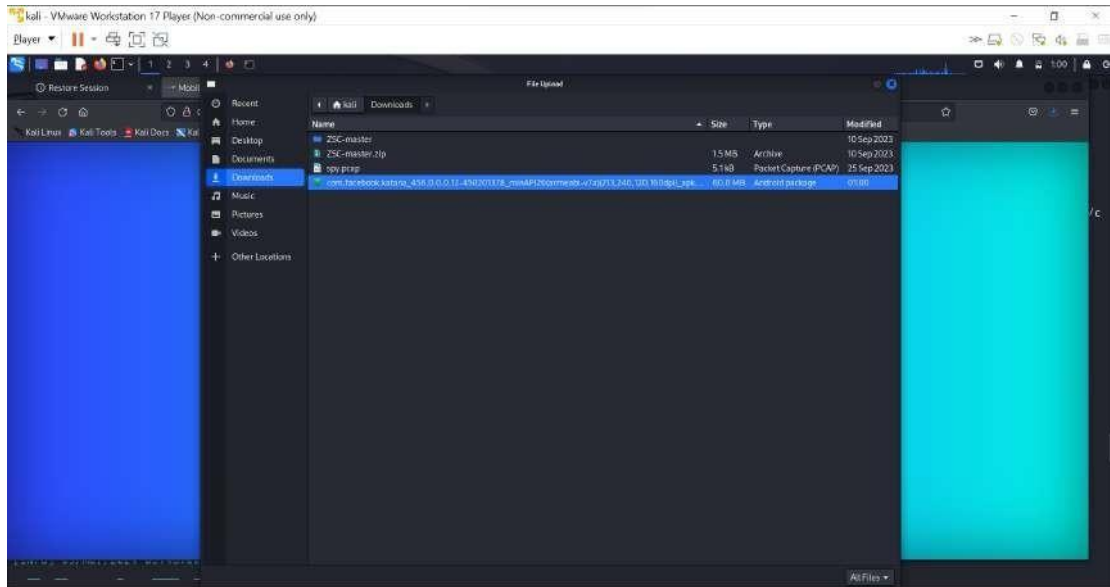


For our testing, we take Facebook Lite's APK file.

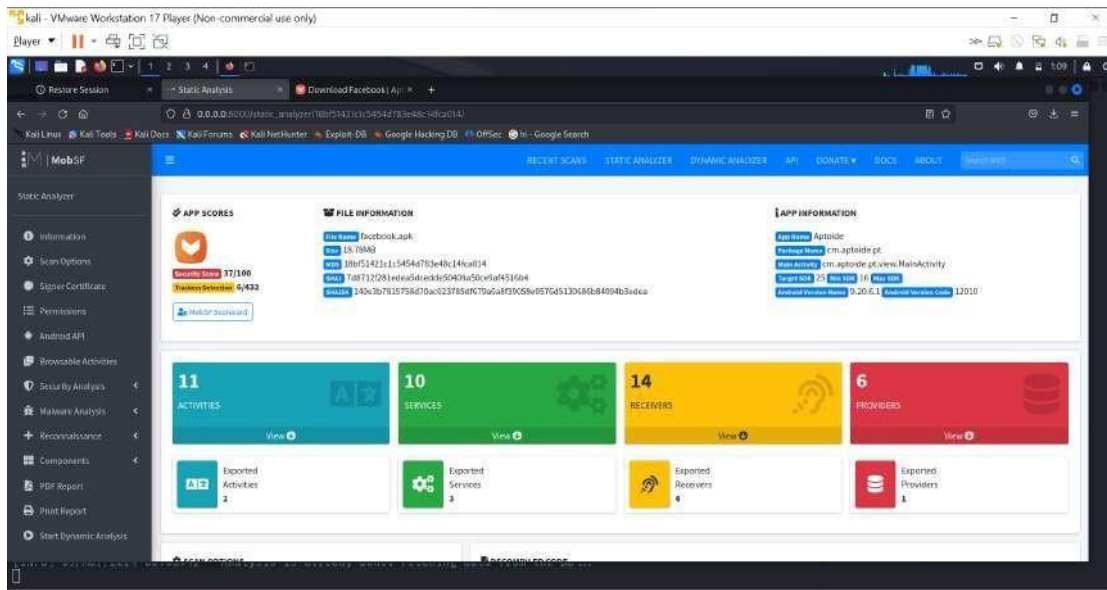




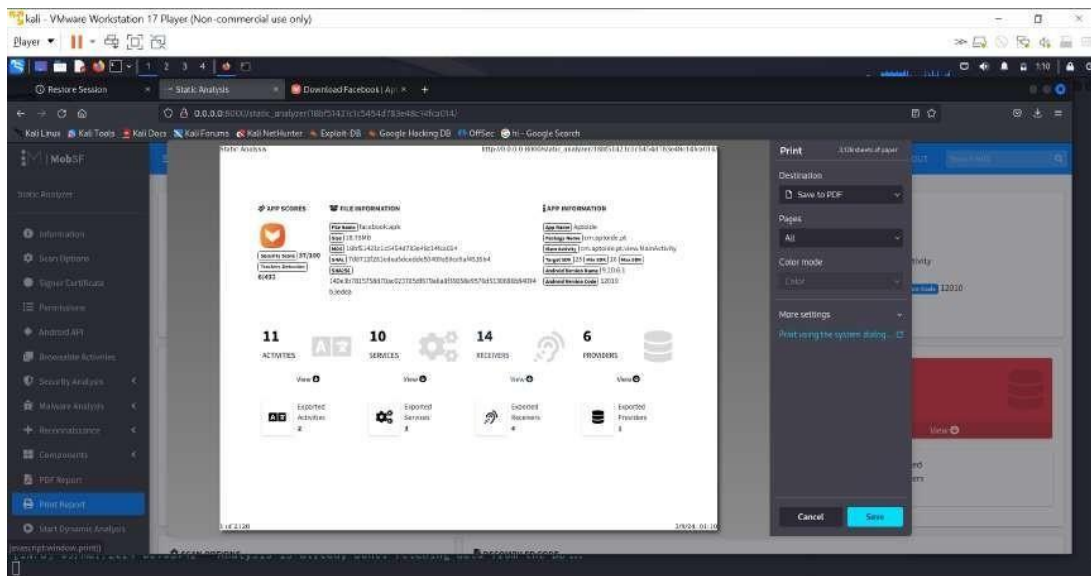
Once you upload the APK file, you would notice the following output being printed on our terminal:



Once the analysis is complete, the browser will show assessment details, such as application description, Android permission analysis, code analysis, CVSS grading, malware analysis etc.



You also notice a brief overview of the application you have reviewed. You can also the report for same.



Practical - 10

AIM: Implementation of Cyber Forensics tools for Disk Imaging, Data acquisition, Data extraction and Data Analysis and recovery.

SCOPE:

The Scope of practical acquisition of digital evidence can encompass a wide range of methods and devices. Common areas are: Hard Drives, USB, Mobile Devices, Network Forensics, Live system etc.

REQUIREMENTS:

The requirements are Hardware and Devices, FTK imager software, Forensics workstations, sample data, Documentation Templates etc.

THEORY:

Data Acquisition:

- The gathering and recovery of sensitive data during a digital forensic investigation is known as data acquisition and collecting and preserving electronic evidence from various sources.
- Digital forensic analysts need to know how to access, recover, and restore that data as well as how to protect it for future management. This involves producing a forensic image from digital devices and other computer technologies.
- The data acquisition process involves several steps: Identification, Preservation, collection, Verification, Documentation.

Disk Imaging:

Disk imaging, also known as forensic imaging or disk cloning, is a crucial process in digital forensics for cybersecurity.

It involves creating a bit-for-bit copy or snapshot of an entire storage device, such as a hard drive, solid-state drive (SSD), or any other media, to preserve its contents for analysis and investigation.

The purpose of disk imaging is to obtain an exact replica of the original storage device, including not only the visible files and folders but also the hidden and deleted data.

Types of Disk Imaging:

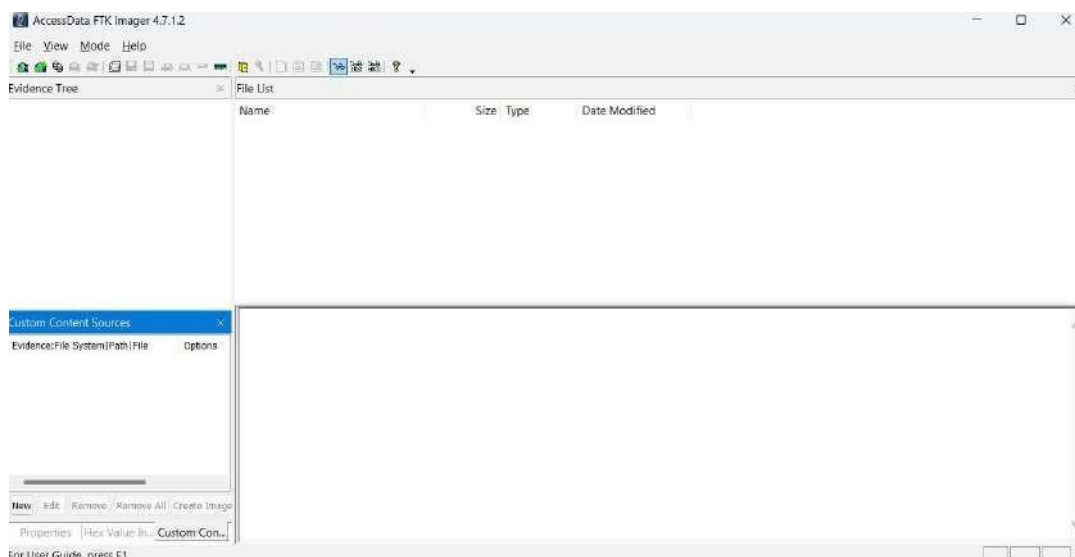
- **Logical Imaging:** Logical imaging, also known as file-level imaging or logical acquisition, involves creating an image or snapshot of the logical structure of a storage device, such as a file system or a partition. This type of imaging captures the files and directories stored on the disk, along with their metadata and attributes, without including unallocated space or low-level disk structures.
- **Physical Imaging:** Physical imaging, also known as bit-for-bit imaging or sector-level imaging, is a comprehensive process that involves creating an exact copy of the entire storage device, including all sectors, regardless of whether they are allocated or unallocated. This type of imaging captures not only the visible files

and folders but also the hidden data, deleted files, slack space, and all other data present on the disk.

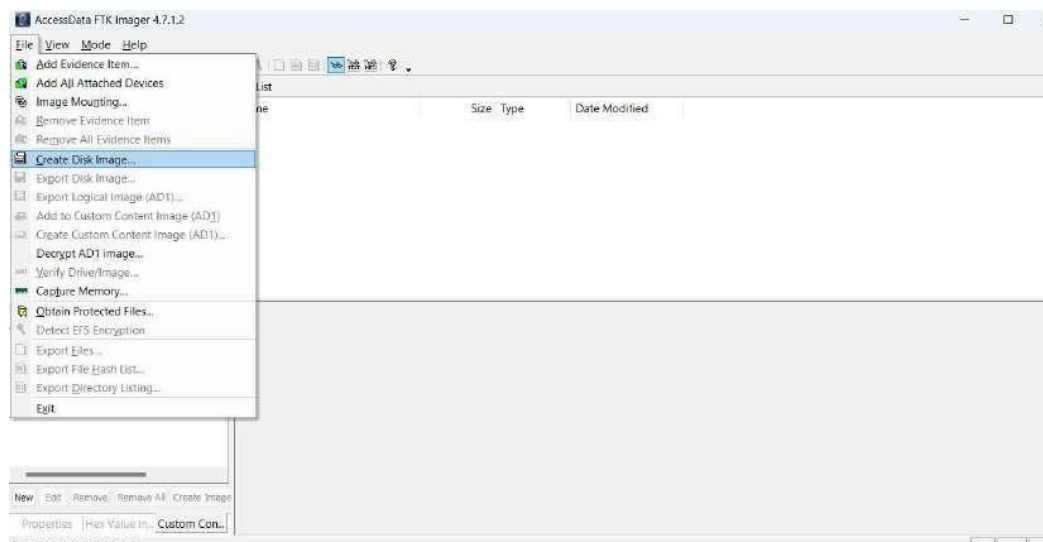
- Forensic Significance of Imaging:
- Forensic imaging prevents the loss of original data. These imaging tools and techniques are the only way to ensure that electronic data can be successfully admitted as evidence in a court or legal proceeding.

EXAMINATION:

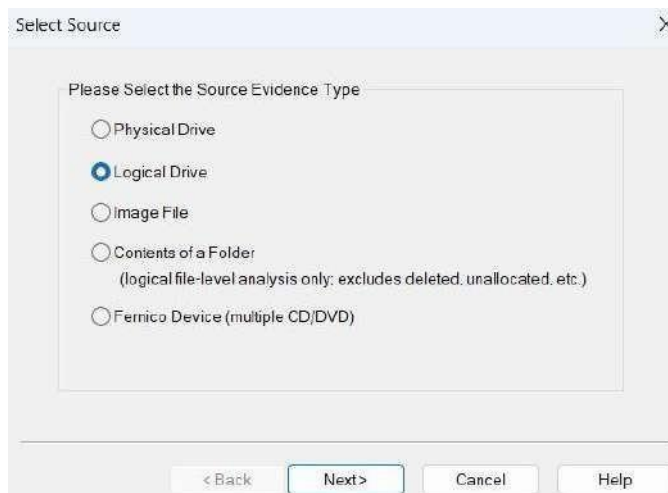
Step 1: Open FTK imager software.



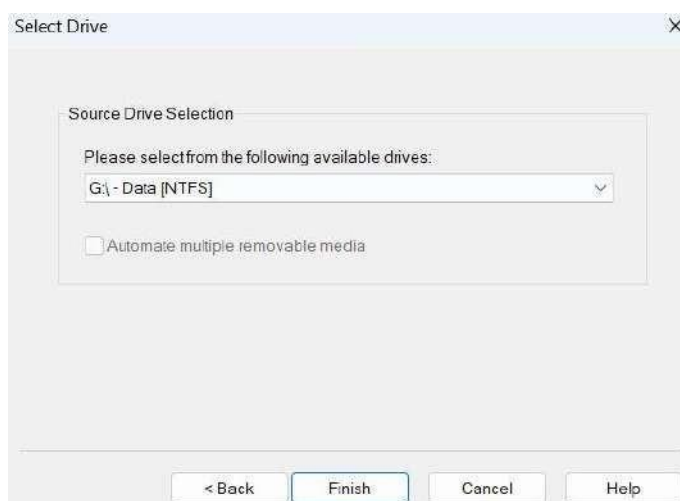
Step 2: Now go to the file option - > choose Create Disk Image.



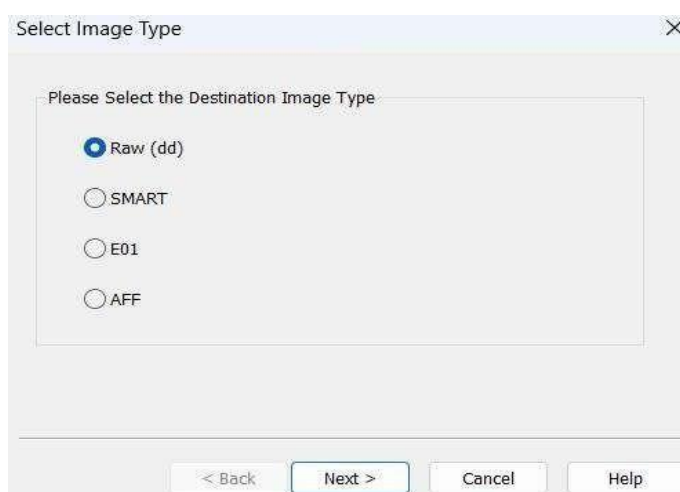
Step 3: After clicking Create Disk image -> select the source of evidence type like Logical Drive, then click next option.



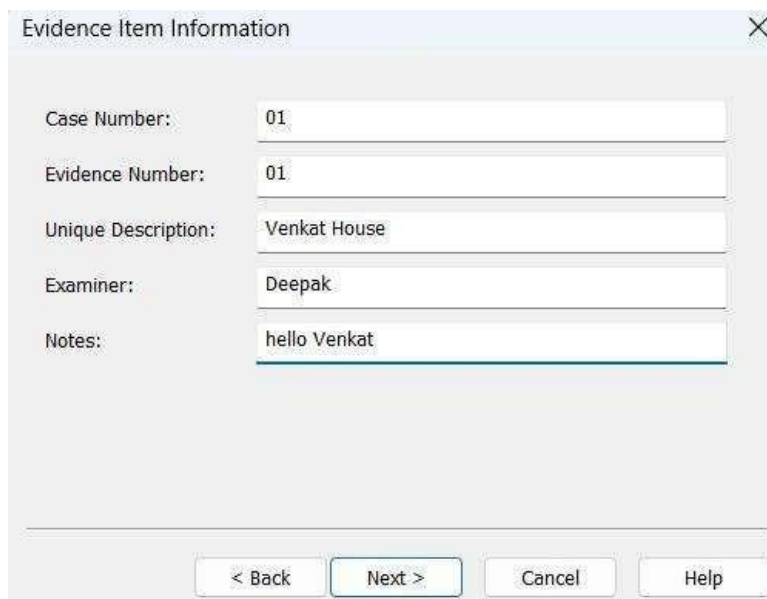
Step 4: Now select source drive path and then press the finish button.



Step 5: Choose destination image type (Raw(dd)) -> Next.



Step 6: Now fill the form of Evidence item information.

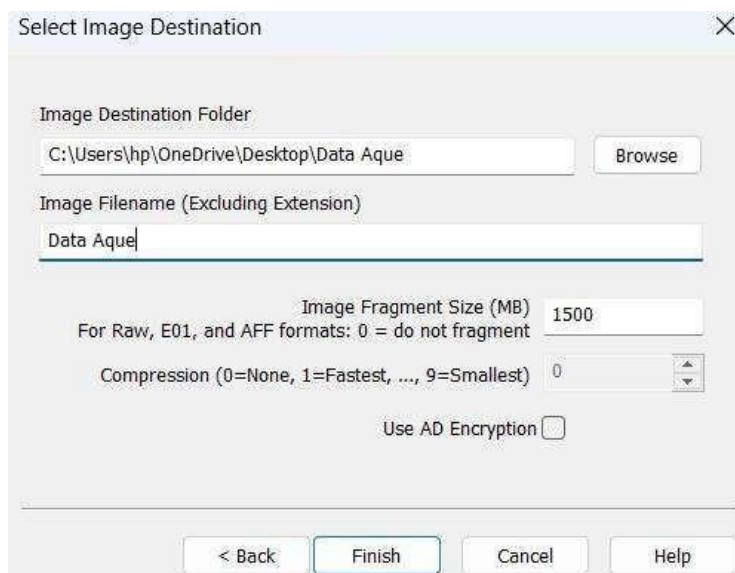


The 'Evidence Item Information' dialog box contains the following fields and values:

Field	Value
Case Number:	01
Evidence Number:	01
Unique Description:	Venkat House
Examiner:	Deepak
Notes:	hello Venkat

Buttons at the bottom: < Back, Next >, Cancel, Help.

Step 7: After filling the information -> select image destination, enter Image filename and Image fragment size then click the finish button.

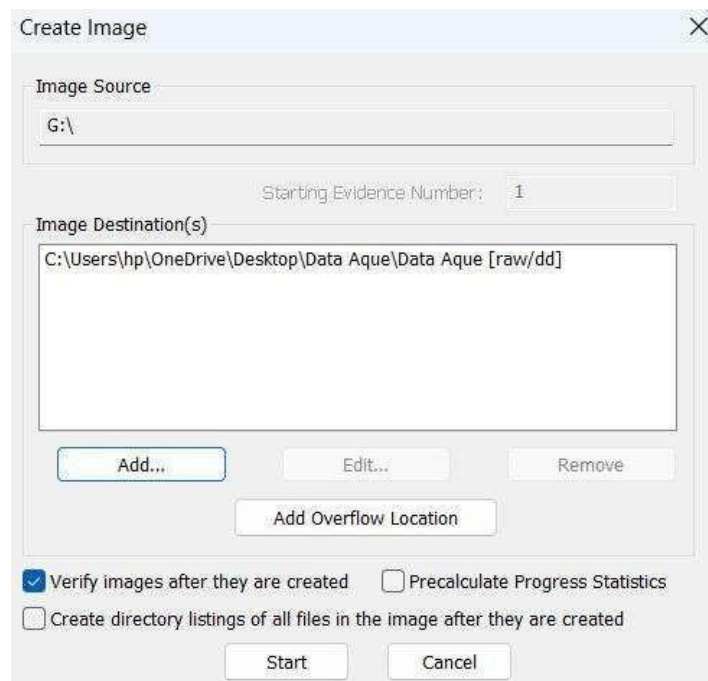


The 'Select Image Destination' dialog box contains the following fields and values:

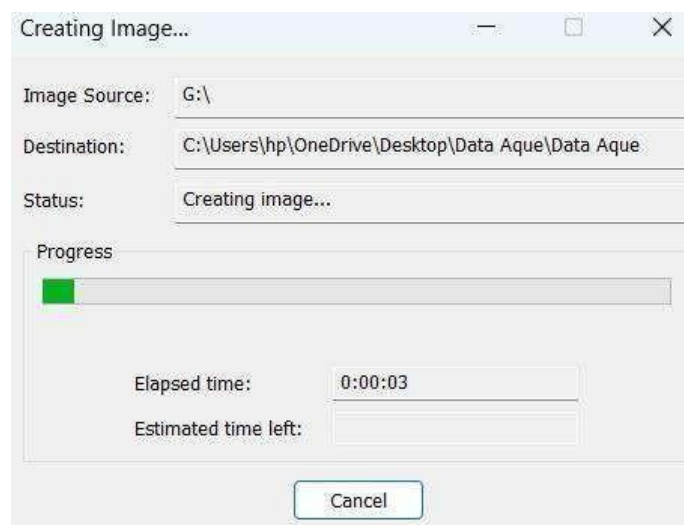
Field	Value
Image Destination Folder	C:\Users\hp\OneDrive\Desktop\Data Aque
Image Filename (Excluding Extension)	Data Aque
Image Fragment Size (MB)	1500
Compression (0=None, 1=Fastest, ..., 9=Smallest)	0
Use AD Encryption	☐

Buttons at the bottom: < Back, Finish, Cancel, Help.

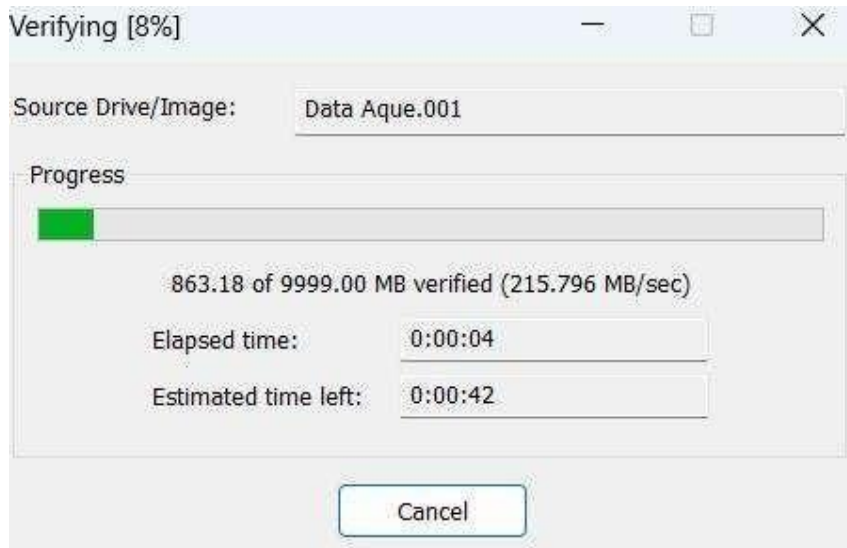
Step 8: After all the processes now, click in the checkbox of verify images after they are created -> start.



Step 9: Progress for creating image.



Step 10: Verifying the source drive/image.



Step 11: Verify result of source drive/image form of hash values.

Drive/Image Verify Results	
Name	Data Aque.001
Sector count	20477952
MDS Hash	
Computed hash	a8b13d1099105de415d7b795374ef1aa
Report Hash	a8b13d1099105de415d7b795374ef1aa
Verify result	Match
SHA1 Hash	
Computed hash	b473b690ef93531639142ee301187aba0947d81
Report Hash	b473b690ef93531639142ee301187aba0947d81
Verify result	Match
Bad Blocks List	
Bad block(s) in image	No bad blocks found in image

OBSERVATION:

Created By AccessData® FTK® Imager 4.7.1.2

Case Information:

Acquired using: ADI4.7.1.2

Case Number: 01

Evidence Number: 01

Unique description: Venkat House

Examiner: Deepak

Notes: hello Venkat

Information for C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque:

Physical Evidentiary Item (Source) Information:

[Device Info]

Source Type: Logical

[Drive Geometry]

Bytes per Sector: 512

Sector Count: 20,477,952

[Physical Drive Information]

Removable drive: False

Source data size: 9999 MB

Sector count: 20477952

[Computed Hashes]

MD5 checksum: a8b13d1099105de415d7b795374ef1aa

SHA1 checksum: b473b690eff93531639142ee301187a8a0947d81

Image Information:

Acquisition started: Wed Jun 28 13:00:17 2023

Acquisition finished: Wed Jun 28 13:01:01 2023

Segment list:

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.001

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.002

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.003

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.004

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.005

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.006

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.007

Image Verification Results:

Verification started: Wed Jun 28 13:01:01 2023

Verification finished: Wed Jun 28 13:01:40 2023

MD5 checksum: a8b13d1099105de415d7b795374ef1aa : verified

SHA1 checksum: b473b690eff93531639142ee301187a8a0947d81 : verified

Autopsy (Data Extraction, Data Analysis and Data Recovery)

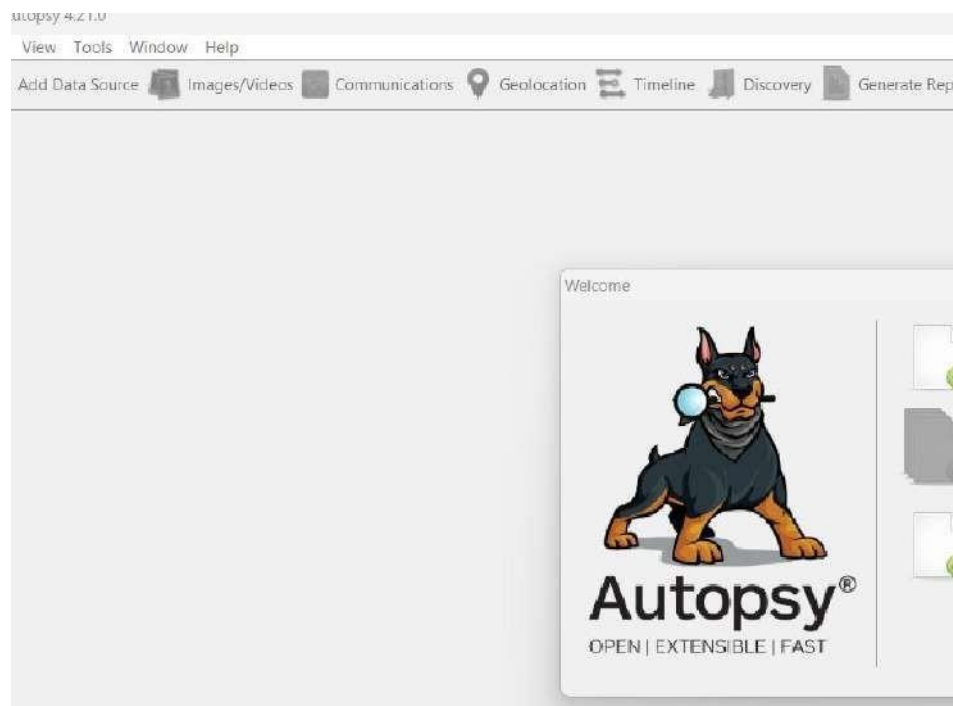
- Autopsy is a digital forensics platform and graphical interface to The Sleuth Kit and other digital
- forensics tools. It is used by law enforcement, military, and corporate examiners to investigate what
- happened on a computer. You can even use it to recover photos from your camera's memory card.
- Basically, the autopsy is a free open-source tool that supports a wide range of other digital forensics
- modules and tools. The tool is largely maintained by Basis Technology Corp. With the assistance of
- programmers from the community. The Autopsy is computer software that makes it simpler to deploy
- many of the open-source programs and plugins used in The Sleuth Kit.

Features of Autopsy

- Timeline Analysis:- Displays system events in a graphical interface to help identify activity.
- Keyword Search: Text extraction and index searched modules enable you to find files that
- mention specific terms and find regular expression patterns.
- Web Artifacts:- Extracts web activity from common browsers to help identify user activity.
- Registry Analysis:-Uses Reg Ripper to identify recently accessed documents and USB devices.
- Email Analysis: Parses MBOX format messages, such as Thunderbird.
- EXIF: Extracts geo location and camera information from JPEG files.
- Robust File System Analysis: Support for common file systems, including NTFS,
- FAT12/FAT16/FAT32/ExFAT, HFS+, ISO9660 (CD-ROM), Ext2/Ext3/Ext4,
- Unicode Strings Extraction: Extracts strings from unallocated space and unknown file types in many languages
- File Type Detection based on signatures and extension mismatch detection. Interesting Files Module will flag files and folders based on name and path.
- Android Support: Extracts data from SMS, call logs, contacts, Tango, Words with Friends, and more.
- Multi-User Cases: Collaborate with fellow examiners on large cases.
- LNK File Analysis:- Identifies shortcuts and accessed documents

EXAMINATION:

Step 1: Download and setup the installation process of Autopsy tool. Open it.



Step 2: Create a case file – Open Autopsy and click New Case > enter a case name and choose a directory.

New Case Information

Steps

1. Case Information
2. Optional Information

Case Information

Case Name:

Base Directory:

Case Type: ☒ Single-User ☐ Multi-User

Case data will be stored in the following directory:

< Back Next > Finish Cancel Help

New Case Information

Steps

1. Case Information
2. Optional Information

Optional Information

Case

Number:

Examiner

Name:

Phone:

Email:

Notes:

Organization

Organization analysis is being done for:

< Back Next > Finish Cancel Help

Step 3: Select data source – Now, add the data source windows pops up > select Logical disk from the drop-down list > choose the targeted drive image.

Add Data Source

Steps

1. Select Host
2. **Select Data Source Type**
3. Select Data Source
4. Configure Ingest
5. Add Data Source

Select Data Source Type

- Disk Image or VM File
- Local Disk
- Logical Files**
- Unallocated Space Image File
- Autopsy Logical Imager Results
- XRY Text Export

< Back Next > Finish Cancel Help

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
3. **Select Data Source**
4. Configure Ingest
5. Add Data Source

Select Data Source

Local files and folders

F:\

Add Delete Clear

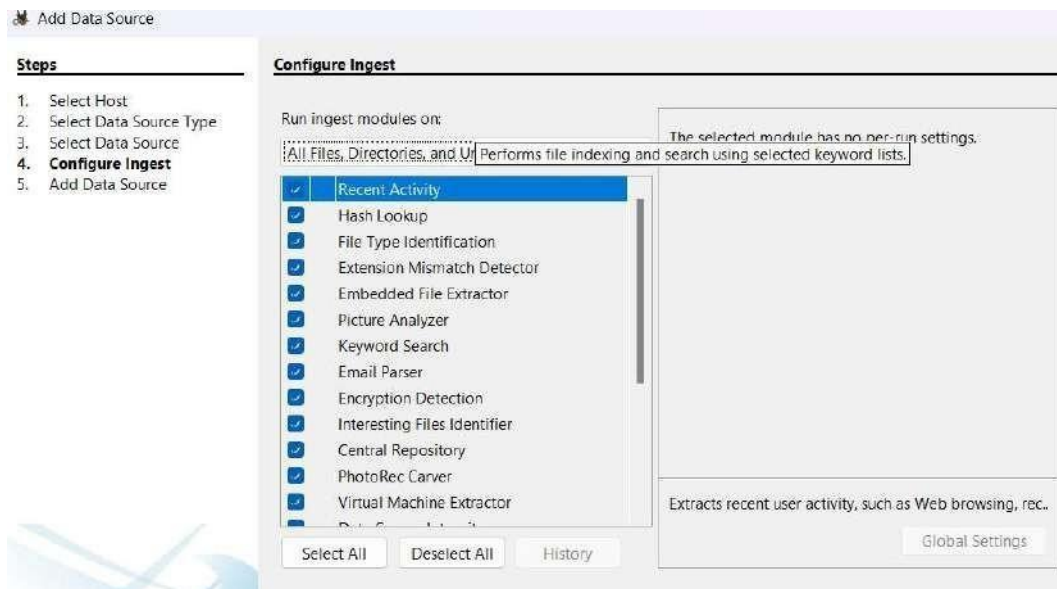
Logical File Set Display Name: Default Change

Timestamps To Include:

- ☐ Modified Time - Often not changed when a file is copied
- ☐ Creation Time - Often changed when a file is copied
- ☐ Access Time - Can be changed when the file is opened

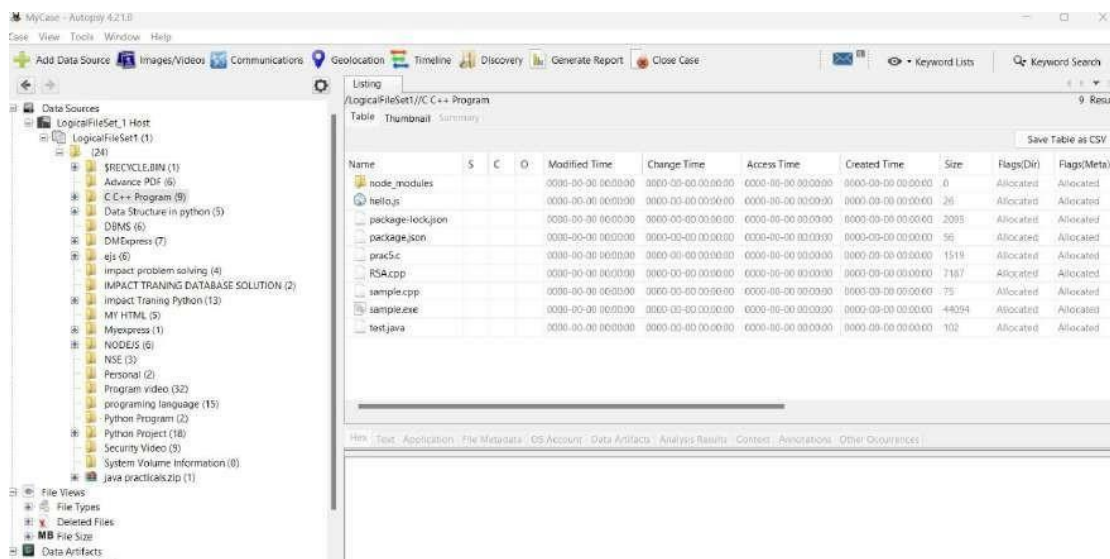
NOTE: Time stamps may have changed when the files were copied to the current location.

< Back Next > Finish Cancel Help



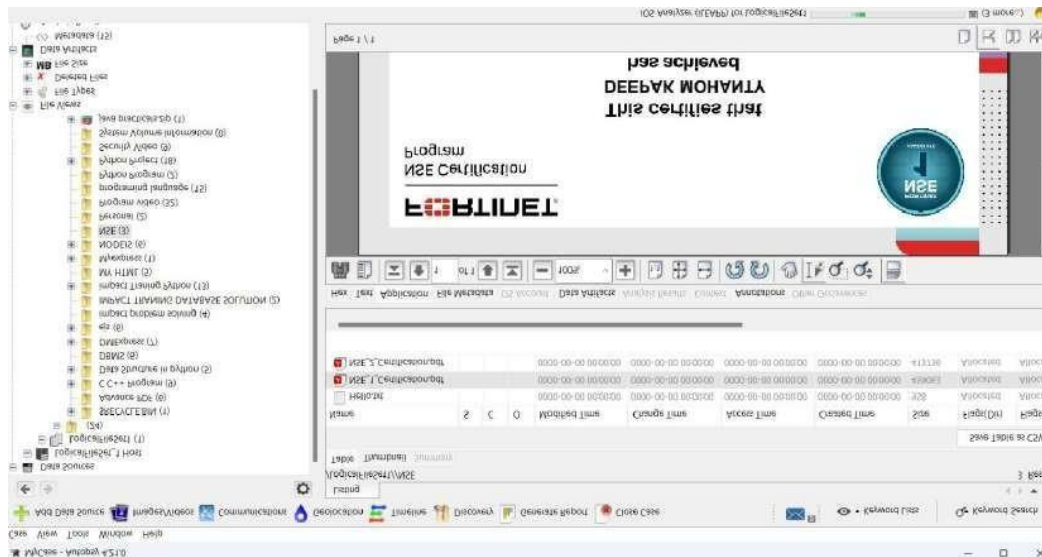
Step 4: Click next to proceed.

Step 5: Wait for the analysis to complete, and the data will be displayed in different categories.



OBSERVATION:

Data Restoration – Open the folder of the files, Right-click on it to restore and select export. Choose a location to export the data , click save.



Practical - 7

Aim: Implementation of IT Audit, malware analysis and Vulnerability assessment and generate the report.

Objective: To know how to find vulnerabilities by using NESSUS

Requirements: Laptop, Kali linux, Nessusd package in kali

Malware analysis

The image shows two screenshots. The top screenshot is a Google search for "phishing links list". It displays a table of phishing URLs with targeted brands and times. The bottom screenshot is the VirusTotal website, showing the search bar and the URL "http://84.109.047.100/wordpress.com/" entered for analysis.

Phishing URL	Targeted Brand	Time
http://84.109.047.100/wordpress.com/	BT Group plc	01/01/18
http://media-clone-elbon-muvenca.com/	Netflix Inc.	01/01/18
http://pplucet.com.blogspot.co.at/	Singtel Optus	01/01/18
https://gocodrogados.edu.br/adm/	Microsoft OneDrive	00/00/00

People also ask: What are the phishing links?

VIRUSTOTAL

Analyze suspicious files, domains, IPs and URLs to detect malware and other threats, automatically share them with the security community.

FILE URL SEARCH

<http://84.109.047.100/wordpress.com/>

By submitting data above, you are agreeing to our Terms of Service and Privacy Policy, and to the sharing of your URL submission with the security community. Please do not submit any personal information; VirusTotal is not responsible for the contents of your submission. Learn more.

Want to automate submissions? Check our API, or access your API key.

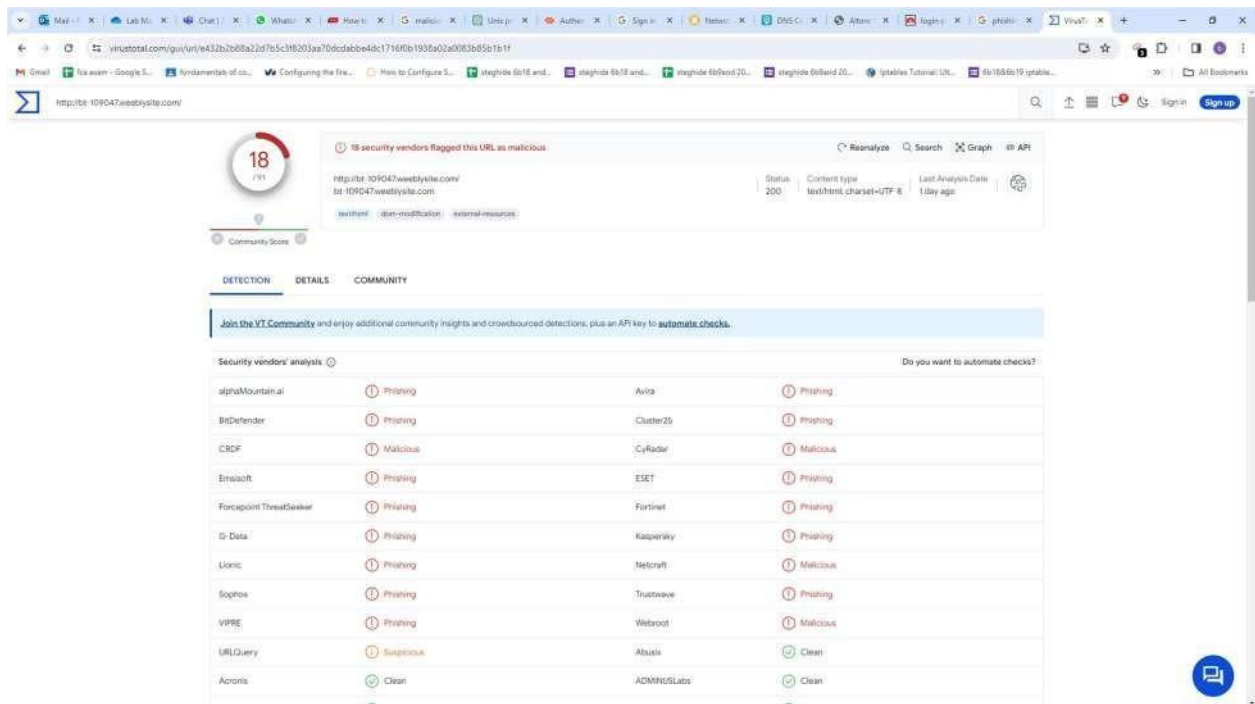
VirusTotal
Contact Us
Get Support
How it Works
TOS | Privacy Policy
Blog | Releases

Community
Join Community
Vote and Comment
Contributors
Top Users
Community Buzz

Tools
API Scripts
YARA
Desktop Apps
Browser Extensions
Mobile App

Premium Services
Get a Demo
Intelligence
Hunting
Graph
API v3 | v2
Use Cases

Documentation
Searching
Reports
API v3 | v2
Use Cases



NESSUS:

Nessus, developed by Tenable Inc, is a widely-used open-source vulnerability scanner. It offers a paid subscription, Nessus Professional, as well as a free version, Nessus Essentials, which is limited to 16 IP addresses per scanner.

Nessus provides a range of services, including vulnerability assessments, network scans, web scans, asset discovery, and more, to aid security professionals, penetration testers, and other cybersecurity enthusiasts in proactively identifying and mitigating vulnerabilities in their networks.

How to install a Nessus in kali

Unlike many security tools, Nessus doesn't come [installed on Kali Linux](#). But it is very easy to download and install. Follow these steps to install Nessus on your Kali:

1. Download the Nessus package for Debian on the [Nessus website](#) and make sure you set the Platform to Linux-Debian-amd64.

Tenable Nessus

Tenable Nessus Agent

Tenable Nessus Network Monitor

Tenable Security Center

Integrations

Sensor Proxy

Tenable Log Correlation Engine

Tenable Core

Tenable OT Security

Tenable Identity Exposure

Frictionless

Tenable Cloud Security

Downloads / Tenable Nessus

Tenable Nessus

1 Download and Install Nessus

Choose Download

Version

Nessus - 10.7.0

Platform

Linux - Debian - amd64

[Download](#) [Checksum](#)

Download by curl

```
curl --request GET \
  --url 'https://www.tenable.com/downloads/api/v2/pages/nessus/files/Nessus-10.7.0-debian10_amd64.deb' \
  --output 'Nessus-10.7.0-debian10_amd64.deb'
```

[View the Downloads API documentation](#)

2. When it's finished downloading, open your Linux terminal and navigate to the location you downloaded the Nessus file to.

```

root@kali: ~
└─$ systemctl status nessusd.service
Unit nessusd.service could not be found.

root@kali: ~
└─$ systemctl status nessusd
Unit nessusd.service could not be found.

root@kali: ~
└─$ nessusd
nessusd: command not found

root@kali: ~
└─$ apt-get install nessusd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package nessusd

root@kali: ~
└─$ curl --request GET \
  --url 'https://www.tenable.com/downloads/api/v2/pages/nessus/files/Nessus-10.7.0-debian10_amd64.deb' \
  --output 'Nessus-10.7.0-debian10_amd64.deb'
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 64.8M    0 64.8M    0     0  6762k      0  0:00:09  0:00:09  0:00:00 14.1M

root@kali: ~
└─$ ls
Desktop  Documents  Downloads  Music  Nessus-10.7.0-debian10_amd64.deb  Pictures  practice  Public  Templates  Videos

```

Install Nessus using this command:

```
(root@kali)~# dpkg -i Nessus-10.7.0-debian10_amd64.deb
Selecting previously unselected package nessus.
(Reading database ... 399450 files and directories currently installed.)
Preparing to unpack Nessus-10.7.0-debian10_amd64.deb ...
Unpacking nessus (10.7.0) ...
Setting up nessus (10.7.0) ...
HMAC : (Module_Integrity) : Pass
SHA1 : (KAT_Digest) : Pass
SHA2 : (KAT_Digest) : Pass
SHA3 : (KAT_Digest) : Pass
TDES : (KAT_Cipher) : Pass
AES_GCM : (KAT_Cipher) : Pass
AES_ECB_Decrypt : (KAT_Cipher) : Pass
RSA : (KAT_Signature) : RNG : (Continuous_RNG_Test) : Pass
Pass
ECDSA : (PCT_Signature) : Pass
ECDSA : (PCT_Signature) : Pass
DSA : (PCT_Signature) : Pass
TLS13_KDF_EXTRACT : (KAT_KDF) : Pass
TLS13_KDF_EXPAND : (KAT_KDF) : Pass
TLS12_PRF : (KAT_KDF) : Pass
PBKDF2 : (KAT_KDF) : Pass
SSHKDF : (KAT_KDF) : Pass
KBKDF : (KAT_KDF) : Pass
```

Start the Nessus service with this command:

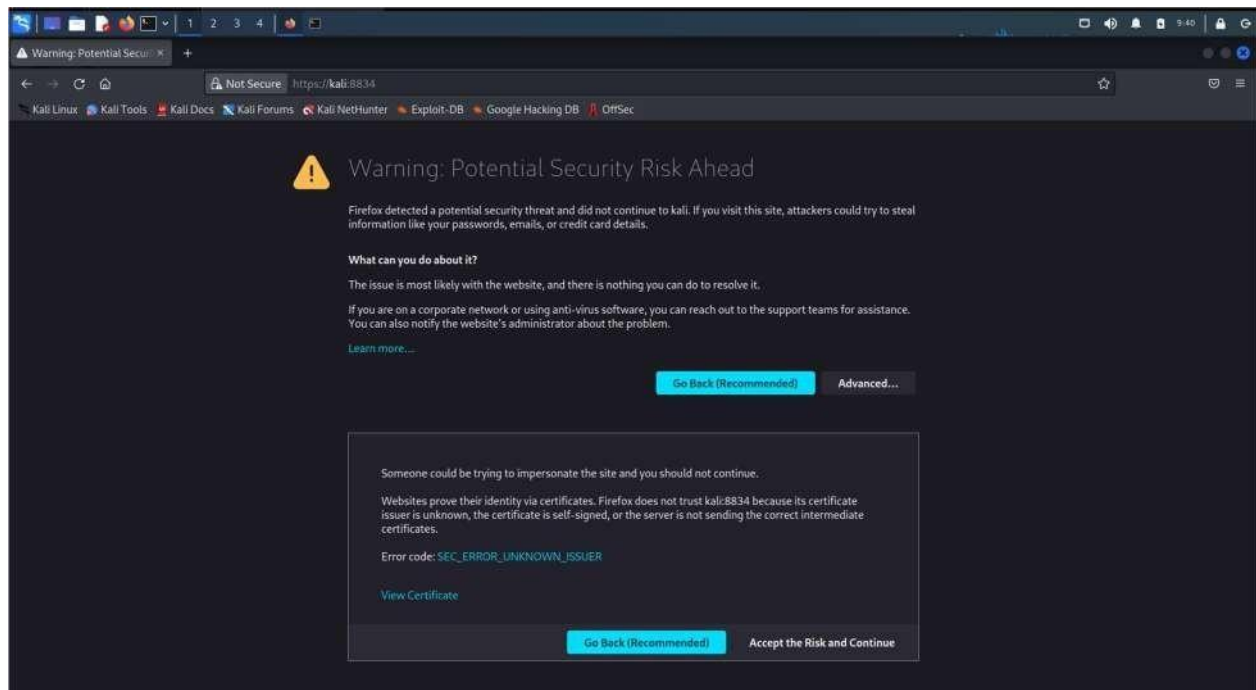
```
(root@kali)~# systemctl start nessusd.service

(root@kali)~# systemctl status nessusd.service
● nessusd.service - The Nessus Vulnerability Scanner
   Loaded: loaded (/lib/systemd/system/nessusd.service; disabled; preset: disabled)
   Active: active (running) since Sat 2024-02-10 00:56:06 EST; 13s ago
     Main PID: 7753 (nessus-service)
        Tasks: 14 (limit: 2260)
           Memory: 114.4M
              CPU: 12.710s
     CGroup: /system.slice/nessusd.service
             └─7753 /opt/nessus/sbin/nessus-service -q
               7754 nessusd -q

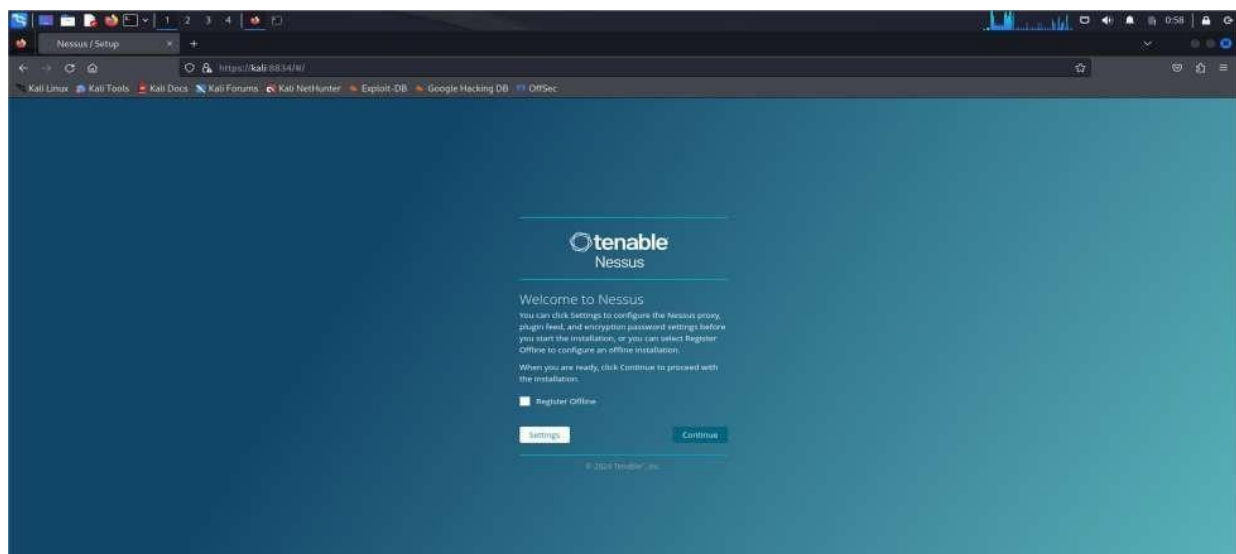
Feb 10 00:56:06 kali systemd[1]: Started nessusd.service - The Nessus Vulnerability Scanner.
Feb 10 00:56:08 kali nessus-service[7754]: Cached 0 plugin libs in 1msec
Feb 10 00:56:08 kali nessus-service[7754]: Cached 0 plugin libs in 0msec

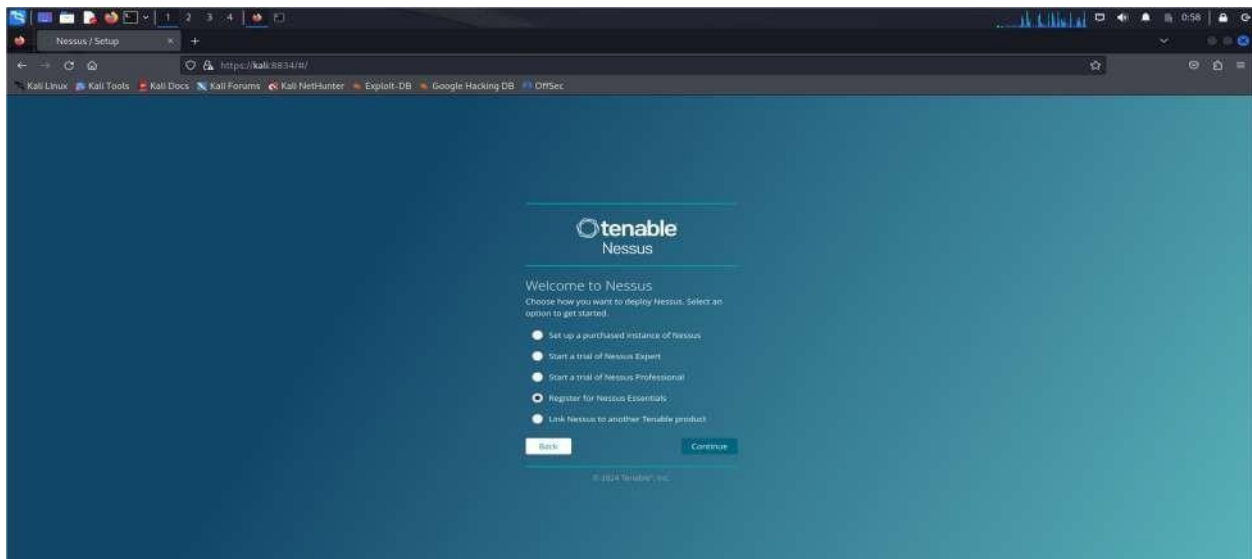
(root@kali)~#
```

On your browser, go to <https://kali:8834/>. It would show a warning page. Click on Advanced. Then, click on Accept Risk and Continue.

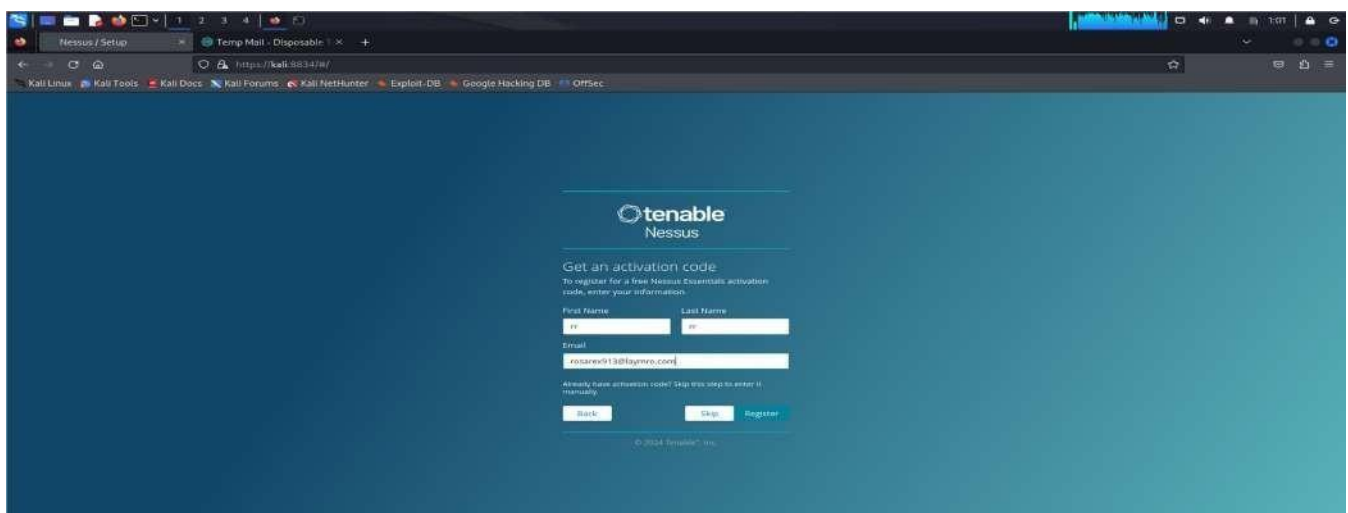


7. Choose the Nessus Product you prefer. If you want the free version of Nessus, click on **Register Nessus Essentials**.

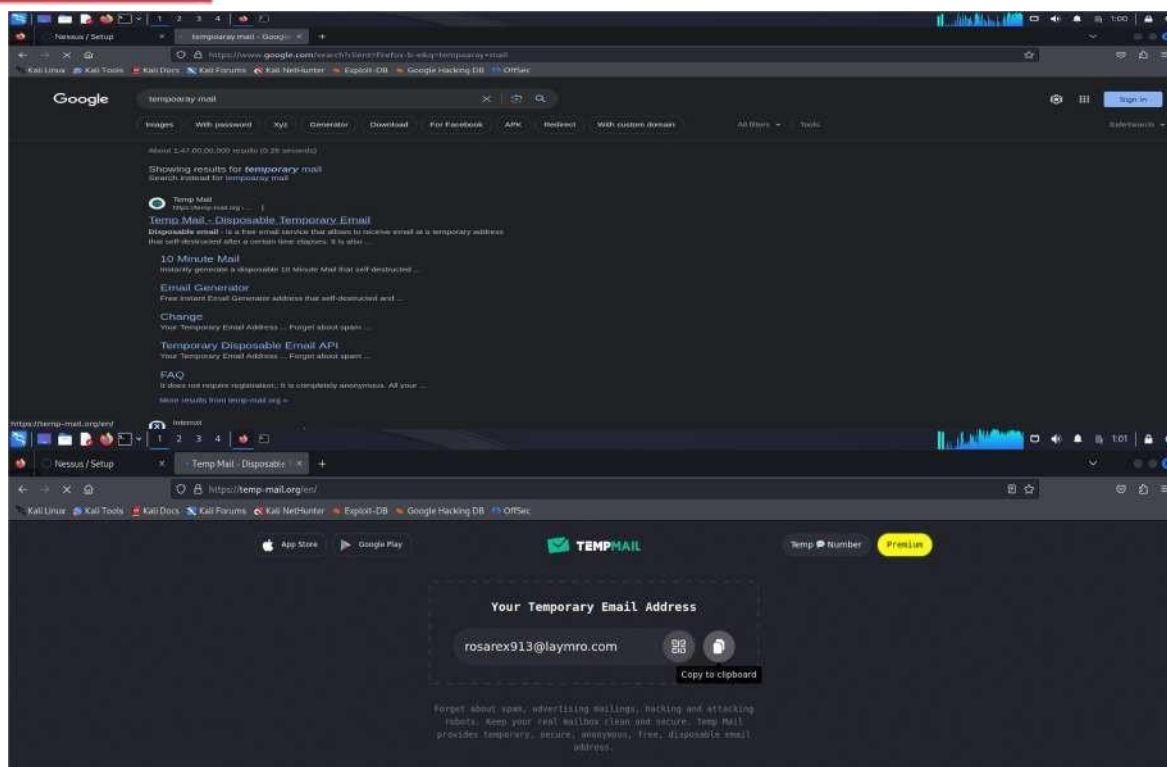




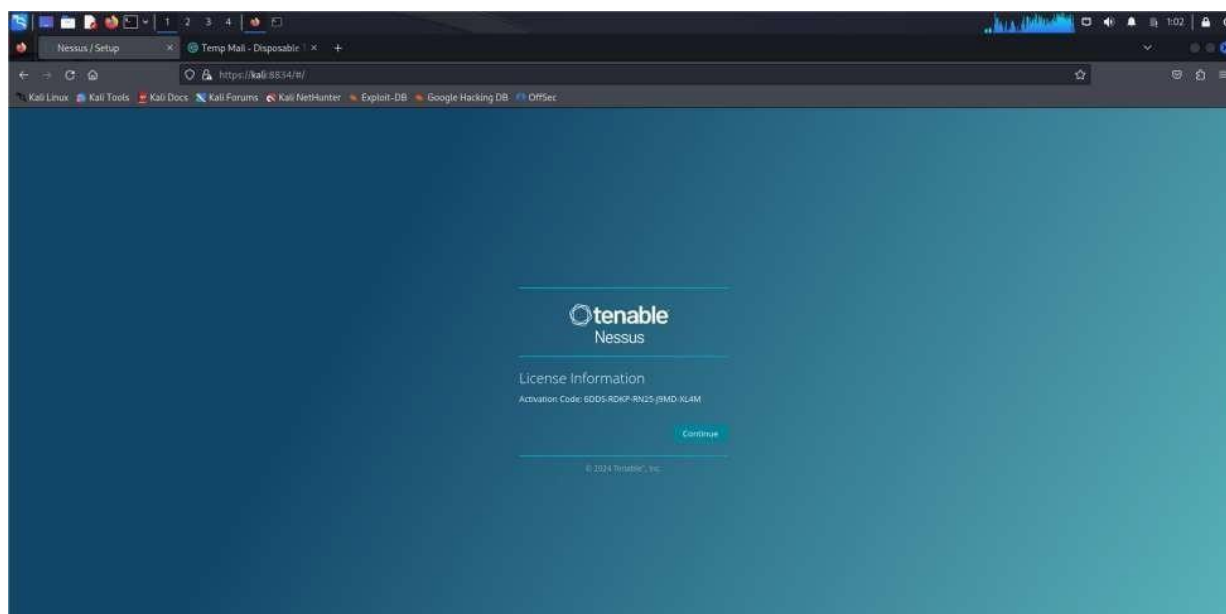
Enter your name and email address to receive an activation code by email. Paste the activation code into the space provided and choose a username and password.

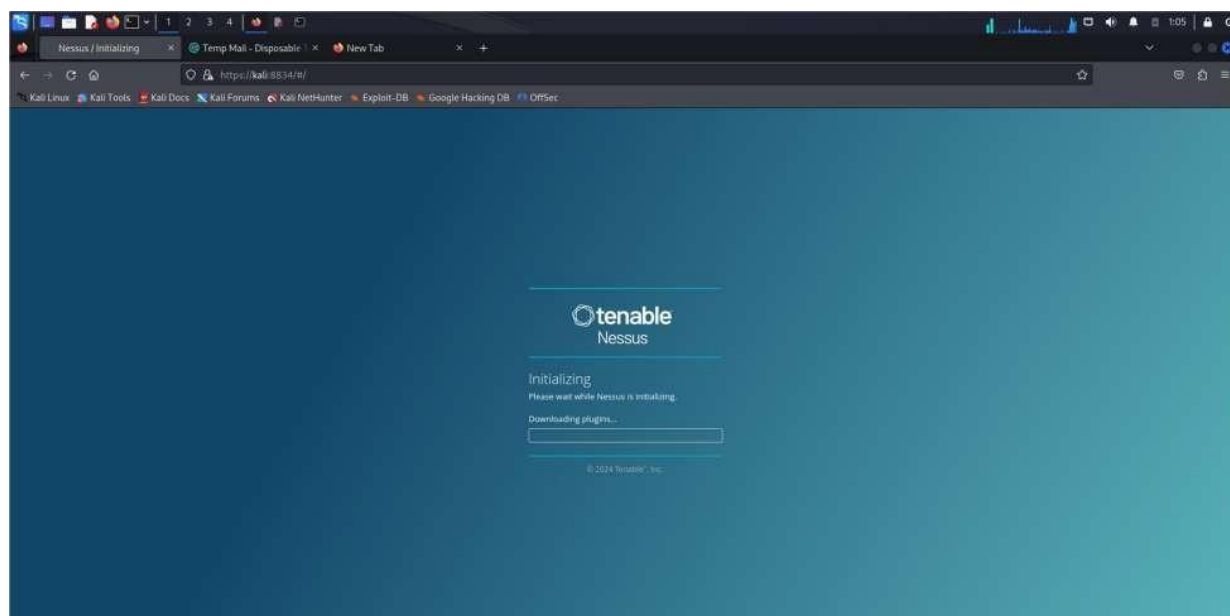
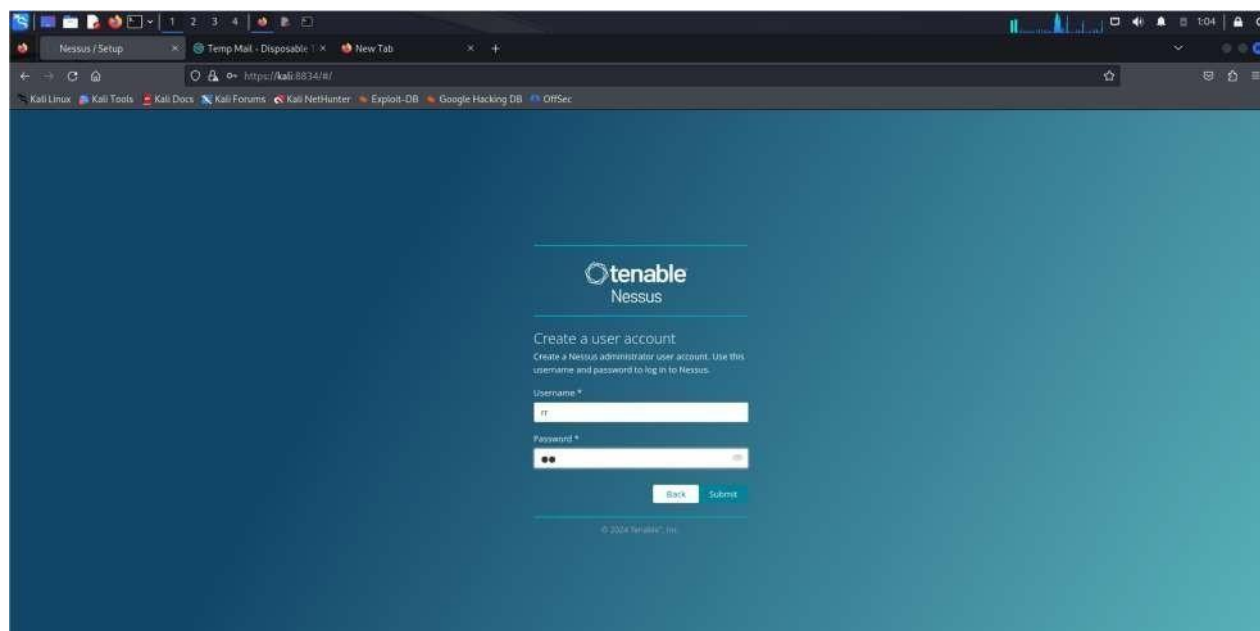


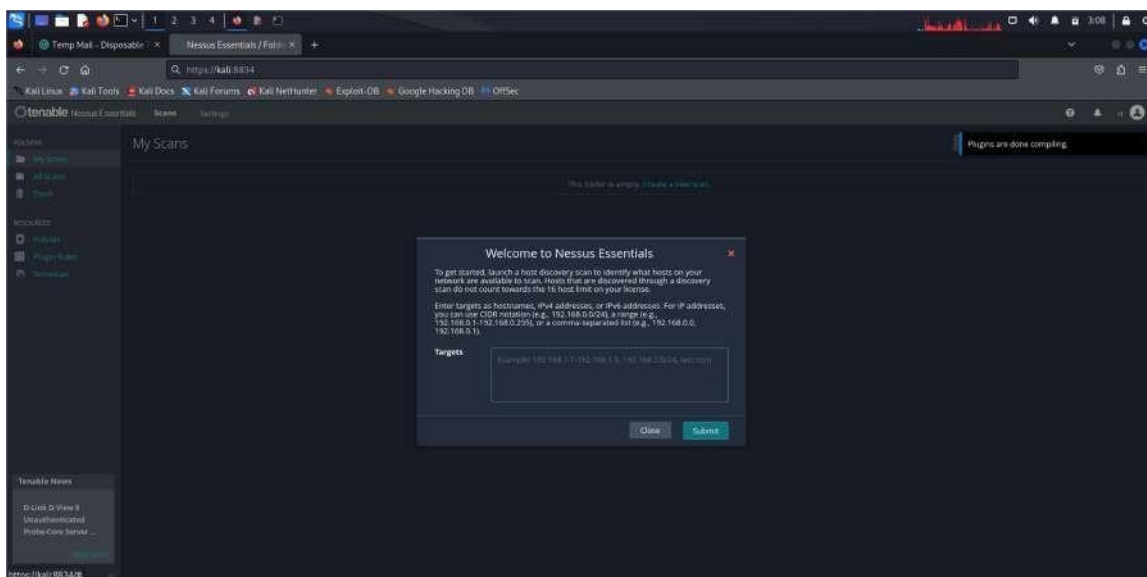
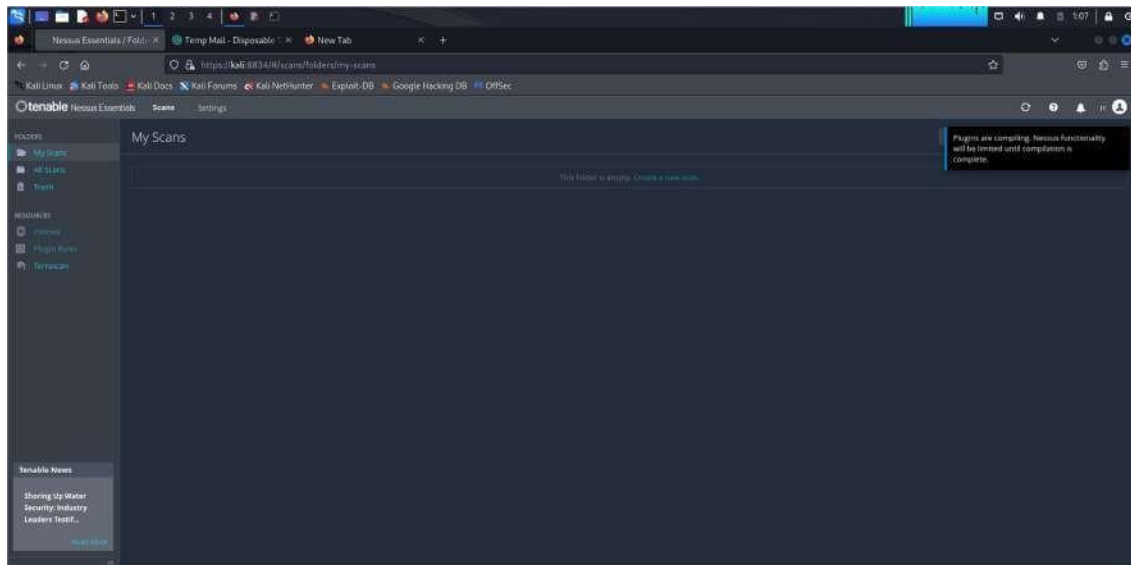
You can use email address as temporary you can visit online temporary email address examples below



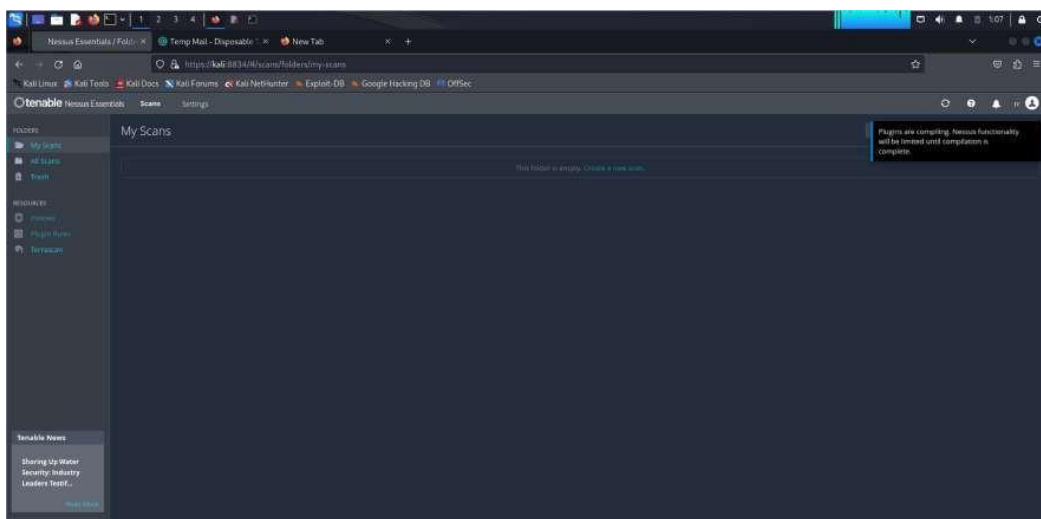
Allow Nessus to download the necessary plugins.



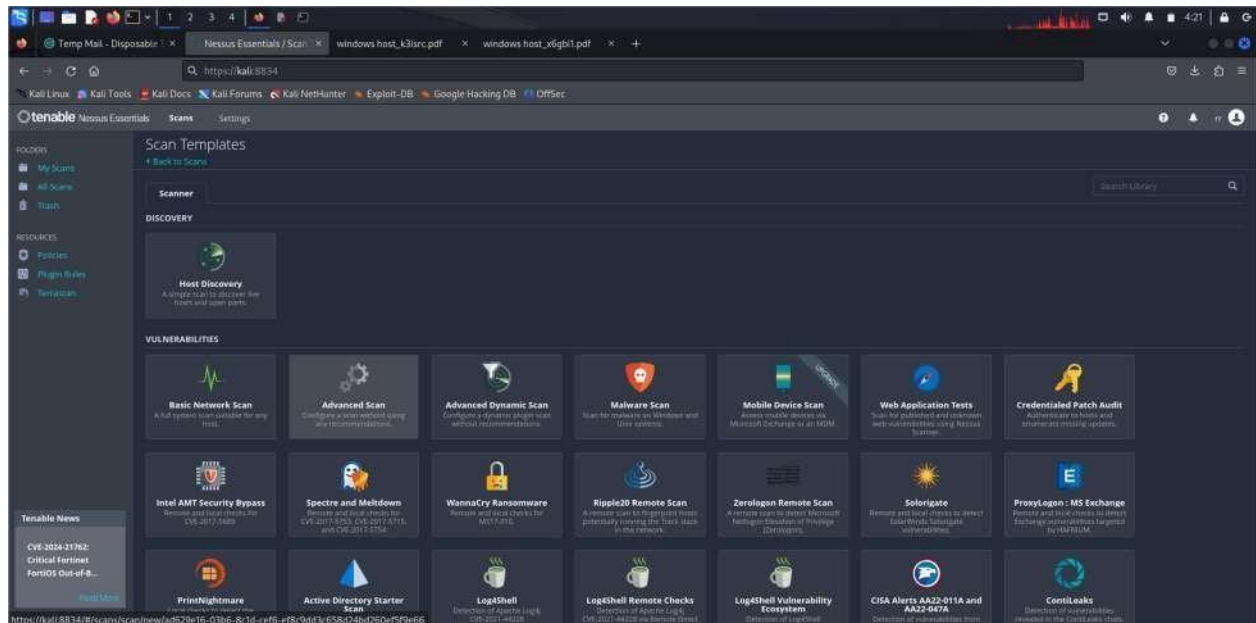




Click on New Scan to begin scanning for vulnerabilities

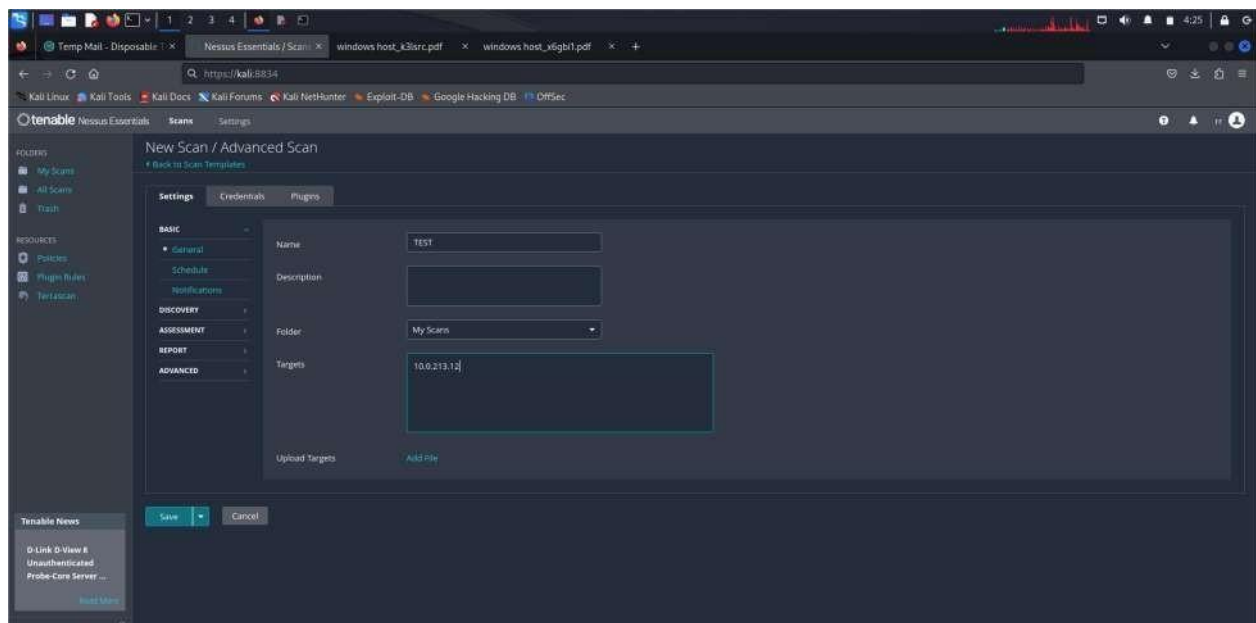


Click on Advance scan



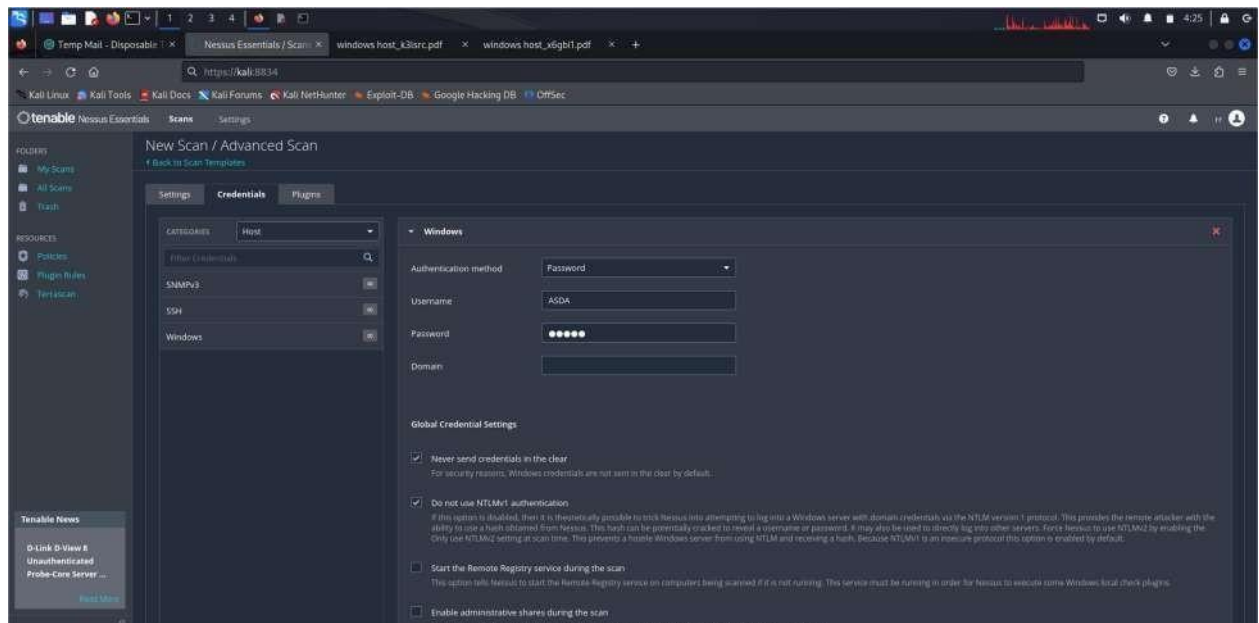
Give name for scanning (e.g test, windows scan. etc)

Give the ip address of your windows that you want to scan for vulnerabilities in
Target box shown in the below

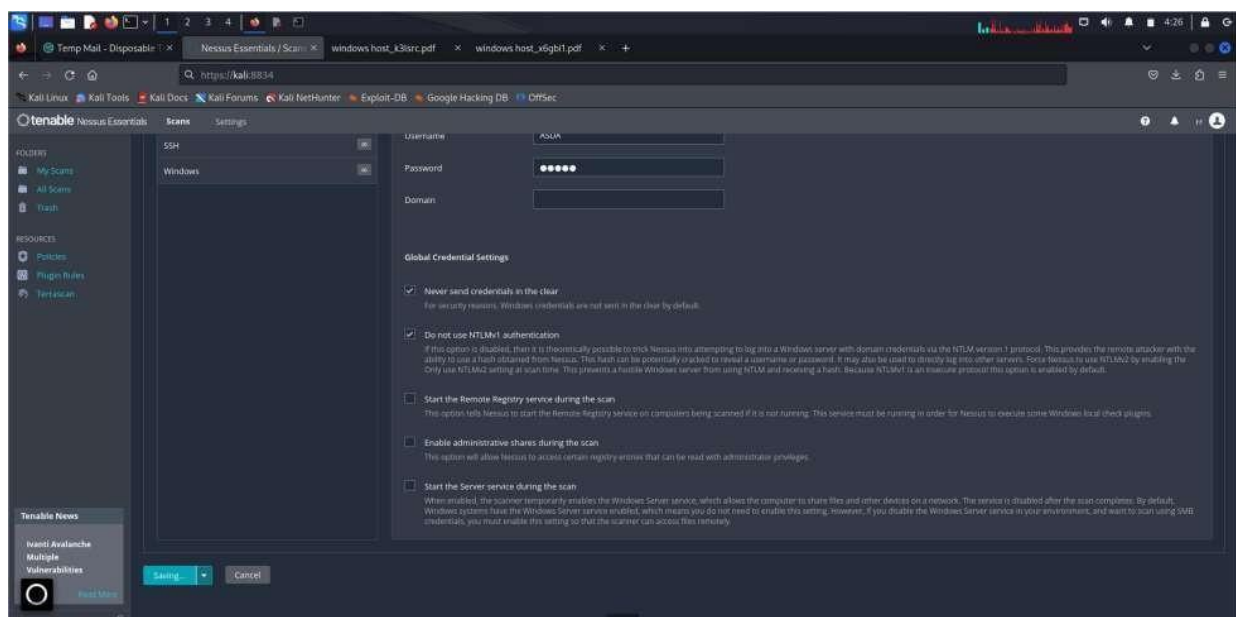


Give the credentials of your windows like username and password if you known
(OPTIONAL).

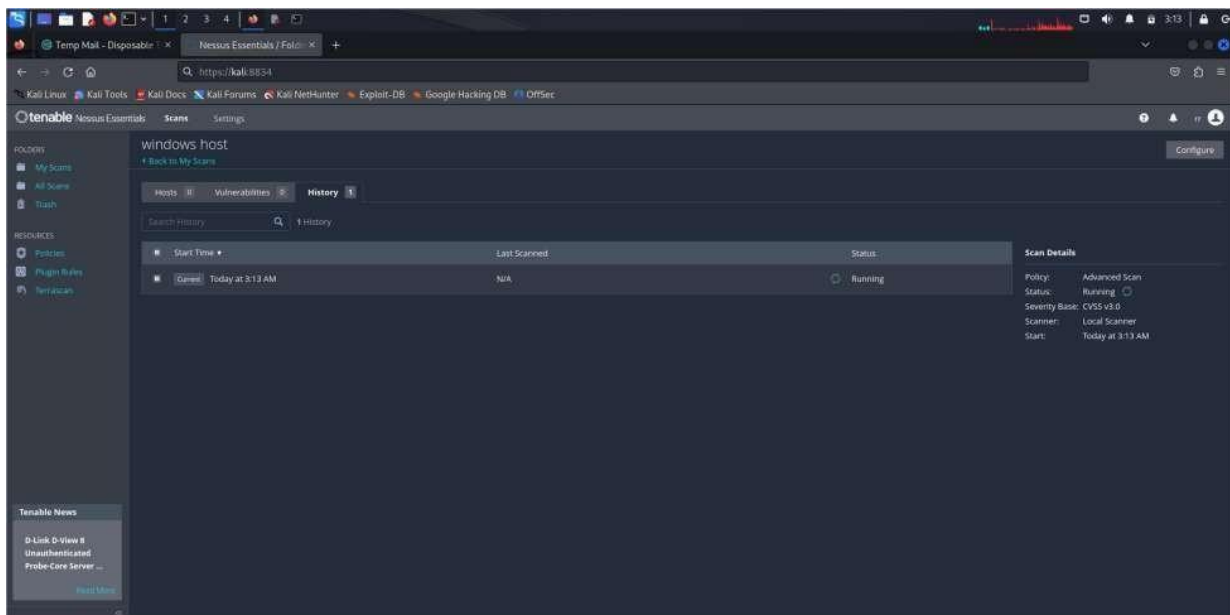
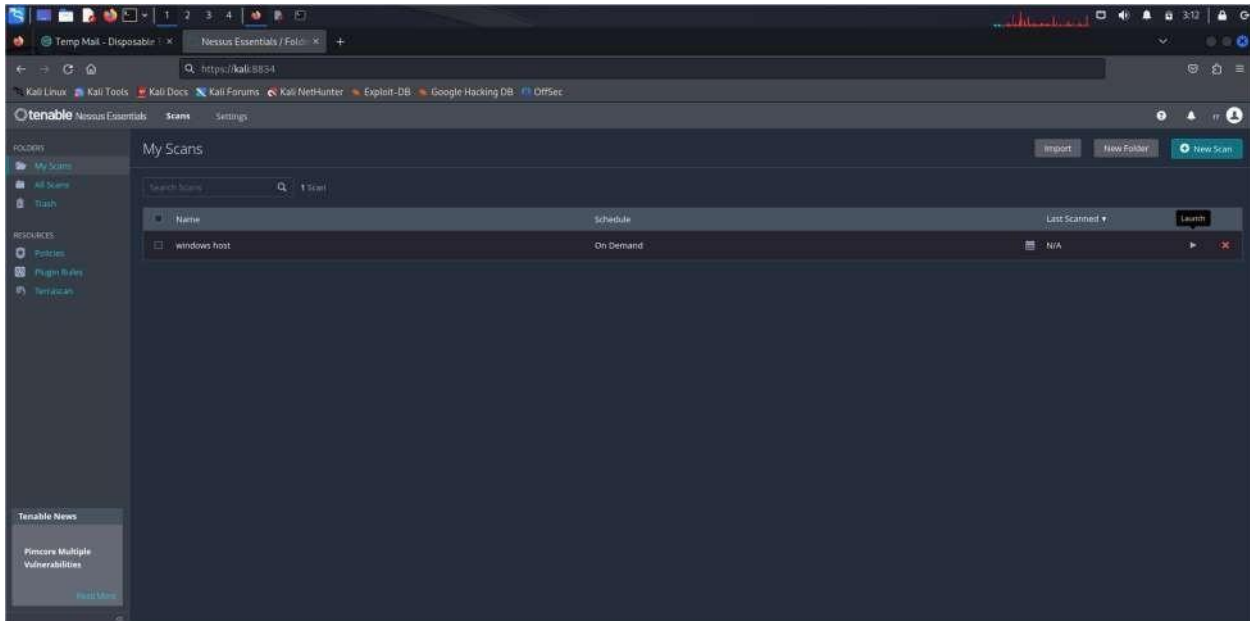
NOTE:- Use of giving credential it will help you scan more into your system.



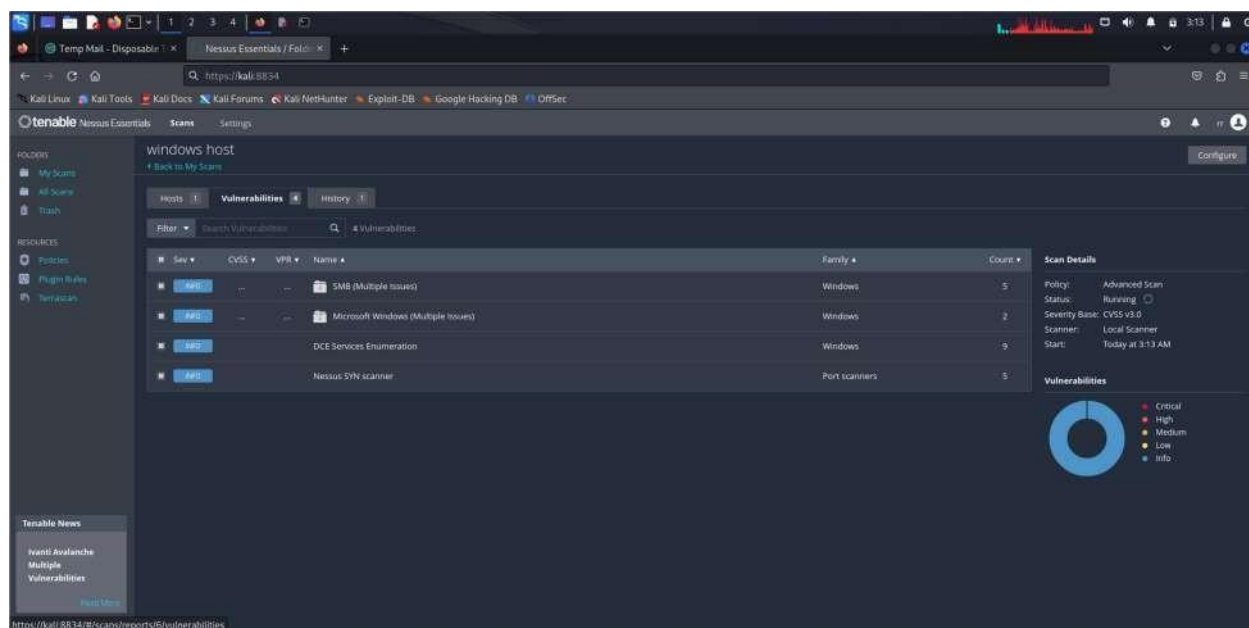
Go down and save the Progress



Click on launch



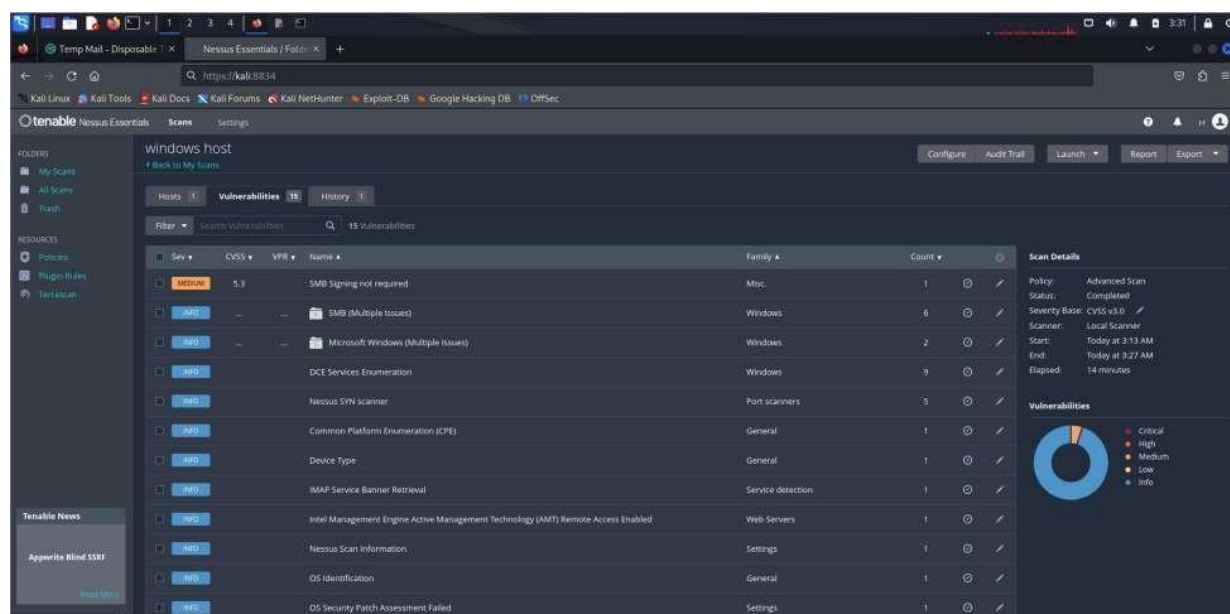
Wait for Some time to get the Output.



The screenshot shows the Nessus Essentials interface. The main panel displays the 'windows host' scan results. The 'Vulnerabilities' tab is selected, showing a table of 4 vulnerabilities. The table has columns for Severity, CVSS, VPR, Name, Family, and Count. The vulnerabilities listed are:

Severity	CVSS	VPR	Name	Family	Count
Info	-	-	SMB (Multiple Issues)	Windows	5
Info	-	-	Microsoft Windows (Multiple Issues)	Windows	2
Info	-	-	DCE Services Enumeration	Windows	9
Info	-	-	Nessus SYN scanner	Port scanners	5

On the right, the 'Scan Details' section shows: Policy: Advanced Scan, Status: Running, Severity Base: CVSS v3.0, Scanner: Local Scanner, Start: Today at 3:13 AM. Below this is a 'Vulnerabilities' donut chart showing the distribution of severity levels: Critical (0), High (0), Medium (0), Low (0), and Info (4).



The screenshot shows the Nessus Essentials interface after the scan is completed. The main panel displays the 'windows host' scan results. The 'Vulnerabilities' tab is selected, showing a table of 15 vulnerabilities. The table has columns for Severity, CVSS, VPR, Name, Family, and Count. The vulnerabilities listed are:

Severity	CVSS	VPR	Name	Family	Count
Medium	5.9	-	SMB Signing not required	Misc.	1
Info	-	-	SMB (Multiple Issues)	Windows	6
Info	-	-	Microsoft Windows (Multiple Issues)	Windows	2
Info	-	-	DCE Services Enumeration	Windows	9
Info	-	-	Nessus SYN scanner	Port scanners	5
Info	-	-	Common Platform Enumeration (CPE)	General	1
Info	-	-	Device Type	General	1
Info	-	-	IMAP Service Banner Retrieval	Service detection	1
Info	-	-	Intel Management Engine Active Management Technology (AMT) Remote Access Enabled	Web Servers	1
Info	-	-	Nessus Scan Information	Settings	1
Info	-	-	OS Identification	General	1
Info	-	-	OS Security Patch Assessment Failed	Settings	1

On the right, the 'Scan Details' section shows: Policy: Advanced Scan, Status: Completed, Severity Base: CVSS v3.0, Scanner: Local Scanner, Start: Today at 3:13 AM, End: Today at 3:27 AM, Elapsed: 14 minutes. Below this is a 'Vulnerabilities' donut chart showing the distribution of severity levels: Critical (0), High (0), Medium (1), Low (0), and Info (14).

ON RIGHT SIDE CLICK ON REPORT TO GENERATE THE ENTIRE REPORT OF YOUR SYSTEM VULNERABILITY

